Policy Research 2024-19-05

2024 K-Innovation Partnership Program with Perú

S&T Planning and Technological Forecasting for Prioritized Sectors in Perú

STEPI SCIENCE AND TECHNOLOGY POLICY INSTITUTE

CONCYTEC CONSEJO NACIONAL DE CIENCIA, TECNOLOGÍA E INNOVACIÓN

2024 K-Innovation Partnership Program with Peru

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Program Investigators:	Jihyun Kim, Research Fellow, Office of Sustainable Innovation Policy Research, STEPI Dong Un Park, Research Fellow, Office of Sustainable Innovation Policy Research, STEPI
Project Manager:	ChiUng Song, Senior Research Fellow, Office of Sustainable Innovation Policy Research, STEPI
Researchers (Authors)	ChiUng Song, Senior Research Fellow, Office of Sustainable Innovation Policy Research, STEPI Sohyun Kwon, Senior Researcher, Division of Global Innovation Strategy Research, STEPI Hwanil Park, Senior Research Fellow, Division of Global Innovation Strategy Research, STEPI Johng-Ihl Lee, Professor, Department of Technology and Society, SUNY Korea Hyun Yim, Senior Research Fellow, Center for Technology Foresight, KISTEP Ahreum Hong, Professor, Kyung Hee University Sehee Park, CTO, Deliverdy Agnes Franco, Director, Research&Studies, CONCYTEC Victor Izaguirre, Director, Policies and Programs, CONCYTEC Felipe Valentín, Senior Specialist, CONCYTEC John Collantes, Senior Specialist, CONCYTEC
Project Coordinator:	Sohyun Kwon, Senior Researcher, Division of Global Innovation Strategy Research, STEPI

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5th-7th Flrs., Building B, Sejong National Research Complex, 370 Sicheong-daero, Sejong-si, 30147.
Republic of South Korea
Tel: +82-44-287-2000 Fax: +82-44-287-2068
www.stepi.re.kr
For more information, visit our website
iicc.stepi.re.kr

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Executive Summary

In 2022, the K-Innovation Partnership Program with Peru was launched to strengthen the capacity of the Peruvian government in science, technology and innovation (STI) policies. To this end, the National Council of Science, Technology & Technological Innvation (CONCYTE) and Science & Technology Policy Institute (STEPI) have agreed to first sign a Memorandum of Understanding and promote the transfer of technology forecasting methodology. The STEPI team initially presented theories of technology forecasting methods and delivered relevant Korean case studies to their Peruvian counterparts via video conferences. In October 2022, the STEPI team visited Lima and held the Dissemination Workship with CONCYTEC to conduct on-site practice about technology forecasting with Peruvian experts.

Based on the 2022 activities and achievement, STEPI and CONCYTEC tried to establish a technology roadmap for 6 priortized sectors indentified by the Peruvian government in 2023. After the kick-off meeting via video conference, STEPI team invited the number of Peruvian experts to Korea and had the Capacity Building Workshop. In this workshop, 6 experts representing each sector participated in the practice and they led the constructing drafts of technology roadmap for 6 priortized sectors. Subsequently, via a series of video conferences, the 6 technology roadmaps were progressed and developed. Ultimately, STEPI team and 6 representative experts along with CONCYTEC team were able to complete the final version of drafts for technology roadmap through the Dissemination Workshop held in Lima on October 2023. The drafts of technology roadmap for 6 priortized sectors were presented at the Media Day by Peruvian experts.

This year, STEPI and CONCYTEC have focused on the theme of R&D planning and technology foresight center. At first, STEPI team invited 5 Peruvian experts from CONCYTEC and other related institutes to Korea and held the Capacity Building Workshop on May. In this workshop, the number of Korean experts has shared experiences of building R&D

governance as well as technology foresight center during the last few decades. In addition, Korean experts explained what kinds of mission and function should be there in the technology foresight center, and specified the current status of the Korean technology foresight center. At the same time. Korean experts tried to explain why the R&D governance would be very important issue, then Peruvian experts shared the current status of R&D budget as well policy isses to be addressed.

After having a series of video conferences, STEPI team made business trip to Lima, Peru on early October. STEPI team and CONCYTEC together reviewed the main outcome and policy implication from the Capacity Building Workshop on May. Then, CONCYTEC has figured out what needs to be done for the next few years in order to make institutional development including technology foresight center setting. At the Media Day held on October 10^{th}, Peruvian experts talked about the future plan for institutional progress and policy agenda to be addressed sooner than later. Finally, STEPI team and CONCYTEC sat down together and set up the follow-up project to be pursued jointly from the year of 2026. STEPI team and CONCYTEC plan to address the agenda of setting mid-to-long term masterplan for institutional development of CONCYTEC, setting government policy for promoting trilateral cooperation among industry, academia and government as well as setting masterplan for establishing bio R&D infrastructure and program.

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Chapter 1. Project Overview

1. Introduction

The K-Innovation Partnership Program represents one of Korea's flagship ODA programs dedicated to advancing science, technology and innovation (STI) policy development. Its primary objective is to deliver customized STI policy consultations that address the specific needs of Korea's partner countries. By drawing on Korea's well-established models of STI advancement and its extensive experience, the program also provides capacity-building training designed to strengthen the ability of partner countries to formulate and implement effective STI policies that contribute to sustainable development. Following the evaluation of project concept paper (PCP) submitted by applicant countries, policy consultations are provided, and Peru is one of the countries whose project was selected.[1]

The National Council for Science, Technology, and Technological Innovation (Consejo Nacional de Ciencia, Tecnología e Innovación Tecnológica, CONCYTEC) is the central agency responsible for formulating national-level policies and strategies in the field of science, technology and innovation (STI) in Peru. It serves as the managing body of the National Science, Technology, and Innovation System (Sistema Nacional de Ciencia, Tecnología e Innovación Tecnológica, SINACYT).

CONCYTEC's authority is derived from the Framework Law on Science and Technology No. 28303, under which it is empowered to regulate, guide, promote, coordinate, supervise, and evaluate scientific, technological, and innovation activities at the national level. It oversees SINACYT, a system composed of academia, national research institutes, business

[1] 2022 K-Innovation Partnership Program with Peru; 2023 K-Innovation Partnership Program with Peru

organizations, local communities, and civil society.

In 2018, the adoption of Peru's National Competitiveness and Productivity Policy further emphasized CONCYTEC's role in fostering innovation, promoting technology adoption, and enabling technology transfer. This policy prioritizes strengthening capacities for science and technology policy planning and foresight development within CONCYTEC.

The National Competitiveness and Productivity Policy aims to stimulate Peru's economic growth and enhance its competitiveness in domestic and international markets. It encompasses a range of strategies and programs designed to encourage sustainable development and social inclusion. The implementation of these strategies relies on the collaboration between government agencies, the private sector, and academic institutions.

Against this backdrop, CONCYTEC has focused on advancing Peru's sustainable growth and innovation through the establishment of a policy framework for science and technology planning and technological foresight in six promising industrial sectors. To achieve this, it seeks policy advice and capacity-building training, leveraging Korea's experiences and expertise in the field of STI.

The six prioritized sectors identified by CONCYTEC are:

1) Agro-industry & food processing
2) Forestry
3) Textiles & apparel
4) Mining & related manufacturing
5) Advanced manufacturing
6) Creative industries

CONCYTEC hopes to benefit from Korea's proven approaches in science, technology and innovation(STI) to build a robust policy framework and enhance its capabilities to drive growth and innovation in these strategic sectors. In particular, CONCYTEC is prioritizing the enhancement of capacities to design and establish a R&D planning and Technology Foresight Center planning.

2. Objectives

The project is designed to support CONCYTEC in improving the capacity of Peruvian policymakers and experts to design R&D programs, by sharing STI planning and technology foresight methodologies and supporting the creation of technology roadmaps. It is also tied to the establishment of a technology foresight-specialized organization, with the aim of building capacity for related policy development.

The objectives of the project include the following activities[2]:

1) Activity 1(2022): Development of CONCYTEC's capacity and institutional building plan to improve the foundations for S&T planning and technological forecasting for the prioritized sectors.

2) Activity 2(2023): Creation of the technology roadmap and formulation of the mid- to long-term technology strategy based on technological forecasting for the prioritized sectors.

3) Activity 3(2024): Development of the R&D program based on a technology roadmap for the prioritized sectors and establishment of a permanent center for technological forecasting.

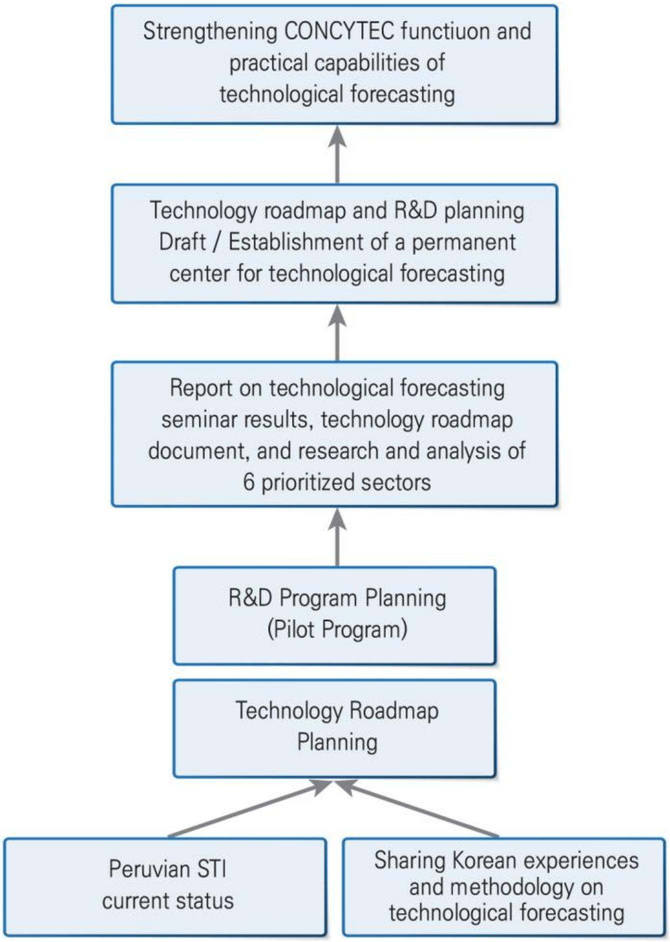
The theme of the third year in 2024 is to support capacity building for Peruvian policymakers and experts, centered around CONCYTEC, in sharing science and technology planning and technology foresight methodologies, establishing technology roadmaps, R&D program planning capabilities, and technology forecasting centers establishing draft.

[2] 2022 K-Innovation Partnership Program with Peru; 2023 K-Innovation Partnership Program with Peru

3. Project Framework

The project framework is designed to enhance Peru's capacity in S&T planning and technological foresight within prioritized sectors. Implemented over a three-year period, the project aims to assess the current status of Peru's STI landscape, transfer methodologies for technological foresight, and support the development of technology roadmaps and guidelines for mid- to long-term strategies. It further includes the design of R&D programs based on these roadmaps and the establishment of a specialized institution dedicated to technological forecasting.

[Figure 1-1] Analytical Framework



Source: 2023 K-Innovation Partnership Program with Peru, p. 6

[Table 1-1] Research Framework

Year	2022 (1st year)	2023 (2nd year)	2024 (3rd Year)
Annual Objectives	Develop CONCYTEC's capacity and institutional building plan to improve the foundations for S&T planning and technological forecasting for the prioritized sectors.	Build the technology roadmap and mid-/long-term technology strategy through technological forecasting for prioritized sectors.	Develop the R&D program based on a technology roadmap for the prioritized sectors and establish a permanent center for technological forecasting.
Specific Objectives	1. Share the technology foresight methodology; 2. Current status of the 6 prioritized industries*: *① Agro-industry & food processing (Agribusiness) ② Forestry ③ Textiles & apparel ④ Mining & related manufacturing ⑤ Advanced manufacturing ⑥ Ecotourism, restaurant & catering, and creative Industries	Establish the technology roadmap for the 6 key sectors	1. Support establishment of the technology foresight center. 2. R&D planning (draft).
Main Activities	• Share Korea's experience of technology foresight; • Analyze the current status of the 6 prioritized sectors; • Enhance the expertise of CONCYTEC researchers, local experts in the 6 key sectors, and other key stakeholders on S&T planning and technological forecasting methodology	• Determine the detailed technology priorities in the 6 key sectors; • Derive the technology acquisition methodology; • Share the R&D planning methodology.	• Practical R&D planning; • Proposal for planning the technology foresight center

Source: 2023 K-Innovation Partnership Program with Peru, p. 7

In the third year of 2024, the goal is to provide policy consultation and capacity building on R&D program planning and on the establishment of a specialized technology foresight center in Peru.

■ **First Year (2022):**

During the first year, the project conducted a comprehensive assessment of Peru's six prioritized industrial sectors and organized capacity-building workshops to enhance practical knowledge of technology foresight methodologies. Using the foundational data collected, an initial draft of technology roadmaps was developed, along with guidelines for medium- and long-term technology strategies.

■ **Second Year (2023):**

The primary objective of the second year was to develop technology roadmaps and establish medium- and long-term technology strategies for the six priority sectors. To achieve this, the project involved:

- Sharing methodologies and case studies on technology roadmap development.
- Building and operating technology roadmap working groups.
- Conducting capacity-building activities, including mini-Delphi surveys.
- Developing and distributing survey questionnaires and ensuring targeted respondent participation.
- Providing methods for analyzing survey data to inform roadmap development.

These activities facilitated the formulation of technology roadmaps for the six prioritized sectors and strengthened the capacity of stakeholders to apply roadmap methodologies effectively.

■ **Third Year (2024):**

Building on the outcomes of the first two years, the 2024 phase emphasizes sharing knowledge and expertise necessary for R&D program planning and the establishment of a specialized Technology Foresight Center. Using the technology roadmaps developed through technology foresight methodology, the project will guide the design of a pilot model for R&D program planning and propose a framework for the creation of the Technology

Foresight Center.

Through collaboration with Korean and Peruvian experts, the project aims to deliver actionable plans for R&D program development and the establishment of a Technology Foresight Center in Peru, thus contributing to the nation's capacity for innovation and strategic planning.

4. Project Team

4.1 Research Team

Korean and Peruvian research teams are assigned to carry out joint studies aimed at developing policy recommendations. In parallel, the Korean side intends to provide practical capacity-building support and help Peru internalize the ability to formulate implementable plans.

[Table 1-2] Korean Research Team

Organization	Position	Name	Main roles
STEPI	Senior Research Fellow	ChiUng Song	PM • Strategies &policy analysis for Peruvian science & technology policy and technology foresight planning; • Planning and operation of a technology roadmap working group to promote the 2nd Delphi Survey; • Building technology roadmap for 6 prioritized sectors in Peru; • Policy consultation and invitation training and local capacity building workshop for deriving R&D planning draft for 6 prioritized sectors and establishment of technology foresight center draft; • Planning of Korea-Peru cooperation project

Organization	Position	Name	Main roles
STEPI	Senior Researcher	Sohyun Kwon	PC, PI • Preliminary analysis of the current state of science & technology in Peru; • Strategies & policy analysis for Peruvian science & technology policy and technology foresight planning; • Planning and operation of a technology roadmap working group to promote the 2nd Delphi Survey; • Supporting to build technology roadmap for 6 prioritized sectors in Peru; • Planning and Operating on invitation training and local capacity building workshop for deriving R&D planning draft for 6 prioritized sectors and establishment of technology foresight center draft; • Planning of Korea-Peru cooperation project
	Senior Research Fellow	Hwanil Park	• Policy consultation for Kickoff meeting, invitational training for deriving R&D planning draft for 6 prioritized sectors and establishment of technology foresight center draft; • Planning of Korea-Peru cooperation project
SUNY Korea (State University of New York, Korea)	Professor	Johng-Ihl Lee	Policy consultation and practical training on R&D planning and utilization methods
KISTEP (Korea Institute of Science & Technology Evaluation and Planning)	Senior Research Fellow	Hyun Yim	Policy consultation and practical training on technology foresight center establishing
Kyung Hee University	Professor	Ahreum Hong	Policy consultation and practical training on the application of policies based on the Delphi method
Deliverdy	CTO	Sehee Park	Policy consultation and practical training on the results analysis of the 2nd Delphi survey

[Table 1-3] Peruvian Research Team

Affiliation	Position	Name	Delegated role
CONCYTEC	Director of Research & Studies	Agnes Franco	Project Manager(PM)
	Senior Specialist	Felipe Valentín	Project Coordinator(PC)
	Senior Specialist	John Collantes	Proposal for Peruvian R&D governance
	Director of Policies and Programs	Victor Izaguirre, Director	Proposal for Peruvian R&D planning and technology foresight center establishing, Discussion of Korea-Peru cooperation project
	Senior Specialist	Camilo Figueroa	Deputy Project Coordinator(PC)
	Head of International Cooperation Office	Rocio Casildo	International Cooperation
National Agrarian University La Molina	(Former) Chief of Budget & Planning Office	Alberto E. Valcarcel	Local consultant for agro-industry & food processing industry (Agribusiness)
Environmental Investigation Agency Inc.	Forestry Specialist	Ines Dhaynee Orbegozo Sanchez	Local consultant for forest industry
National University of Engineering of Peru	Professor	Beatriz Gloria Orcón Basilio	Local consultant for textiles & apparel
Pontifical Catholic University of Peru	Lecturer	Jorge Dulanto Carbajal	Local consultant for mining & related manufacturing industry
Pontifical Catholic Universityof Peru	Professor	Marco Antonio Carbajal	Local consultant for advanced manufacturing industry
ESAN Graduate School of Business (AACSB–AMBA)	Professor	Otto Regalado-Pezúa	Local consultant for creative industries
Pontifical Catholic Universityof Peru	Professor	Carlos Hernandez	Proposal for establishment of Peruvian Technology Forecasting Center (plan)
Development Consultant	Consultant	Marisela Benavides	Measures for linking with the private sector to promote industrial innovation

5. Main Activities

1) Kickoff Meeting (February 13, 2024/ Online)

There was a kickoff meeting held in February 13, 2024, when the Korean expert group met with the relevant government officials and experts online to provide a clear understanding of the 2024 project and clarify its scope.

During the kickoff meeting in February 2024, the Korean and Peruvian research teams had an online discussion by delineating the 2024 project scope and direction and an overview of R&D planning and technology foresight center establishing through technology foresight methodology application. The meeting objective was to cover the following topics:

- To review the 2023 work progress and the 2024 work plan;
- To set a concrete direction for the 2024 project;
- To discuss on 2024 Delphi survey for 6 prioritized sectors;
- To prepare for Capacity-Building Training Workshop in South Korea

2) Capacity-Building Workshop (Invitational Training) in South Korea (May 18~24, 2024)

The Policy Practitioners' Capacity-Building Training Workshop was held from May 18~24, 2024 in South Korea to support planning capabilities for R&D program and technology foresight center for CONCYTEC and local experts, with the objectives as follows:

- To cultivate practical skills required for planning Peruvian R&D program and establishing technology forecasting center
- To share knowledge and experience required for R&D program planning and technology forecasting center establishment based on technology roadmap for 6 prioritized sectors in 2022-2023

- To draw up a R&D planning draft;
- To draw up a technology foresight center
- To take a field trip and experience Korean culture

The program was structured as follows:

- Lectures and discussions: 1) Results of the 1st Delphi survey analysis and the 2nd Delphi survey method, 2) R&D planning using Korea's industrial technology history and the Delphi method, 3) Korea's R&D governance, 4) In-depth discussion on R&D planning, 5) Korea's experience in operating a technology foresight center, 6) Delphi method and policy making
- Presentations by Peruvian experts: R&D program planning (draft), CONCYTEC's technology foresight center planning (draft)

3) 2nd Delphi Survey (August 1~October 7, 2024)

The 2nd Delphi survey was conducted from August 1 to October 7, 2024, targeting experts from academia, industry, and research in six prioritized sectors in Peru.

4) Dissemination Workshop & Media Day in Peru (October 5-13, 2024)

The Dissemination Workshop & Media Day was held in Lima from October 5 - 13, 2024. This workshop was held for CONCYTEC policy practitioners and stakeholders (government policymakers, research institutes, universities, private sector, general public, etc.) to share knowledge and experience necessary for planning R&D programs and establishing technology foresight center based on the Technology Roadmap for six prioritized sectors and to strengthen the policy makers' capacities.

- In-depth discussion on the results of the technology roadmap analysis with the Technology Roadmap Working Group (STEPI+CONCYTEC Research Team) and the respondents of the 2nd Technology Foresight Delphi Survey

- Draft for R&D planning and establishment of a technology forecast center (plan) through in-depth theory and practice

In addition, a media day co-hosted by STEPI-CONCYTEC-PUCP was held to share the R&D program plan and the establishment plan of the Technology Foresight Center and to disseminate related research results.

- Sharing the R&D program and technology forecast specialized agency planning process
- 2024 business review and dissemination of the results of the 2024 K-Innovation Partnership Program with Peru
- Discussion of the 3-year project wrap-up
- To disseminate the outcome of the 2024 K-Innovation Partnership Program with Perú;
- To discuss follow-up actions and a new scale-up project in the future

5) Publication of the Report

After each yearly project is completed, STEPI will issue final reports with policy recommendations to be shared with participants. The reports will be posted on the STEPI website in both English and Korean, except when classified as confidential.

6) Monitoring and Evaluation of Project Results

During and after the project, both sides will actively participate in monitoring and managing project results. This entails conducting interviews with related stakeholders from Peru, collecting data and evidence of the results, and reporting the findings that concern the following:

- Implementation of the policy recommendations proposed through the project with Peru, such as introduction or modification of policies, laws, institutions, systems, etc.;
- Prospects for follow-up collaboration.

6. Project Schedule

In 2024, the project was carried out according to the following schedule and its activities were carried out in either an offline or online format.

[Table 1-4] 2024 Project Schedule

Activities	Jan.	Feb.	Mar.	Apr.	May	June	July	Aug.	Sep.	Oct.	Nov.	Dec.
Preliminary Meeting (online)												
Kickoff Meeting (online)												
Policy Practitioners' Capacity-Building Workshop in Korea												
Interim Reporting												
Dissemination Workshop & Media Day in Peru												
Final Reporting												
Exploring a new project												

7. Deliverables

- **Deliverable 1:** To provide policy consultation through joint research to support Peru's capability to conduct S&T planning and technological forecasting in the policy areas agreed upon by the participants;
- **Deliverable 2:** To form a research team composed of experts, which may employ "local consultants" from Peru to support the research;
- **Deliverable 3:** To conduct meetings, workshops, interviews, and field studies on the previously agreed-upon topics and provide logistical support to coordinate the visits of Peruvian public officials & experts to South Korea to attend seminars, workshops, and training courses for the purpose of exchanging knowledge & experience;
- **Deliverable 4**: To exchange information & data on matters of common interest upon request;
- **Deliverable 5:** To seek opportunities to influence policy, decision-making processes, and follow-up projects & programs to achieve the mutually agreed objectives;
- **Deliverable 6:** To monitor and evaluate the implementation of the policy recommendations, and draft an annual report on the progress of the project results;
- **Deliverable 7:** To establish a technology roadmap(TRM) working group to effectively conduct a Delphi survey
- **Deliverable 8:** To submit the final report version including proposals for R&D program planning and technology foresight center establishing based on a draft of the technology roadmap for the six prioritized sectors

CHAPTER 2

TRM 2nd Delphi Survey Result

S&T Planning and Technological Forecasting for Prioritized Sectors in Perú

Chapter 2. TRM 2nd Delphi Survey Result

1. Summary[3]

- **Data collection**: Online survey form using Google Forms
- Questionnaires are identical to those of the 1st round.
- **Survey period**: 2024-08-01 and 2024-09-23
- Closed survey to gather 2nd round experts' opinion
- Questionnaires were distributed to 270 respondents from the 1st year Delphi survey.
- **Respondents**: 235
- This report only analyzed 185 responses from those who provided email addresses and identified as first-year Delphi survey respondents.
- Given the time gap between the first and second survey rounds, some respondents reported changes in their organization, location, or specialty. To ensure consistency in the analysis, the information from respondents in the first round was applied to their second-year responses

[3] This report presents the results of the 2nd year Delphi survey conducted by the Peru TRM project. The survey was conducted to gather experts' second opinions on the technology readiness of various industries in Peru. The survey was conducted online using Google Forms and was distributed to 270 respondents from the 1st year Delphi survey. A total of 235 responses were received, and this report analyses 185 responses from those who provided identifiable information to confirm first-year Delphi survey respondents.

<Chapter 2. TRM 2nd Delphi Survery Result> was written by Sehee Park, CTO, Deliverdy.

2. Descriptive Statistics

2.1 Response Rate

The valid second respondents were 185 out of 270 first-year respondents. The overall response rate was 69%. Response rate by experts' specialty is presented in Table 2-1.

[Table 2-1] Response Rate by Specialty

Specialty	First Year	Second Year	Response Rate
Advanced Manufacturing	47	29	62%
Agribusiness	66	42	64%
Creative Industry	39	24	62%
Forestry	24	19	79%
Mining And Materials	47	33	70%
Textile And Apparel	47	38	81%
Total	**270**	**185**	**-**

3. Comparing Two Delphi Survey Results

3.1 Technology Capability

Peruvian experts assessed the technological capacity of six industries across five different technologies using the following categories:

- HR: the competency of human resources
- Infra: the state of R&D infrastructure
- Invest: the current level of R&D investment
- Eco: the current economic significance

Responses from the first and second years were compared using a two-sided t-test to detect changes in the technology readiness level. A p-value of less than 0.05 indicates that the means of the two groups are significantly different, with a 95% confidence level.

3.1.1 Key Findings

■ **Average Scores: The mean scores across the categories are:**

- HR: 2.77
- Infrastructure: 2.57
- Investment: 2.51
- Economic: 3.27
- Total: 7.85

■ **Score Ranges:**

- The lowest score (2.08) is in Investment, indicating that Investment may be a relatively weaker area across industries.
- The highest individual area score (3.69) is in Economic Impacts, suggesting that the economic outlook or performance is generally viewed more favorably than in other areas.

3.1.2 Indications

1. Higher Economic Scores: Consistently higher scores in the Economic area across industries indicate that overall economic stability or growth may be a focal point or strength relative to other categories like HR or Infrastructure.
2. Potential Areas for Improvement: HR and Investment scores are lower, suggesting that industries could focus on strengthening capabilities or resources in these areas.
3. Total Score Variation: Industries' total scores vary widely, ranging from 6.43 to 9.66. This indicates significant differences in overall performance or satisfaction across sectors and technologies.

In summary, no significant differences were found between the means of the first and second-year responses across all industries and technologies. The detailed results of the t-test are shown in Table 2-2.

3.1.3 Industry Comparison in Technology Levels

- Mining and Materials: This industry scores the highest overall, with a total of 8.75, reflecting high confidence in its technological level. The strongest areas are Economic Impacts (3.57) and HR (3.05), suggesting a perceived robust economic impact and human resources capacity in this field.
- Textile and Apparel: With a total score of 8.61, this industry is also viewed confidently, particularly in HR (3.09) and Economic Impacts (3.42). This implies that experts see solid technological capabilities and economic output in the workforce.
- Creative Industry: With a score of 8.14, this industry shows moderate confidence levels, especially in HR (2.90) and Infrastructure (2.63). Investment (2.60) and economic impacts (3.21) suggest stable but lower perceived capabilities in these areas.
- Agribusiness: With a total of 7.33, Agribusiness has higher confidence in Economic aspects (3.26) but scores lower in HR (2.60) and Investment (2.34), indicating room for improvement in these technological areas.
- Advanced Manufacturing: At 7.20, this industry has moderate confidence. Economic Impacts (3.19) are slightly more substantial than other categories, while low HR and Investment scores (2.34) indicate perceived technological gaps.
- Forestry: With the lowest score (7.05), Forestry's scores in HR (2.51) and Investment (2.24) are notably low, while economic confidence (2.97) suggests a need for development across all technology levels.

3.1.4 Consensus Changes in Technology Levels

The changes in consensus between the two Delphi surveys were measured using standard deviation. This approach measures consensus based on the change in standard deviation (SD), indicating "consensus up" when SD decreases (indicating stronger agreement) and "consensus down" when SD increases (indicating weaker agreement) from the first to the second year.

The results, summarized in Table 2-2, suggest that consensus generally improved across all categories, as indicated by the higher counts for "Consensus UP" compared to "Consensus DOWN." For instance, there were 23 instances in the HR category where consensus increased versus 7 where it decreased, showing that expert opinions became more aligned in most cases. Infrastructure shows an even more substantial alignment improvement, with 27 consensus increases compared to just 3 decreases. Investment and Economic Importance also reflect positive trends with 29 and 20 "Consensus UP" cases, respectively. However, Economic Importance has a relatively higher proportion of "Consensus DOWN" cases, indicating some topics within this category may still have divergent opinions.

The analysis reveals an overall trend toward increased consensus among experts from the first to the second year, especially within the Investment and Infrastructure categories, where nearly all sub-categories showed improved agreement. The HR and Economic Importance categories also show overall positive trends and have more variability and areas where consensus remains lower. This indicates a strengthening alignment in expert perceptions of technological leadership, though certain areas, particularly in Economic Importance, may still require further clarity or convergence of expert opinions.

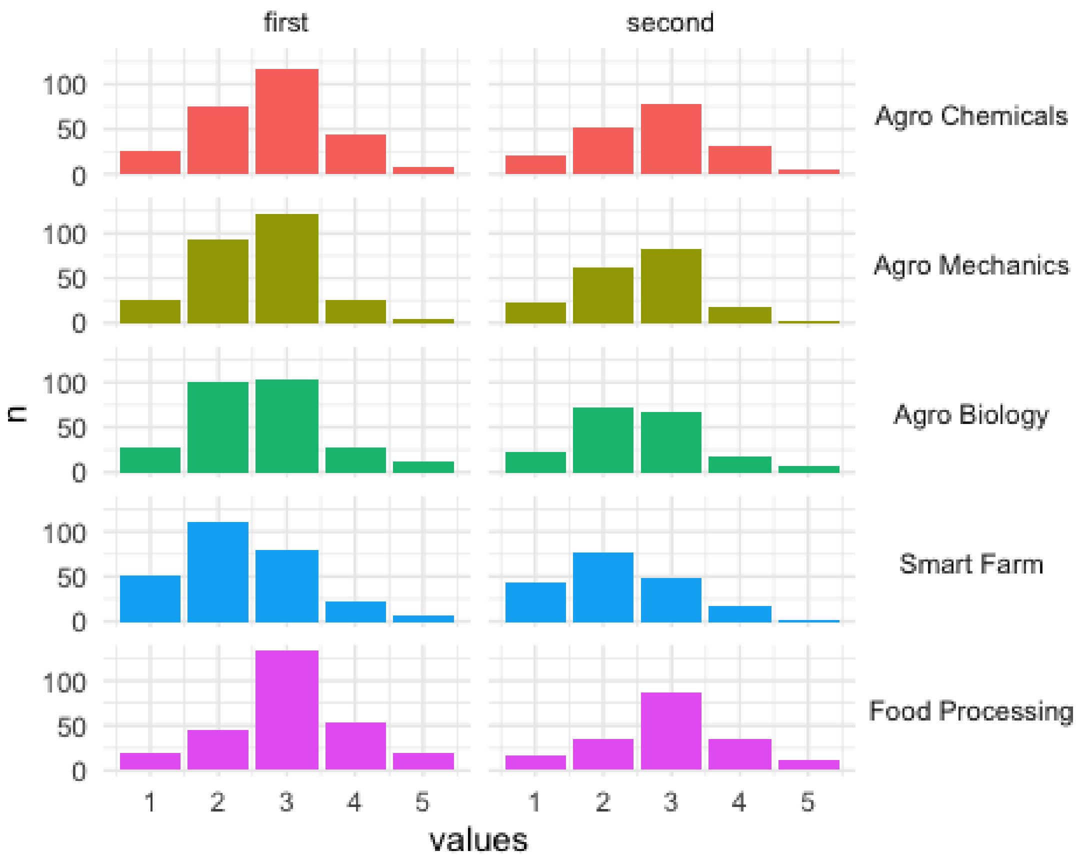
[Table 2-2] Consensus on Technology Levels

down	up
HR	
7	23
Infra	
3	27
Invest	
1	29
Eco	
10	20

3.1.5 Snapshot analysis of HR capability in Agribusiness

The general distribution of responses remains similar between first and second-year surveys. However, there was a minor shift in Agro Biology, where the maximum number of responses moved from value 3 to 2. This shift indicates a slight decrease in the perceived HR capability in Agro Biology in the second year. Figure 2-1 shows the distribution of responses for HR capability in Agribusiness.

[Figure 2-1] HR Capability in Agribusiness



3.2 Technology Realization

This section presents findings from a Delphi survey focused on assessing perceptions of technology realization in Peru compared to leading countries globally. Experts provided insights on three key areas: opinions regarding technology realization in Peru and leading markets, anticipated leadership in technology development, and estimated timelines for achieving technological milestones in both leading nations and Peru. This section offers a comprehensive outlook on Peru's positioning within the global technological landscape by capturing expert views on technological advancement's relative pace and scope. It identifies potential gaps or opportunities for accelerated growth in specific sectors.

3.2.1 Leading Country

The following question with pre-set categories measures the level of Peruvian experts' perception of the leading country in each technology field. The analysis aggregated responses that selected China, Japan, or Korea as a leading country as one variable C/J/K to conduct the chi-square test for comparing the difference between the first-year and second-year responses.

Note

Q. Which country or market area will lead the technology development? Select one country that holds the most advanced technology in each field.

1. US/EU
2. China
3. Japan
4. Korea
5. Others

The summary of leading country responses is given in Table 2-3 in Appendix B. The analysis reveals an overall trend toward increased consensus among experts from the first to the second year, especially within the Investment and Infrastructure categories, where nearly all sub-categories showed improved agreement. The HR and Economic Importance categories also show overall positive trends and have more variability and areas where consensus remains lower. This indicates a strengthening alignment in expert perceptions of technological leadership, though certain areas, particularly in Economic Importance, may still require further clarity or convergence of expert opinions.

The chi-squared test results (summarized in Table 2-3) indicate no statistically significant difference between the first and second-year survey responses across all industries and technologies, as all p-values are greater than 0.05. For instance, the p-value for "Agro Chemicals" is 0.924, and for "Agro Mechanics," it is 0.977, showing no significant change.

In short, Peruvian experts' opinions on the leading countries in each industry and technology remained unchanged between 2023 and 2024.

[Table 2-3] Chi-squared statistics on Leading Country between first and second year observations

Industry	Technology	P-value	Statistics	Statistical Significance
Advanced Manufacturing	Automation	0.9910534	0.01797371	FALSE
Advanced Manufacturing	Clean Manufacturing	0.9758093	0.04897624	FALSE
Advanced Manufacturing	Digital Production	0.9956223	0.00877458	FALSE
Advanced Manufacturing	Engineering Service	0.9849043	0.03042160	FALSE
Advanced Manufacturing	Robotics	0.7436076	0.59248372	FALSE
Agribusiness	Agro Biology	0.6327468	0.91536977	FALSE
Agribusiness	Agro Chemicals	0.9235683	0.15902100	FALSE
Agribusiness	Agro Mechanics	0.9773876	0.04574400	FALSE
Agribusiness	Food Processing	0.6533825	0.85118521	FALSE
Agribusiness	Smart Farm	0.9017430	0.20685137	FALSE
Creative Industry	Copyright System	0.9047827	0.20012087	FALSE

Industry	Technology	P-value	Statistics	Statistical Significance
Creative Industry	Digital Contents	0.8584521	0.30524886	FALSE
Creative Industry	Financial Service	0.8969840	0.21743440	FALSE
Creative Industry	Marketing Service	0.8806442	0.25420321	FALSE
Creative Industry	Media Service	0.9917223	0.01662435	FALSE
Forestry	Environment Management	0.7269448	0.63780950	FALSE
Forestry	Furniture Manufacturing	0.7464668	0.58480818	FALSE
Forestry	Precision Machineries	0.8539871	0.31567836	FALSE
Forestry	Saw Milling	0.8916498	0.22936363	FALSE
Forestry	Surface Treatment	0.8372908	0.35516757	FALSE
Mining And Materials	Casting And Welding	0.8216323	0.39292465	FALSE
Mining And Materials	Clean Energy Mining	0.9286541	0.14803800	FALSE
Mining And Materials	Geotechnical Engineering	0.7924122	0.46534704	FALSE
Mining And Materials	Material Processing	0.5558320	1.17457828	FALSE
Mining And Materials	Resource Exploration	0.9024960	0.20518195	FALSE
Textile And Apparel	Apparel Manufacturing	0.6715541	0.79632131	FALSE
Textile And Apparel	Digital Design	0.9265042	0.15267333	FALSE
Textile And Apparel	Logistics	0.8978709	0.21545786	FALSE
Textile And Apparel	Textile Manufacturing	0.5695987	1.12564652	FALSE
Textile And Apparel	Textile Materials	0.6873492	0.74982561	FALSE

3.2.2 Time of Realization: Leading Countries

Peruvian experts indicate the time of technology realization in leading by pre-set categories as below:

- Before 2023
- Between 2023 and 2027
- Between 2028 and 2032

- Between 2033 and 2038
- After 2038

Experts expect technologies to be realized imminently in all industries, as shown in Figure 2-2. However, the expected level of technology is not very futuristic. For example, the Creative Industry has the highest early realization counts, suggesting a faster technology lifecycle. Forestry and Agrobusiness showed near-future (2023-2027) projects, reflecting ongoing advancements. However, Advanced Manufacturing and Mining and Materials industries received more long-term projects than others.

This summarizes the Peruvian experts’ view that the leading countries have realized or will realize the technologies soon.

[Figure 2-2] Time of Realization (2024) in Leading Countries

The chi-square test revealed no significant difference between the first-year and second-year responses (p-value >= 0.05). Table 2-4 summarizes the chi-square test result. The ranks of each category did not change at all between the first- and second-year responses. In summary, the time of realization in leading countries did not change between 2023 and 2024.

[Table 2-4] Chi-squared statistics on 'Time of Realization' by leading countries

Industry	Technology	P-value	Statistics	Statistical Significance
Advanced Manufacturing	Automation	0.9440960	0.75716990	FALSE
Advanced Manufacturing	Clean Manufacturing	0.9172346	0.95031050	FALSE
Advanced Manufacturing	Digital Production	0.9839580	0.38156738	FALSE
Advanced Manufacturing	Engineering Service	0.9516873	0.69709014	FALSE
Advanced Manufacturing	Robotics	0.9969552	0.16028399	FALSE
Agribusiness	Agro Biology	0.8768934	1.20731135	FALSE
Agribusiness	Agro Chemicals	0.9722107	0.51312547	FALSE
Agribusiness	Agro Mechanics	0.9889960	0.31245504	FALSE
Agribusiness	Food Processing	0.7540155	1.90070473	FALSE
Agribusiness	Smart Farm	0.9901224	0.29519171	FALSE
Creative Industry	Copyright System	0.9994850	0.06488653	FALSE
Creative Industry	Digital Contents	0.9869114	0.34245354	FALSE
Creative Industry	Financial Service	0.9864999	0.34811848	FALSE
Creative Industry	Marketing Service	0.9921891	0.26102666	FALSE
Creative Industry	Media Service	0.9950994	0.20484935	FALSE
Forestry	Environment Management	0.9891962	0.30944670	FALSE
Forestry	Furniture Manufacturing	0.9621406	0.60826692	FALSE
Forestry	Precision Machineries	0.9868833	0.34284190	FALSE
Forestry	Saw Milling	0.9899110	0.29849620	FALSE
Forestry	Surface Treatment	0.9341552	0.83165034	FALSE
Mining And Materials	Casting And Welding	0.9120846	0.98484900	FALSE

Industry	Technology	P-value	Statistics	Statistical Significance
Mining And Materials	Clean Energy Mining	0.9897609	0.30082398	FALSE
Mining And Materials	Geotechnical Engineering	0.9826501	0.39788018	FALSE
Mining And Materials	Material Processing	0.9381351	0.80233241	FALSE
Mining And Materials	Resource Exploration	0.9995515	0.06050771	FALSE
Textile And Apparel	Apparel Manufacturing	0.9184792	0.94186537	FALSE
Textile And Apparel	Digital Design	0.8902184	1.12541956	FALSE
Textile And Apparel	Logistics	0.9843995	0.37593340	FALSE
Textile And Apparel	Textile Manufacturing	0.8529394	1.34940387	FALSE
Textile And Apparel	Textile Materials	0.9189581	0.49969637	FALSE

3.2.3 Time of Realization: Peru

Peruvian experts also provided their opinions on the time of technology realization in Peru. The following categories were used to measure the time of technology realization in Peru:

- Before 2023
- Between 2023 and 2027
- Between 2028 and 2032
- Between 2033 and 2038
- After 2038

Figure 2-3 shows the distribution of responses for the time of realization in Peru. The highest responses are in “Between 2028 and 2032,” indicating that Peru’s technologies are 5-9 years behind those of the leading countries. The distribution of responses is consistent across industries. Advanced Manufacturing showed higher numbers of responses after 2028, indicating that it is Peru's least developed technology.

■ **Key findings**

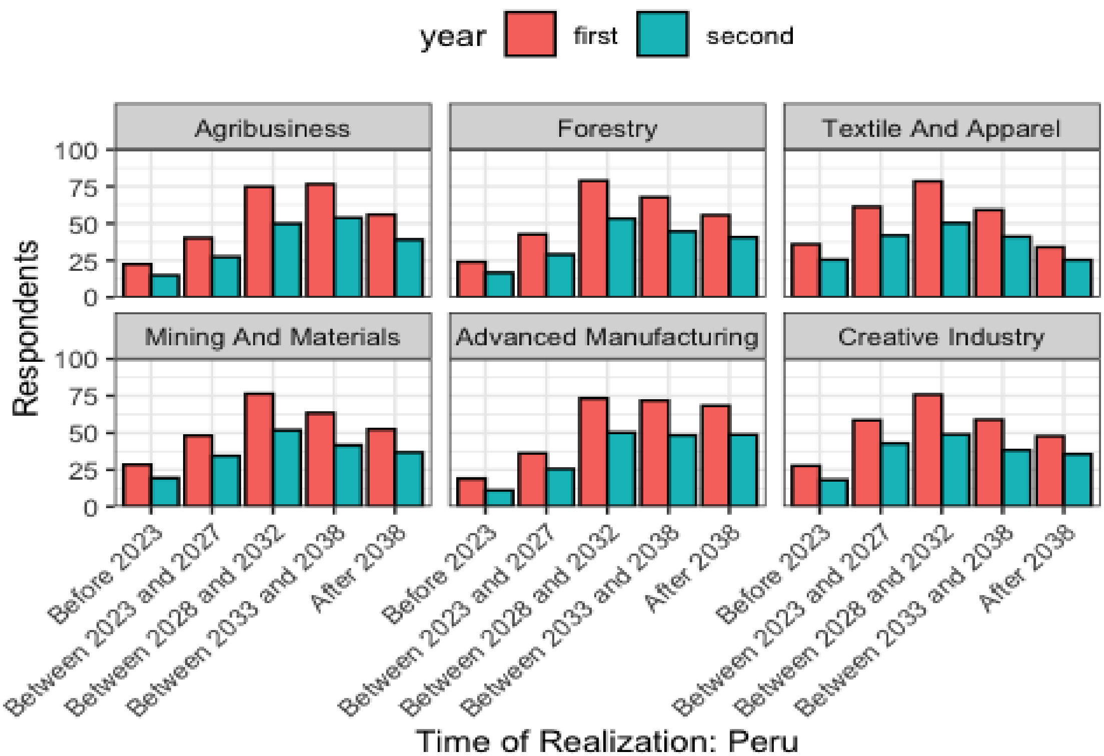
No Significant Change: There is no shift of trends in responses between the first and second years. This suggests that expectations around the timing of technological advancements remained stable across both years.

Time Realization Consistency: The expected periods didn't shift significantly between the two years for most industries. The proportions of Peruvian experts predicting advancements in each period remained roughly the same.

General Stability in Technological Forecasting: The consistency in responses suggests that respondents may feel confident in their original predictions or that there haven't been significant market, technological, or policy changes that would drastically shift expectations.

These findings indicate a stable outlook on technological advancements across industries and suggest that respondents' expectations did not fluctuate between the first and second years.

[Figure 2-3] Time of Realization in Peru



The chi-squared test also confirmed no significant difference between the first-year and second-year responses (p-value >= 0.05). The results are summarized in Table 5. The ranks of each category have slightly changed between the first- and second-year responses. Table 6 shows the filtered data (26 rows out of 150) and how the response ranks changed between 2023 and 2024. The results suggest that the time of realization in Peru has not changed between 2023 and 2024.

[Table 2-5] Chi-squared statistics on 'Time of Realization' for selected technologies by Peru

Industry	Technology	P-value	Statistics	Statistical Significance
Advanced Manufacturing	Automation	0.9367704	0.81245569	FALSE
Advanced Manufacturing	Clean Manufacturing	0.9862598	0.35138864	FALSE
Advanced Manufacturing	Digital Production	0.9992484	0.07856030	FALSE
Advanced Manufacturing	Engineering Service	0.9818448	0.40766177	FALSE
Advanced Manufacturing	Robotics	0.8356946	1.44866267	FALSE
Agribusiness	Agro Biology	0.9816507	0.40999233	FALSE
Agribusiness	Agro Chemicals	0.9995268	0.06216998	FALSE
Agribusiness	Agro Mechanics	0.9981093	0.12558098	FALSE
Agribusiness	Food Processing	0.9977424	0.13749673	FALSE
Agribusiness	Smart Farm	0.9915497	0.27199201	FALSE
Creative Industry	Copyright System	0.9861841	0.35241503	FALSE
Creative Industry	Digital Contents	0.9701705	0.53338642	FALSE
Creative Industry	Financial Service	0.9319549	0.84760149	FALSE
Creative Industry	Marketing Service	0.9896194	0.30300512	FALSE
Creative Industry	Media Service	0.9435009	0.76175212	FALSE
Forestry	Environment Management	0.9763815	0.46971420	FALSE
Forestry	Furniture Manufacturing	0.9853491	0.36357332	FALSE
Forestry	Precision Machineries	0.9750675	0.48370776	FALSE
Forestry	Saw Milling	0.9899872	0.29731005	FALSE

Industry	Technology	P-value	Statistics	Statistical Significance
Forestry	Surface Treatment	0.9976317	0.14090810	FALSE
Mining And Materials	Casting And Welding	0.9820154	0.40560578	FALSE
Mining And Materials	Clean Energy Mining	0.9606993	0.62102200	FALSE
Mining And Materials	Geotechnical Engineering	0.9937951	0.23151974	FALSE
Mining And Materials	Material Processing	0.9762533	0.47109393	FALSE
Mining And Materials	Resource Exploration	0.9972909	0.15095495	FALSE
Textile And Apparel	Apparel Manufacturing	0.9788815	0.44216495	FALSE
Textile And Apparel	Digital Design	0.9310158	0.85435683	FALSE
Textile And Apparel	Logistics	0.9941691	0.22416300	FALSE
Textile And Apparel	Textile Manufacturing	0.9896346	0.30277124	FALSE
Textile And Apparel	Textile Materials	0.9891302	0.31044090	FALSE

[Table 2-6] Changes in the Time of Realization in Peru

industry	technology	period	Percent in 1st survey	Percent in 2nd survey	Rank in 1st	Rank in 2nd	Same rank	Difference in percentage
Agribusiness	Agro Mechanics	Between 2028 and 2032	0.2888889	0.2918919	4	5	FALSE	0.003003003
Agribusiness	Agro Mechanics	Between 2033 and 2038	0.2925926	0.2864865	5	4	FALSE	-0.006106106
Agribusiness	Agro Biology	Between 2028 and 2032	0.2740741	0.2486486	4	3	FALSE	-0.025425425
Agribusiness	Smart Farm	After 2038	0.2592593	0.2702703	3	4	FALSE	0.011011011
Forestry	Environment Management	After 2038	0.2342007	0.2500000	3	4	FALSE	0.015799257
Forestry	Precision Machineries	Between 2028 and 2032	0.2565056	0.2663043	3	4	FALSE	0.009798772
Forestry	Precision Machineries	Between 2033 and 2038	0.2602230	0.2391304	4	3	FALSE	-0.021092614
Textile And Apparel	Textile Manufacturing	Before 2023	0.1263941	0.1250000	2	1	FALSE	-0.001394052
Textile And Apparel	Textile Manufacturing	Between 2028 and 2032	0.2713755	0.2500000	5	4	FALSE	-0.021375465
Textile And Apparel	Textile Manufacturing	After 2038	0.1226766	0.1304348	1	2	FALSE	0.007758203
Textile And Apparel	Digital Design	Before 2023	0.1338290	0.1413043	2	1	FALSE	0.007475352
Mining And Materials	Clean Energy Mining	Between 2033 and 2038	0.2453532	0.2173913	4	3	FALSE	-0.027961856
Mining And Materials	Clean Energy Mining	After 2038	0.2379182	0.2500000	3	4	FALSE	0.012081784

industry	technology	period	Percent in 1st survey	Percent in 2nd survey	Rank in 1st	Rank in 2nd	Same rank	Difference in percentage
Mining And Materials	Geotechnical Engineering	Between 2028 and 2032	0.2565056	0.2445652	5	4	FALSE	-0.011940359
Mining And Materials	Geotechnical Engineering	After 2038	0.2007435	0.1956522	3	2	FALSE	-0.005091321
Mining And Materials	Casting And Welding	Between 2023 and 2027	0.1821561	0.1684783	3	2	FALSE	-0.013677873
Mining And Materials	Casting And Welding	After 2038	0.1672862	0.1793478	2	3	FALSE	0.012061581
Mining And Materials	Material Processing	After 2038	0.1970260	0.2065217	3	2	FALSE	0.009495717
Advanced Manufacturing	Automation	Between 2033 and 2038	0.2602230	0.2391304	4	3	FALSE	-0.021092614
Advanced Manufacturing	Automation	After 2038	0.2490706	0.2608696	3	4	FALSE	0.011798933
Advanced Manufacturing	Engineering Service	Between 2028 and 2032	0.2825279	0.2663043	5	4	FALSE	-0.016223533
Advanced Manufacturing	Engineering Service	Between 2033 and 2038	0.2750929	0.2880435	4	5	FALSE	0.012950541
Creative Industry	Financial Service	Between 2023 and 2027	0.1784387	0.2065217	2	3	FALSE	0.028083077
Creative Industry	Financial Service	After 2038	0.1895911	0.2010870	3	2	FALSE	0.011495878
Creative Industry	Copyright System	Between 2023 and 2027	0.2081784	0.2228261	3	4	FALSE	0.014647648
Creative Industry	Copyright System	Between 2033 and 2038	0.2081784	0.2010870	3	2	FALSE	-0.007091482

3.3 Policy Measures

Experts selected **TWO** policy measures that the government should prioritize for the realization of technologies within the chosen category:

- Manpower Training
- Cooperation
- Infrastructure Establishment
- Investment Expansion
- Deregulation

3.3.1 Key Findings

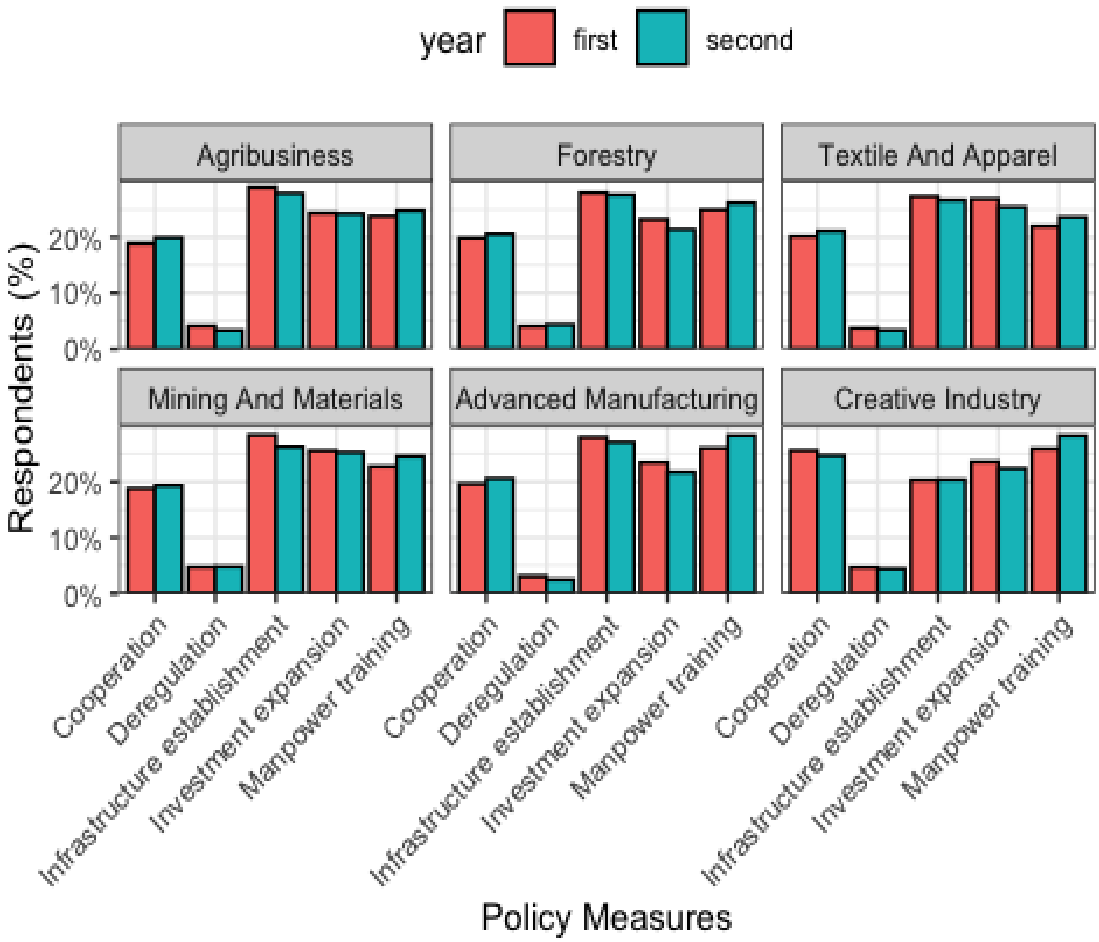
Across all six industries, Deregulation received the fewest policy requirements responses. Figure 2-4 shows that the infrastructure Establishment received fewer responses in the second year, whereas Manpower training and Cooperation received more. The results suggest that experts are more inclined to consider Manpower training and Cooperation as policy requirements for technology realization in Peru.

- Emphasis on Manpower Training: The responses for Manpower Training increased in the second year, indicating a growing focus on human resource development for technology realization across all industries (min 1.1%, max 2.3%). The most significant increase was in the Creative Industry, where 2.3% more respondents selected Manpower Training in the second year.
- Stable priority for Cooperation: The proportion of responses for Cooperation slightly increased in the second year (< 1.0%), except Creative Industry, where second-year responses were less than first-year responses.
- Reduced Infrastructure and Investment focus: The proportion of responses for Infrastructure Establishment and Investment expansion decreased in the second year, with the most significant decrease in the Mining and Materials industry (-2.1%).

Investment Expansion also decreased in the second year, with the most significant decrease in the Forestry (-1.8%). This suggests a shift in priority from building or acquiring physical resources to more soft-power measures like training and cooperation.

- Low demand for Deregulation: Deregulation received the fewest responses in both years, indicating that experts do not consider it a significant policy requirement for technology realization in Peru.

[Figure 2-4] Policy Measures in Peru



Despite the proportional shifts in policy requirements, the chi-squared test results (summarized in Table 2-7) indicate no statistically significant difference between the first- and second-year responses across all industries and technologies, as all p-values are greater than 0.05. Hence, we can conclude that the Peruvian experts retained their original views on policy requirements between 2023 and 2024.

[Table 2-7] Chi-squared statistics on 'Policy Measures' in Peru

Industry	Technology	P-value	Statistics	Statistical Significance
Advanced Manufacturing	Automation	0.7558298	1.8908274	FALSE
Advanced Manufacturing	Clean Manufacturing	0.8000392	1.6485599	FALSE
Advanced Manufacturing	Digital Production	0.9365872	0.8138091	FALSE
Advanced Manufacturing	Engineering Service	0.7043901	2.1707370	FALSE
Advanced Manufacturing	Robotics	0.9584841	0.6402804	FALSE
Agribusiness	Agro Biology	0.8109072	1.5882234	FALSE
Agribusiness	Agro Chemicals	0.9513029	0.7002110	FALSE
Agribusiness	Agro Mechanics	0.9524382	0.6909668	FALSE
Agribusiness	Food Processing	0.9375995	0.8063149	FALSE
Agribusiness	Smart Farm	0.9050956	1.0307657	FALSE
Creative Industry	Copyright System	0.9714646	0.5206027	FALSE
Creative Industry	Digital Contents	0.9499746	0.7109272	FALSE
Creative Industry	Financial Service	0.8245782	1.5116368	FALSE
Creative Industry	Marketing Service	0.9504256	0.7073003	FALSE
Creative Industry	Media Service	0.9691286	0.5435179	FALSE
Forestry	Environment Management	0.9813036	0.4141325	FALSE
Forestry	Furniture Manufacturing	0.9175354	0.9482730	FALSE
Forestry	Precision Machineries	0.9349527	0.8258249	FALSE
Forestry	Saw Milling	0.9435065	0.7617094	FALSE
Forestry	Surface Treatment	0.9867577	0.3445787	FALSE

Industry	Technology	P-value	Statistics	Statistical Significance
Mining And Materials	Casting And Welding	0.7868123	1.7214794	FALSE
Mining And Materials	Clean Energy Mining	0.9750650	0.4837340	FALSE
Mining And Materials	Geotechnical Engineering	0.9826469	0.3979192	FALSE
Mining And Materials	Material Processing	0.8803590	1.1862392	FALSE
Mining And Materials	Resource Exploration	0.9018165	1.0519658	FALSE
Textile And Apparel	Apparel Manufacturing	0.8713762	1.2405623	FALSE
Textile And Apparel	Digital Design	0.9846914	0.3721692	FALSE
Textile And Apparel	Logistics	0.9711021	0.5242060	FALSE
Textile And Apparel	Textile Manufacturing	0.9408224	0.7821693	FALSE
Textile And Apparel	Textile Materials	0.9334105	0.8370687	FALSE

3.4 Technology Acquisition

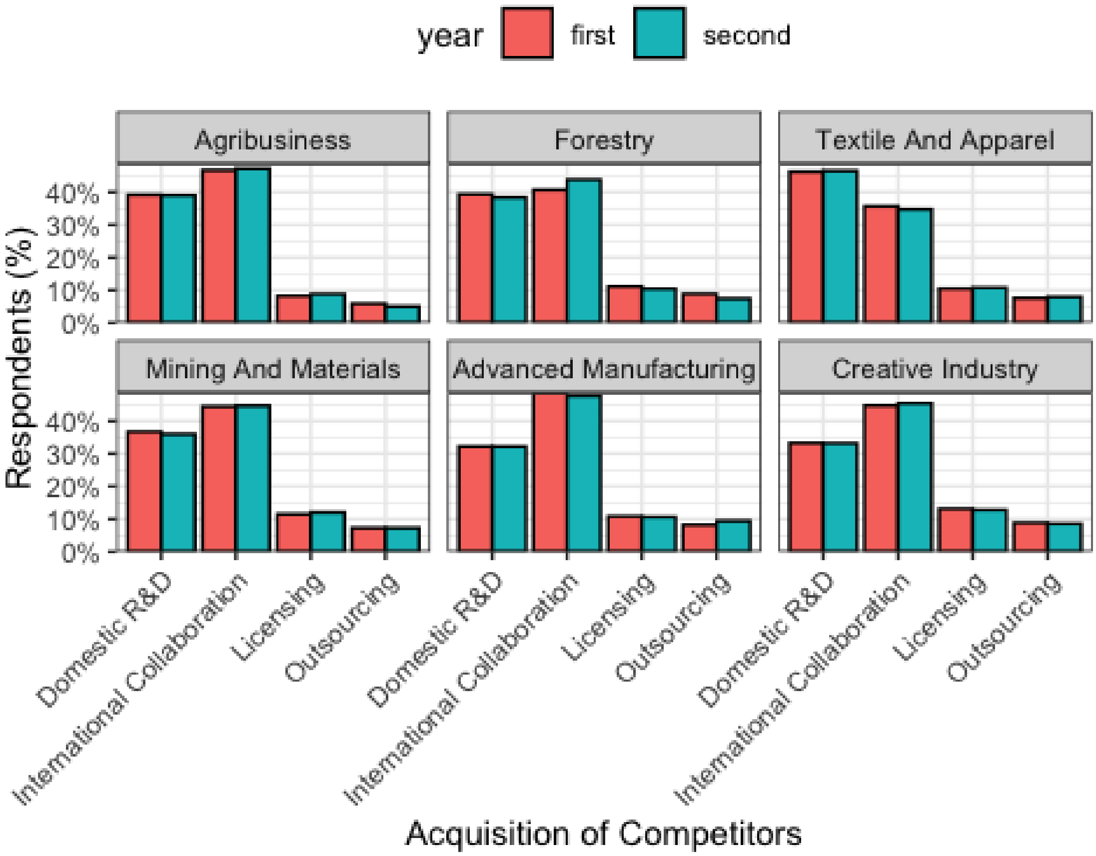
Experts selected **ONE** suitable technology acquisition method for Peru to realize technologies in Peru.

- License
- R&D
- International collaboration
- Outsourcing

3.4.1 Key Findings

- Stable preferences for acquiring technologies: The proportion of responses remained stable between the first and second years, with no significant changes. The results suggest that experts' views on technology acquisition methods did not change between 2023 and 2024.
- Potential for International Collaboration: International Collaboration received the highest number of responses in both years, indicating that experts see this as the most suitable method for technology acquisition in Peru. The proportion of responses for International Collaboration increased in the second year, suggesting a growing emphasis on international partnerships for technology realization.
- Confident domestic R&D in Textile and Apparel: In both years, the Textile and Apparel industry responded most strongly to domestic R&D, indicating that experts see it as a suitable method for acquiring technology in this industry. R&D programs need to reflect this view and invest in domestic R&D capabilities in this industry.
- Low preference for Outsourcing and Licensing: Experts prefer domestic or international collaboration over outsourcing or licensing. There are demands to internalize Peru's R&D capability.

[Figure 2-5] Technology Acquisition in Peru



The chi-squared test results (summarized in Table 2-8) indicate that there is no statistically significant difference between the first and second-year responses across all industries and technologies (p >= 0.05). The Peruvian experts' opinions on technology acquisition methods remained unchanged between 2023 and 2024.

[Table 2-8] Chi-squared statistics on 'Technology Acquisition' method in Peru

Industry	Technology	P-value	Statistics	Statistical Significance
Advanced Manufacturing	Automation	0.9641708	0.27767612	FALSE
Advanced Manufacturing	Clean Manufacturing	0.9190672	0.49919977	FALSE
Advanced Manufacturing	Digital Production	0.9609090	0.29530529	FALSE
Advanced Manufacturing	Engineering Service	0.9272392	0.46161772	FALSE
Advanced Manufacturing	Robotics	0.9614918	0.29218344	FALSE
Agribusiness	Agro Biology	0.8270344	0.89334124	FALSE
Agribusiness	Agro Chemicals	0.8713413	0.70791237	FALSE
Agribusiness	Agro Mechanics	0.9687439	0.25224883	FALSE
Agribusiness	Food Processing	0.9031694	0.57041976	FALSE
Agribusiness	Smart Farm	0.9742033	0.22056353	FALSE
Creative Industry	Copyright System	0.9831888	0.16394331	FALSE
Creative Industry	Digital Contents	0.9985591	0.03103873	FALSE
Creative Industry	Financial Service	0.9996798	0.01134375	FALSE
Creative Industry	Marketing Service	0.9923462	0.09571848	FALSE
Creative Industry	Media Service	0.9881622	0.12886056	FALSE
Forestry	Environment Management	0.6988893	1.42840859	FALSE
Forestry	Furniture Manufacturing	0.8410005	0.83532487	FALSE
Forestry	Precision Machineries	0.6465406	1.65697759	FALSE
Forestry	Saw Milling	0.8896064	0.62967588	FALSE
Forestry	Surface Treatment	0.8996529	0.58589837	FALSE

Industry	Technology	P-value	Statistics	Statistical Significance
Mining And Materials	Casting And Welding	0.8176664	0.93213653	FALSE
Mining And Materials	Clean Energy Mining	0.9417764	0.39255377	FALSE
Mining And Materials	Geotechnical Engineering	0.9583654	0.30879792	FALSE
Mining And Materials	Material Processing	0.9948866	0.07281806	FALSE
Mining And Materials	Resource Exploration	0.9934384	0.08621780	FALSE
Textile And Apparel	Apparel Manufacturing	0.8261395	0.89705077	FALSE
Textile And Apparel	Digital Design	0.8257405	0.89870418	FALSE
Textile And Apparel	Logistics	0.7170741	1.35093901	FALSE
Textile And Apparel	Textile Manufacturing	0.9865497	0.14064031	FALSE
Textile And Apparel	Textile Materials	0.9622458	0.28812678	FALSE

3.5 Leading Organization

Experts selected **ONE** organization that will lead the technological development in Peru.

- Academia
- Research Institution
- Industry
- Other

3.5.1 Key Findings

Figure 2-6 shows the distribution of responses for leading organizations in Peru. The key findings are as follows:

- Industry leadership across sectors: Industry organizations consistently lead technology development across all sectors, accounting for roughly 45-65% of leadership roles. The Textile And Apparel industry shows the most robust industry leadership (62-65%), while Forestry has the lowest (45-50%).

- Research institute participation: Research institutes are the second most important players after industry. Their contribution ranges from 17% to 35% across sectors. Agribusiness has the highest involvement of research institutes (35%), while Textile And Apparel have the lowest (17-21%).
- Role of Academia: Academic institutions contribute 12-19% of technology leadership across sectors. Academia showed slightly higher leadership roles than Research Institutions. The highest leadership role for Academia is in Advanced Manufacturing (18%), while the lowest is in Textile and Apparel (12%).

[Figure 2-6] Leading Organizations in Peru

- Changes between years: Most changes between years were relatively small, suggesting stable institutional roles. The chi-squared test also confirmed no significant difference between the first-year and second-year responses (p-value >= 0.05). The chi-squared test result is summarized in Table 2-9.

[Table 2-9] Chi-squared statistics on 'Leading Organization' in each industry and technology in Peru

Industry	Technology	P-value	Statistics	Statistical Significance
Advanced Manufacturing	Automation	0.8083968	0.97047021	FALSE
Advanced Manufacturing	Clean Manufacturing	0.8333296	0.86722287	FALSE
Advanced Manufacturing	Digital Production	0.9358954	0.42088366	FALSE
Advanced Manufacturing	Engineering Service	0.8907815	0.62458662	FALSE
Advanced Manufacturing	Robotics	0.9246005	0.47383724	FALSE
Agribusiness	Agro Biology	0.8757389	0.68920982	FALSE
Agribusiness	Agro Chemicals	0.7069740	1.39386241	FALSE
Agribusiness	Agro Mechanics	0.6044996	1.84821404	FALSE
Agribusiness	Food Processing	0.4435578	2.68046184	FALSE
Agribusiness	Smart Farm	0.8938514	0.61125352	FALSE
Creative Industry	Copyright System	0.9930409	0.08972808	FALSE
Creative Industry	Digital Contents	0.9476388	0.36367994	FALSE
Creative Industry	Financial Service	0.9358185	0.42125038	FALSE
Creative Industry	Marketing Service	0.9322743	0.43805244	FALSE
Creative Industry	Media Service	0.8994681	0.58670943	FALSE
Forestry	Environment Management	0.9898917	0.11567868	FALSE
Forestry	Furniture Manufacturing	0.9427586	0.38776316	FALSE

Industry	Technology	P-value	Statistics	Statistical Significance
Forestry	Precision Machineries	0.9295499	0.45084512	FALSE
Forestry	Saw Milling	0.9721236	0.23282721	FALSE
Forestry	Surface Treatment	0.9836022	0.16115524	FALSE
Mining And Materials	Casting And Welding	0.7728167	1.11764334	FALSE
Mining And Materials	Clean Energy Mining	0.7627834	1.15929186	FALSE
Mining And Materials	Geotechnical Engineering	0.6623872	1.58679699	FALSE
Mining And Materials	Material Processing	0.9465649	0.36902143	FALSE
Mining And Materials	Resource Exploration	0.5788576	1.96903910	FALSE
Textile And Apparel	Apparel Manufacturing	0.4109081	2.87745227	FALSE
Textile And Apparel	Digital Design	0.8630104	0.74315130	FALSE
Textile And Apparel	Logistics	0.8511310	0.79304004	FALSE
Textile And Apparel	Textile Manufacturing	0.6772853	1.52163820	FALSE
Textile And Apparel	Textile Materials	0.7969453	1.01779866	FALSE

3.5.2 Insights

The Textile and Apparel industry shows the strongest confidence in its R&D capability. This suggests that experts see the industry as capable of leading technology development within Peru.

Peru has an industry-led innovation system with significant contributions from research institutes and academia. The relative stability between years suggests these are well-established patterns rather than rapidly changing dynamics. The variation across sectors likely reflects different technological needs and institutional capabilities in each area.

3.6 Obstacles

Experts selected **TWO** possible obstacles that may hinder the technological development in Peru.

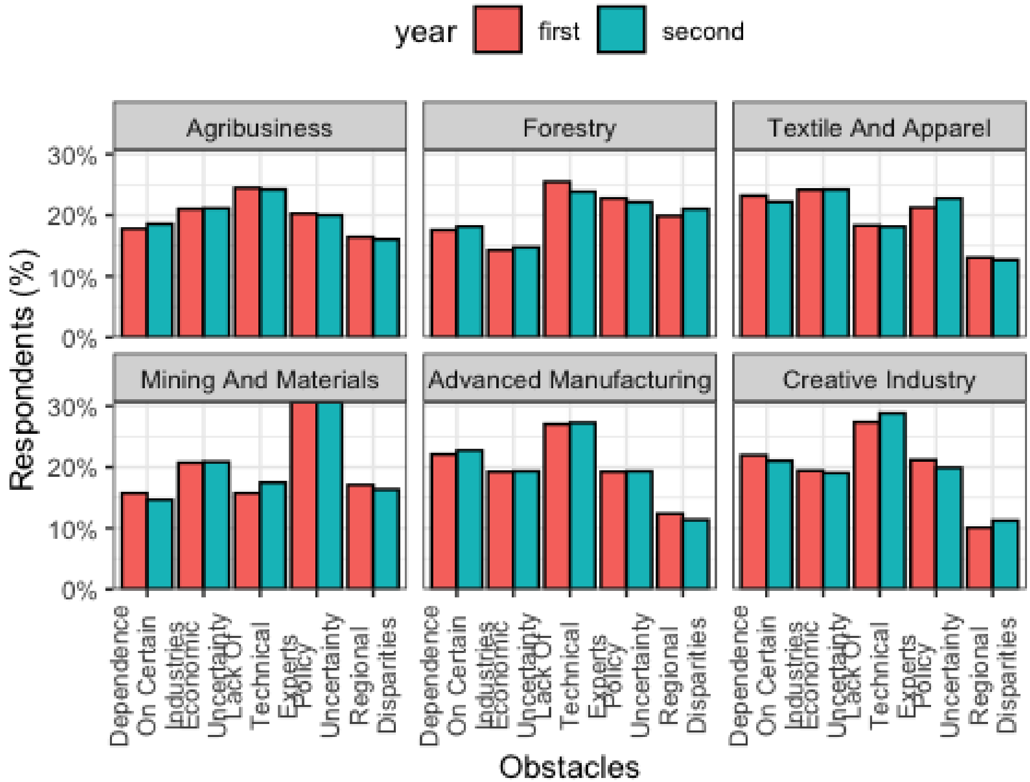
- Uncertain economic situation
- Lack of technical experts
- Dependence on certain industries
- Regional disparities
- Policy uncertainty

3.6.1 Key Findings

The two-year observations are summarized in Figure 2-7. The key findings are as follows:

- Most critical obstacles: 'Lack of Technical Experts' is the biggest challenge for Agribusiness, Forestry, Advanced Manufacturing, and Creative Industries. 'Policy Uncertainty' is the biggest challenge (31%) for Mining and Materials. The Textile and Apparel industry is more concerned about 'Economic Uncertainty' (24%) than other industries. Agribusiness appears to have a more balanced distribution of challenges, with no single obstacle dramatically outweighing others.
- Least concerning obstacles: 'Regional Disparities' consistently rank as one of the lower concerns across most industries. This is particularly true for the Creative Industry and Advanced Manufacturing, where only about 11-12% of responses identified this as a significant obstacle. Forestry has the lowest concern about 'Economic Uncertainty' (14-15%).

[Figure 2-7] Obstacles in Peru



Stability in obstacle perceptions: The chi-squared test results (summarized in Table 2-10) indicate that there is no statistically significant difference between the first and second-year responses (p-value >= 0.05). The proportions remain relatively stable between 2023 and 2024.

[Table 2-10] Chi-squared statistics on ‘Obstacles’ in each industry in Peru

Industry	p_value	statistics	significance
Agribusiness	0.9985477	0.1097740	FALSE
Forestry	0.9985477	0.1097740	FALSE
Textile And Apparel	0.9985477	0.1097740	FALSE
Mining And Materials	0.9985477	0.1097740	FALSE
Advanced Manufacturing	0.9985477	0.1097740	FALSE
Creative Industry	0.9764140	0.4693647	FALSE

3.6.2 Insights

Technical Expertise Gap needs attention from policymakers:

- Strengthen STEM education programs at universities and technical schools
- Create specialized technical training programs aligned with each industry’s specific needs, especially for Advanced Manufacturing and Creative Industries where this is most acute
- Develop partnerships with international educational institutions for knowledge transfer and capacity building

Industry-Academia Collaboration will increase the pool of technical experts:

- Establish internship programs for university students
- Utilize the industry expertise to design curriculum and training programs
- Create joint research projects to bridge the gap between academic knowledge and industry needs

Policy Uncertainty is a major concern for Mining and Materials:

- Create a stable regulatory framework for the mining sector, given its high sensitivity to policy uncertainty (31% identified this as an issue)
- Establish transparent processes for technology-related permits and licenses to reduce uncertainty

3.7 Contribution of Technology Forecasting

Experts selected TWO contributions technology forecasting can make to promoting S&T policy, infrastructure, and implementation in Peru.

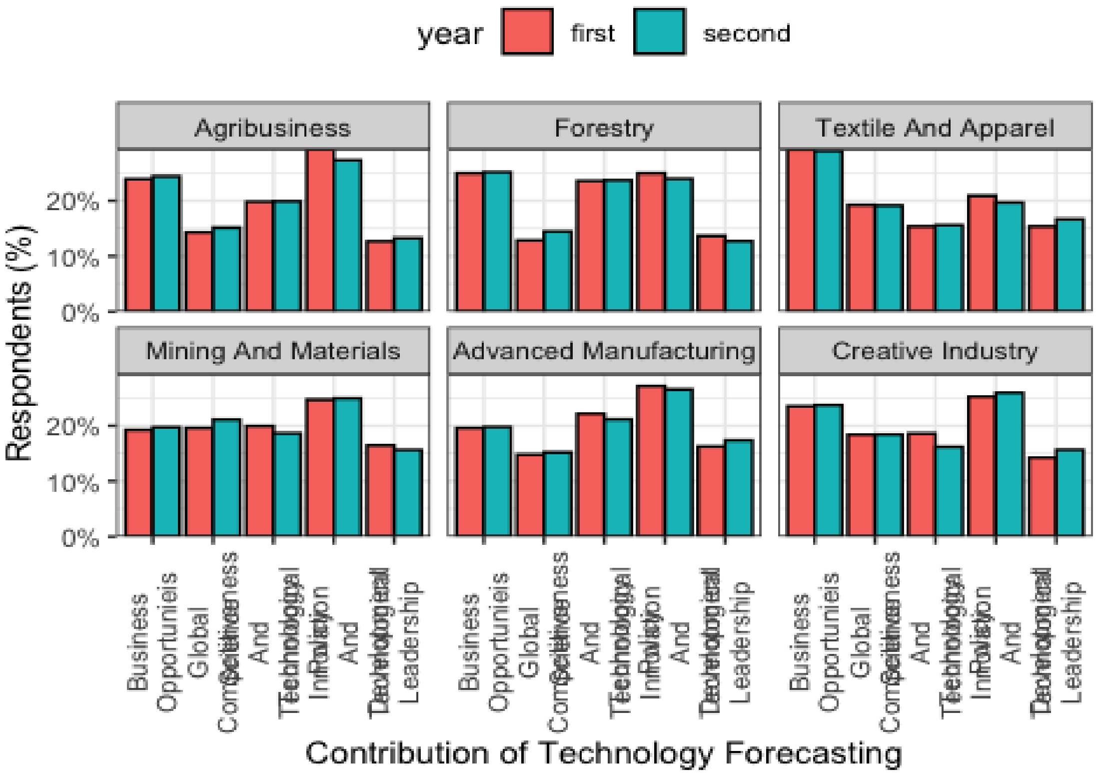
- Leading technological innovation and development
- Increasing business opportunities
- Developing science and technology policy
- Gaining technological leadership
- Increasing global competitiveness

3.7.1 Key Findings

Figure 2-8 shows the distribution of responses for the contribution of technology forecasting in Peru. The key findings are as follows:

- Business Opportunities and Technological Innovation consistently rank highest: 'Business Opportunities' range from 19-29% across industries. Technological Innovation and Development ranges from 20-29%. This suggests that experts see technology forecasting as a critical driver of economic growth and innovation in Peru.
- Industries have different priorities: Textile and Apparel industry strongly prefers 'Business Opportunities' (29%). Advanced Manufacturing and Mining & Materials industries focus most on 'Technological Innovation' (26-27%). Technical leadership and Global competitiveness have lower priorities.

[Figure 2-8] Contribution of Technology Forecasting in Peru



The responses remain relatively stable between the first and second years. There is no statistically significant difference between them (p-value >= 0.05). The chi-squared test results are summarized in Table 2-11.

[Table 2-11] Chi-squared statistics on 'Contribution of Technology Forecasting' in each industry in Peru

Industry	P-value	Statistics	Statistical Significance
Advanced Manufacturing	0.9741321	0.4934849	FALSE
Agribusiness	0.9741321	0.4934849	FALSE
Creative Industry	0.9629541	0.6009860	FALSE
Forestry	0.9741321	0.4934849	FALSE
Mining And Materials	0.9741321	0.4934849	FALSE
Textile And Apparel	0.9741321	0.4934849	FALSE

3.7.2 Insights

Focus on commercialization in the textile and apparel industry:

- This industry sees technology forecasting as a market development and business growth tool. Technology Forecasting and Roadmap will enhance the industry's ability to identify new business opportunities and develop innovative products.

Trust in Technological Innovation:

- Agribusiness, Mining and Materials, Creative Industry, and Advanced Manufacturing see technology forecasting as a technological innovation and development driver. A clear technology roadmap will help these industries identify potential areas for innovation and development. Where international collaboration is preferred, such as in Advanced Manufacturing, technology forecasting can help identify potential partners and areas for cooperation.

3.7.3 Strategic R&D Investment

Experts set the order of R&D investment priority to allocate a limited budget to the five technologies in each industry as below:

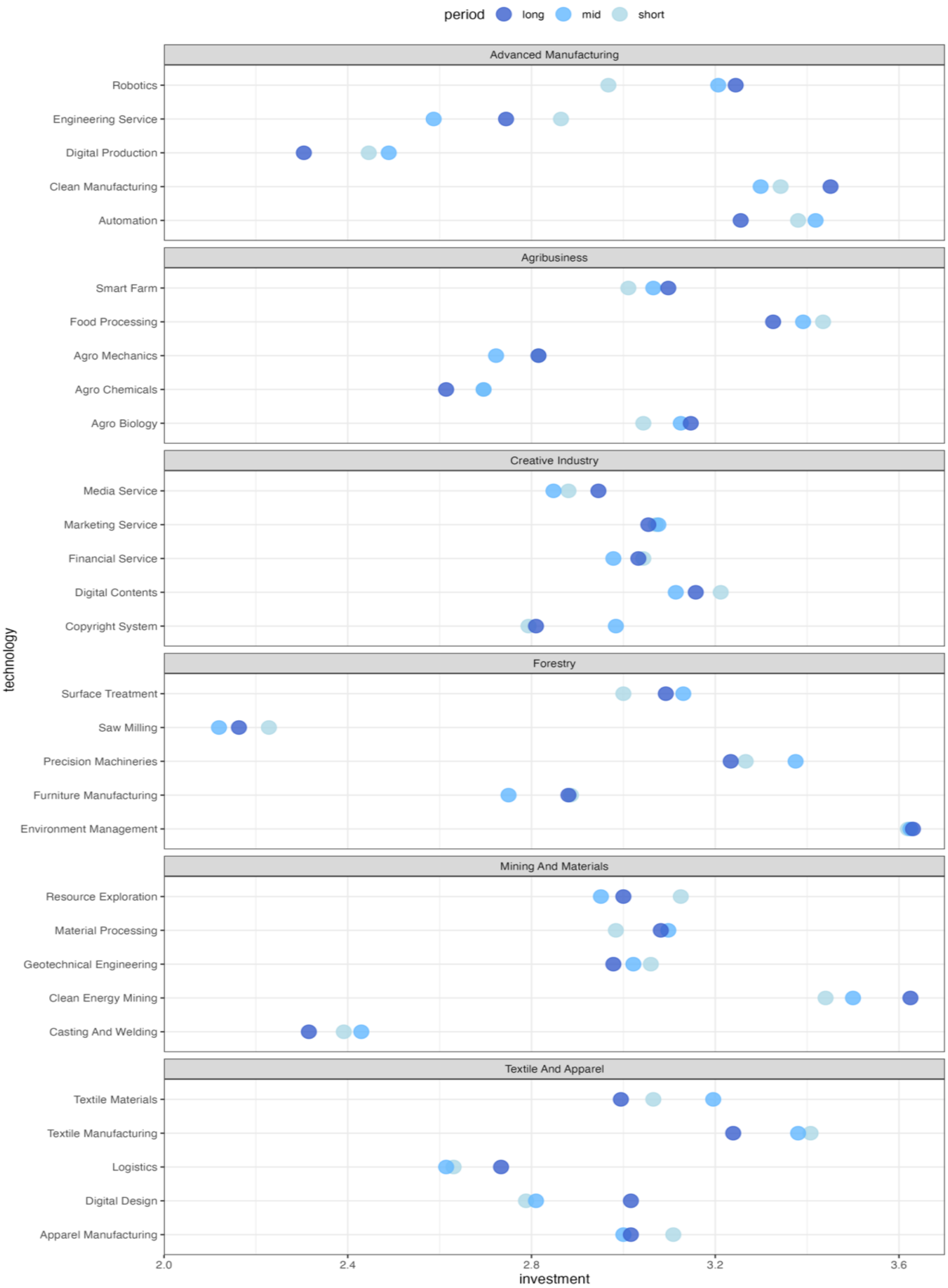
- Very low: Likert scale = 1
- Low: Likert scale = 2
- Medium: Likert scale = 3
- High: Likert scale = 4
- Very high: Likert scale = 5

3.7.4 Summary of weighted mean for each technology

The strategic R&D investment priorities are summarized in Figure 2-9. The key findings are as follows:

- Advanced Manufacturing: Automation and Clean Manufacturing are top priorities for short-, mid-, and long-term investments. Robotics gained more attention toward long-term investment.
- Agribusiness: Food Processing is the top priority for all periods. Smart Farm and Agro Biology gained second and third priority, respectively. There are fewer disparities between short-, mid-, and long-term investments in Agribusiness than in Advanced Manufacturing.
- Creative Industry: All five technologies received similar investment priorities, suggesting that experts consider all technologies equally important for developing the Creative Industry.
- Forestry: Saw Milling is the lowest priority across all periods. Environmental Management gained the highest priority consistently across all periods. There are distinctive orders in investment priority in Forestry compared to other industries.
- Mining and Materials: Clean Energy Mining is the top priority, whereas 'Casting and Welding' technology is the lowest priority. Resource Exploration saw decreased priority in mid-term planning (3.04 to 2.95). Geotechnical Engineering showed slight increases in short and mid-term priorities.
- Textile and Apparel: Textile Manufacturing gained the highest priority in the short term. There are fewer disparities between long-term investment preferences among technologies. Digital Design showed interesting shifts with decreased short/mid-term priority but increased long-term focus. Logistics maintained relatively stable, lower priority levels. Apparel Manufacturing showed minimal changes across timeframes.

[Figure 2-9] Strategic R&D Investment in Peru (2024)



3.7.5 Key Findings

Technologies with environmental sustainability showed consistent increases in priority across time frames. Traditional technologies, such as Saw Milling or Casting and Welding, maintained low priority levels. Digital transformation technologies, such as Digital Design in Textile and Apparel or Digital Content in the Creative Industry, gained relatively higher priority in the long-term investment strategy. However, Digital production showed the least priority within Advanced Manufacturing.

This analysis suggests a clear shift toward sustainability-focused technologies across industries while maintaining balanced investment across different time horizons. The data also indicates a more nuanced approach to digital transformation, with varying priorities across various sectors and applications.

The summarized comparison data between the first-year and second-year surveys is given in Appendix.

CHAPTER 3

Developing R&D Programs: Planning and Practice

S&T Planning and Technological Forecasting for Prioritized Sectors in Perú

Chapter 3. Developing R&D Programs: Planning and Practice

1. Introduction[4]

Planning is about optimizing, minimizaing or maximizing, to reach a business or public goal and is also related with developing or understanding something new, uncertain and priporty. R&D planning is no exception to this general scope. This chapter deals with the goals and scope of R&D planning including the 10 principles of developing R&D programs and the practice of developing R&D programs for Peruvian economy. Also a framework for the collaborative R&D programs for Peru and Korea has been suggested for a further study.

Many aspects should be considered to develop R&D programs. The 10 principles of R&D program are introduced in consideration of developing a program. While developing an R&D program, the fact that this program is being built for Peru is always emphasized to introduce the 10 principles of R&D program. It is naturally true that every country has its own different principles.

Considering every phase of a policy cycle, the participants in an R&D program have their own roles and responsibilities to reduce the R&D risk and secure the first hand outcome. The borders of participants are not clear and even are overlapped. For example, the stage of planning and RFP's is overlapped with the participation of government and agency.

This chapter especially carries two practices of developing R&D programs from the government perspectives. One is to develop a technology development program for the SME's(small and medium sized companies) in Peru, which should consider the characteristics of Peruvian SME's as well as the properties of SME's in general. The other is

[4] <Chapter 3. Developing R&D Programs: Planning and Practice> is written Jong-Ihl Lee, Professor, SUNY Korea.

about developing an international R&D program between Peru and Korea. In general, planning is related with developing or understanding something new, uncertain and pripority. R&D planning is no exception to this general scope: the following elements of R&D planning activities. This chapter deals with the goals and scope of R&D planning and the practice of developing R&D programs for Peruvian economy. Also a framework for the collaborative R&D programs for Peru and Korea has been suggested for a further study.

There are six elements which affect R&D planning and the consequences: newness, priority setting, time, geographical and cultural diversity, strategy and government.

Firstly, newness is the first element of R&D planning and is invoved in some degree of uncertainty. R&D activities are always carried out with technical and human uncertainty. The second is about priority setting because resources are always are limited and policies should have been implemented based on the priority. Time is the third factor which can be a short term or a long term. Whether time can be defined by economics standard or not does not matter as long as the period of time is clearly showm for R&D assistance. Though technical aspects are important, geographical and cultural diversity as the fourth element of R&D planning should not be neglected because it is the baseline for the potential markets. The next element is strategy which can be a crucial tool for a nation to promote successful R&D activities. Whether enjoying the benefit of a late comer or a beginner's luck depends on a nation's R&D strategy. The last element of R&D planning is the role of government. Governments play a significant role in investing on R&D area. How and where governments are investing tax payers' money can decide the strategic industrial areas for the economy.

2. The Principles of R&D Programs

Many aspects should be considered to develop R&D programs. In this section, the 10 principles of R&D program are introduced in consideration of developing a program. While developing a R&D program, the fact that this program is being built for Peru is always emphasized to introduce the 10 principles of R&D program.

1. Goals: All the investment should have its own golas which are clear and punctual. Especially the target of R&D program should be shown outright in terms of specification. The policy goals as well as technical goals should be based on the results of feasibility study and announced to the public in advance. Goals are to be set after considering comparative advantage domestic and global.

2. Planning functions: Governments can themselves plan R&D programs. However, it is natural that governments endorse their roles and responsibilities to other organizations such as government agencies. The existence and degree of agencies' activities are the barometer of how mature and well established R&D programs are utilized. The roles and responsibilities on planning can be coordinated between governments and agencies. Whether common laws or special laws are suitable depends on the characteristics of nations and innovation circumstances.

3. Implementation agencies: As technologies advance further, the level of expertise is getting higher. The role and importance of technology agencies are being emphasized. Along with the R&D cycle, the consequent roles are to be furnished and the budget and personel are mobilized. This agency also has the functions to manage R&D fund and the regulations.

4. Legal infrastructure: The bottom line for the active R&D program is to establish and keep the legal infrastructure. The existence of the right and appropriate laws and acts can secure the optimal time period and budget necessary to sustain successful R7D

activities. Having laws and acts once is not enough and keeping updated them is indispensable to promoting R&D activities.

5. Financial resources: Financial resources are the ultimate tool to expedite R&D activities and to promote technology development in the short run. The combination of central and local governments' participation is also crucial to implement R&D programs. This coordination between central and local governments can play a major role in finding an optimal level of R&D investment through competition among local governments. Financial resources can be divided into public and private, and cash and inkind. The ratio of cash and inkind investments can be decided by the properties of a R&D program. Despite its crucial role, financial resource should play a role to mitigate any possible moral hazard.

6. Demand for a program: It is imperative that a demand for the R&D program should be investigated and confirmed. In order to further the process, a top down approach is adopted when strategic and tacit technologies are necessary and a bottom up approach when application technologies are necessary. A survey can be carried out to find a demand for R&D programs. To meet the desirable outcomes, the sampling population for a survey can be different from researchers to technicians in industries.

7. Intermediaries: Government and technology agencies cannot guarantee the perfect constitution of a successful R&D program. There must be a function to connect or lead R&D programs to program participants such as researchers in companies, research institutes, academic associations or universities. Especially when the R&D infrastructure is not well established, it is required for governemnts to facilitate the activities of intermediaries related with industries.

8. Evaluation: In all the countries having various technology programs, being a grantee of R&D subsidy is usually competitive so that evaluation and its process has become extremely important. Keeping the fairness and objectivity of selection process is crucial to the success of R&D programs. The types of evaluating methods vary over

countries: peer review and committee evaluation. Evaluation method is not only applied for selection but also monitoring and final evaluation.

9. Dissemination: The ultimate policy goal of all the R&D programs is the commercialization of developed technology. More broadly speaking, dissemination of all the outcome and output of R&D programs is second to none. The method and scope of dissemination should be noticed publicly to all the grantees of R&D subsidy. Dissemination-oriented R&D activities is a way to maximize the positive externality of R&D programs to fix a market failure of under-investment.

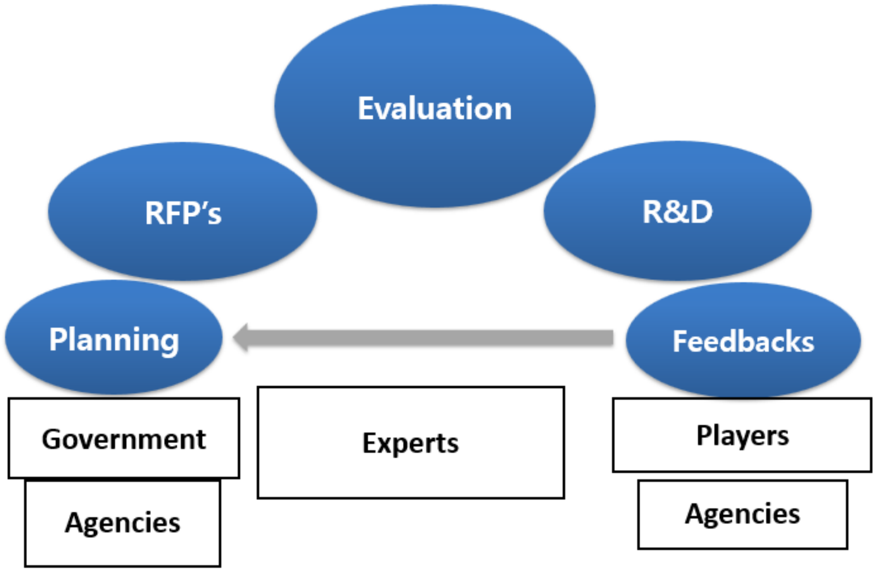
10. Internal tensions: Internal tension leads R&D programs to a success through a desire for individual and organizational motivation and behavior. Internal tension is not the result of a single input, but the aggregation of individual effort, organizational consistency and legal frameworks. Financial resources are the blood, legal structure is the bones and internal tension is for the soul. If not for internal tensions, R&D programs are just consuming financial resources without any economic impact.

3. The Roles and Responsibilities

3.1 The Roles and Responsibilities by R&D Players

Considering the policy cycle, the participants on an R&D program have their own roles and responsibilities to reduce the R&D risk and secure the first hand outcome. The borders of participants are not clear and even are overlapped. For example, the stage of planning and RFP's is overlapped with the participation of government and agency. When it comes to the evaluation stage, experts are directly involved and agency and government are indirectly involved by keeping consistency and objectivity.

[Figure 3-1] The R&R's of innovative players and the work flow



Governments, central and local, are the major provider of financial resources as well as of the regulation for R&D implementation. As a monopoly of laws and regulations, governments should keep the integrity and consistency in the evaluation process. Especially in the selection evaluation, the objectivity and fairness should be kept under any circumstances. It is no doubt that governments should provide the legal assistance to promote R&D activities.

Regarding the agencies, their autonomy and independence should be kept by the governments. With the neutrality secured by their autonomy and independence, the agencies can be fair on selecting projects and allocating R&D budget by keeping their integrity. Agencies is the only entity which is involved in the every stage of R&D program. And that is why agencies have to keep cooperative relations with governments and industry. Agencies have to operate program management tasks effectively and efficiently by maintaining robust relations with experts.

[Table 3-1] The Participations of innovative players over policy cycle

	Universities	PRI's	Companies
Planning	△	△	△
RFP's	○	○	○
Evaluation	△	△	△
R&D	○	○	○
Feedbacks	○	△	○

[Table 3-2] The List of R&D program packages

Structures	Author	Purpose	Contents
Plan for R&D Program	Governments	Policy implementation	Definitions, goals, department, policy target, duration, budget, outcome, IPR management
Public Notices	Agencies	Public announcement	Program title, qualification of application, contents of supports (financial, technical and others), preferential conditions, deadline, introductory documents
Application		Enhancing accessibility	RFP contents (title, expected outcome, tool, period, list of documents, budget), Required information and documents
Regulations	Governments	Legal background and guidelines	Legal documents on qualifications for applications and government grants, and incentives and penalty

3.2 The Contents for R&D Programs

Governments and agencies are providing the contents for implementation of an R&D program as a result of collaboration over R&D process. The contents are usually established in the process of collaborative discussions and feedbacks between governments and agencies. The structure of the contents for R&D programs consists of the summary of programs goals and budget, program operating systems, budget and royalty related guidelines, qualifications for application, and evaluation criteria and process, and related laws and acts. The Q&A's which are commonly asked and answered are also listed at the applicants' convenience. The typical contents can be shown as follows:

<Contents for R&D Program>

1.Summary
  1.1 Goal and Support target
  1.2 Budget and Duration

2. Program Operation system

3. R&D support and Royalty

4. Qualifications

5. Evaluation procedure and criteria

6. Related laws and guidelines

7. Notes

8. Inquiry and Q&A

The contents for R&D programs are the elements of the program operations and the components of the public notice to be announced afterwards. The contents can play a guideline or manual for operating R7D programs. Even though the contents are circulated and utilized as the formal documents among government bodies and public sectors, these can be transformed into a public notice for the public including companies, universities, research institutes and others. This is why the contents of R&D programs had better be well planned, established, and agreed among all the public authorities inside the governments.

sectors, these can be transformed into a public notice for the public including companies, universities, research institutes and others. This is why the contents of R&D programs had better be well planned, established, and agreed among all the public authorities inside the governments.

4. The Practices of Developing R&D Programs

This section carries two practices of developing R&D programs from the government perspectives. One is to develop a technology development program for the SME's(small and medium sized companies) in Peru. In this regard, this program should consider the characteristics of Peruvian SME's as well as the properties of SME's in general. The other is about developing an international R&D program between Peru and Korea. Collaborating with other country as a partner for R&D program surely adds complexity in all the aspects of planning, implementing and reviewing the consequent program. Many others factors which domestic programs do not have to consider should be considered with a partner country. For example, language and budget are the main subjects to be coordinated and adjusted, according to the results of negotiating with other country.

4.1 Industrial Technology Development Programs for SME's (ITDP)

First of all, it is easy and possible to develop an industry technology development program for domestic SME's in Peru. The content for 'the industry technology development program for SME's (ITDP hereafter)' has a short list to operate the program. The ITDP is for SME's to develop industrial technology and consequently should have a guideline of IPR (intellectual property rights) such as how to pay royalty and licensing.

<The Contents for the ITDP>

1. Summary
2. Program Operation system
3. R&D support and Royalty
4. Qualifications
5. Evaluation procedure and criteria
6. Related laws and guidelines
7. Notes
8. Inquiry and Q&A

4.2 Peru and Korea Collaborative R&D Program (CRP)

Similar to the previous ITDP, Peru and Korea Collaborative R&D Program (CRP hereafter) should consist in the contents for R&D program. The difference between CRP and ITDP is that the agreement between Peru and Korea is a precondition to proceed the CRP. The agreements can be made at a government level and at an agency level. Another issue to be resolved is the IPR to be shared between Peru and Korea. Using a common language such as English can be an option both countries choose.

<Detailed Contents for the CRP>

1. Summary
    - 1.1 Goal and Support target
    - 1.2 Budget and Duration
    - 1.3 Terms and conditions for joint projects
2. Program Operation system
    - 2.1 Government ministries
    - 2.2 Agencies
    - 2.3 R&R of Peru and Korea
3. R&D support and royalty
    - 3.1 Amount and types of grants
    - 3.2 Share of cash and in-kind investment
      : by partners, by countries
    - 3.3 Payment of royalty: ratio and period
4. Qualifications
    - 4.1 Companies, universities, PRI’s
    - 4.2 Financial standards, history of penalty
    - 4.3 Identical or different qualifications
    - 4.4 Restrictions

5. Evaluation procedure and criteria
   5.1 Committee, peer review or mixed
   5.2 Possibility, marketability, commercialization
   5.3 Evaluation sheet
6. Related laws and guidelines
   6.1 CRP related laws
      : agreements between two countries
   6.2 Laws and guidelines on qualification
      : Priority and principles
7. Notes
   7.1 Bidding: method, deadline, URL
   7.2 List of documents to be submitted
8. Inquiry and Q&A
   8.1 Contact points
   8.2 Q&A

Depite the complexity of planning and implementing international R&D programs, many countries have been cooperative in various bilateral and multilateral R&D programs. For example, Korea has many bilateral R&D programs with many advanced countries like USA and Germany, and even participates in the multilateral programs like EUREKA and New Horizon.

The outstanding merit of international R&D programs is that innovative players can join the big technology project with a relatively small amount of budget, once the government of a country becomes a member of R&D networks.

5. Remarks

The achievable R&D programs should include tangible and intangible resources. Considering that financial resources and experts are the tangibles resources on the universities, research institutes and the private sectors, there are usually a lot more intangible asset a nation has and Peru is no exception. The crucial factor for a successful R&D program may consist of intangible assets such as legal frameworks and degree of networking among stakeholders among innovative players.

Korea's experiences on developing R&D programs give many meaningful lessons to Peru which is desperate on promoting innovative activities through R&D investments. Knowledge on how to manage and operate tangible asset can be easily transferred. How to lead innovative players and voluntarily participate on R&D programs requires various socioeconomic and even cultural approach for Peru's own.

In this sense, a trial to develop an international R&D program may be a good challenge which is worth investing. Even though it will be complicated and slow, it has less risky if Peru can take advantage of a handy country like Korea.

CHAPTER 4

Role of the Technology Foresight Center

S&T Planning and Technological Forecasting for Prioritized Sectors in Perú

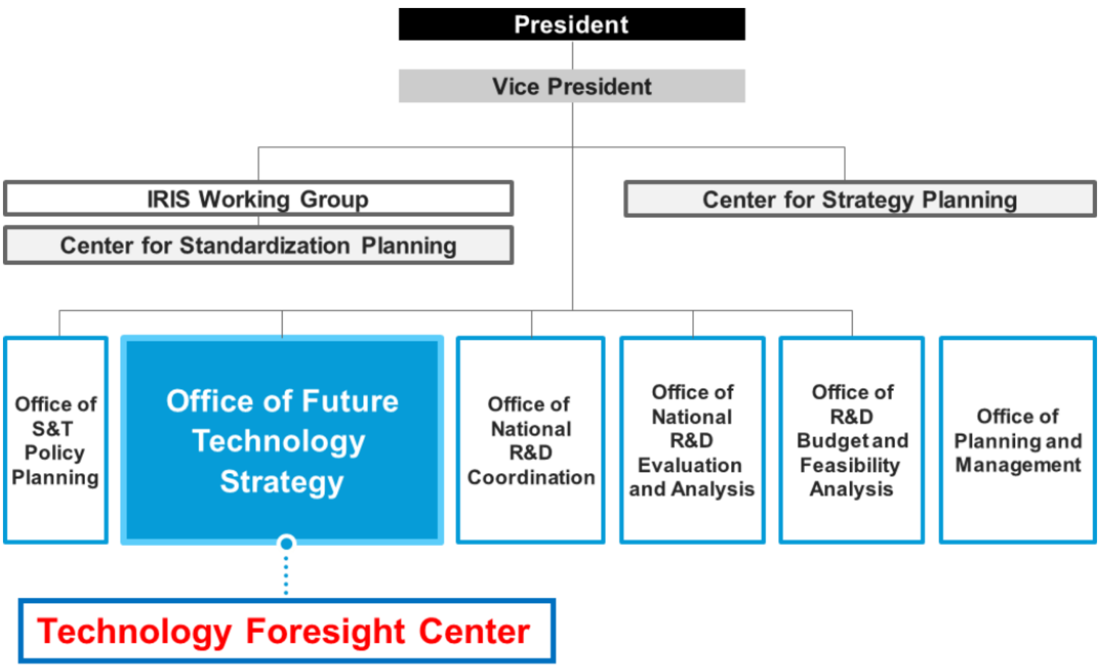
Chapter 4. Role of the Technology Foresight Center

1. Organization of Technology Foresight Center in Korea[5]

1.1 Technology Foresight Center in KISTEP

Since KISTEP was designated as an organization to conduct science and technology foresight programs under the Framework Act on Science and Technology enacted in 2001, the KISTEP Technology Foresight Center has expanded in size and role. Currently, the Technology Foresight Center is located under the Office of Future Technology Strategy and is in charge of S&T foresight program, technology assessment and technology level evaluation.

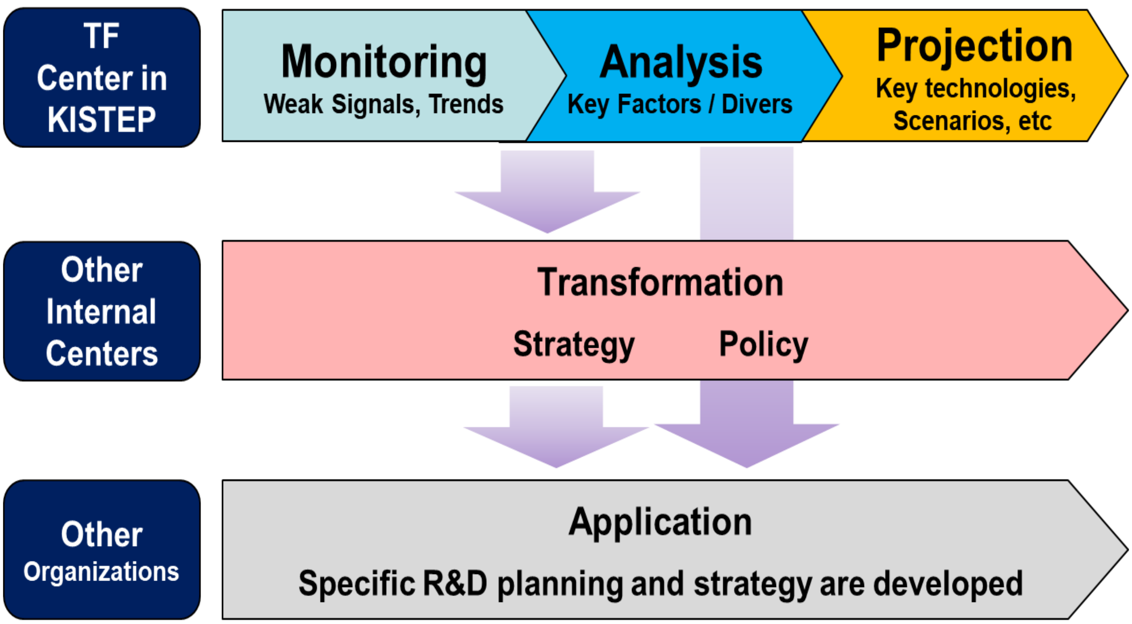
[Figure 4-1] Organizational structure of KISTEP



[5] <Chapter 4. Role of the Technology Foresight Center> is written by Hyun Yim, Senior Research Fellow, KISTEP.

Technology foresight is the prediction and exploration of future technological developments to provide countries and companies with the knowledge and information they need when formulating R&D plans and strategies. In other words, technology foresight allows us to design specific policies for the development of science and technology by predicting changes that will occur in the future and linking the results to economic and social needs. The results of technology foresight conducted by KISTEP's Technology Foresight Center are used in multiple ways. First, the policy formulation centers within KISTEP use the foresight results to formulate cross-ministerial policies such as the S&T Master Plan. In addition, relevant ministries use the results generated by KISTEP to formulate more specific policies such as R&D program planning. In other words, by sharing technology foresight results, KISTEP's Technology Foresight Center is expected to contribute to strengthening the capacity of ministries in R&D planning and strategy formulation.

[Figure 4-2] Multi-layered application of technology foresight results

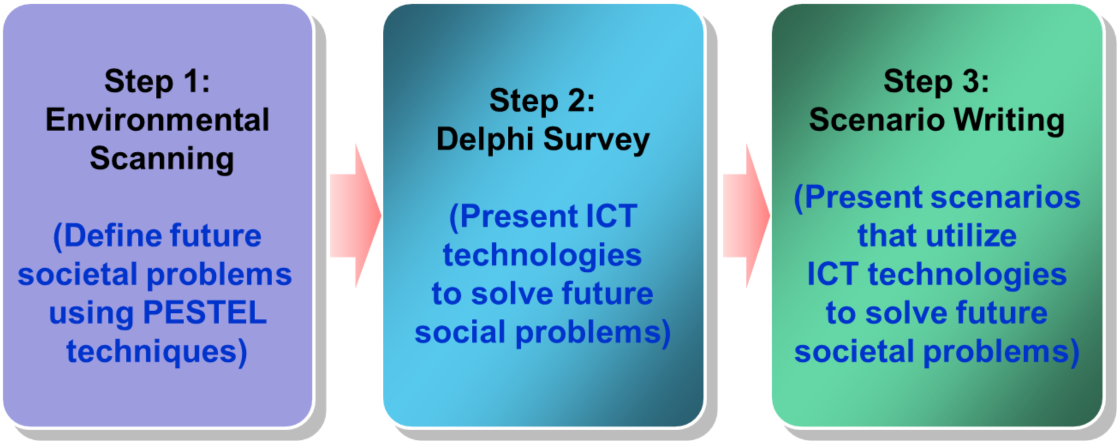


1.2 Technology Foresight Activities of Other Organizations in Korea

Apart from KISTEP's S&T foresight programs for all fields of science and technology, each ministry's affiliated organizations conducted technology foresight programs for ministry-related fields. Some examples are as follows.

In 2015, IITP (ICT Planning and Evaluation Institute) that manages ICT research and development under the Ministry of Science and ICT conducted the “ICT Technology Foresight Study 2030 & Selection of Top 10 Future Promising Technologies”. The study was conducted to select promising ICT technologies with high need for state support and to organize development scenarios and present a systematic future picture in consideration of various situations and impacts that may occur in the future society. Based on 200 ICT promising technologies, 10 promising technologies were derived through portfolio analysis of future technology importance and urgency of development, which can contribute to efficient R&D planning and promotion by planning national R&D projects based on verified future demand and forecast data. This study applied a three-step process using the scenario technique based on Delphi survey results.

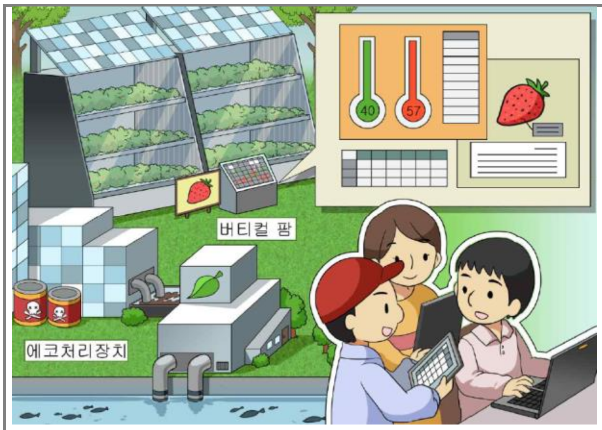
[Figure 4-3] IITP's process for conducting ICT technology foresight



First, environmental scanning using the PESTEL technique was used to define the social problems that will arise in the future and derive ICT future technologies to solve them. Next, to identify the characteristics of various ICT technology needs and identify promising ICT future technologies, IITP conducted a Delphi survey among ICT-related industry, academia, and research experts. ICT promising technologies were selected by considering the importance of the technology, the time of market launch, the relative market size, and the ripple effect. Finally, key keywords for future social issues and detailed technologies related to the promising technologies were selected to create ICT future scenarios, and the scenarios were validated by a group of core experts.

In 2016, IPET (Korea Institute of Planning and Evaluation for Technology in Food, Agriculture, and Forestry) that manages R&D under the Ministry of Agriculture, Food and Rural Affairs conducted the "2040 Agri-Food Future Technology Foresight Study". The study was conducted in three steps: (1) deriving future technology needs, (2) selecting promising technologies, and (3) presenting the future society vision. First, the study, which was conducted to predict future issues related to the agriculture and food sector and analyze the changing trends of science and technology, resulted in six megatrends, 14 sub-trends, and 100 future technologies for agriculture and food. The research results were utilized as the basis for creating a technology roadmap, establishing a mid- to long-term development strategy, and identifying outstanding projects for the next year. The methodology included integrated analysis of linear megatrend forecasts, big data analysis to identify the latest issues, patent activity analysis, V-S (Virtual Scenario) analysis, and Delphi survey. Data mining was conducted based on domestic and foreign patent data, and for V-S analysis, the bottom-up method of finding a feasible business at the present time and the top-down method of realizing a distant future lifestyle were used in parallel. Next, delphi survey was conducted on agriculture and food experts and a total of 10 step-by-step questions were asked to investigate the timing of realization of future technologies, competitiveness, and future development directions. Finally, future scenarios and illustrations were presented to describe how the 100 identified agri-food futures technologies will change our lives in 2040.

[Figure 4-4] Illustrations using agri-food future technologies



Agricultural technologies for environmental adaptation and pollution removal

Personalized medical technology and livestock disease management

2. Role of Technology Foresight Center in KISTEP

The Technology Foresight Center was established to look into the future through technology foresight. The role of the Technology Foresight Center is to provide the basis for the government's mid- to long-term strategic direction, which can be divided into four main areas.

[Figure 4-5] Role of Technology Foresight Center

The first role of KISTEP's Technology Foresight Center is to provide an environmental scan that allows you to anticipate the future without panicking when sudden changes occur. The second role is to identify key technologies As public spending is limited by rising welfare expenditures, major countries around the world are identifying key technologies through prioritization and focusing their budgets on key technologies. In response, KISTEP is conducting the S&T Foresight Program and identifying 10 emerging technologies. The third role is to develop the vision, which can help provide long-term strategic direction for a country. South Korea’s national S&T vision has been established every ten years. The final role is to support policymakers' decision-making through analysis of uncertain and complex issues. The technology foresight center conducts foresight studies on these issues every year.

[Table 4-1] Major roles of technology foresight center in KISTEP

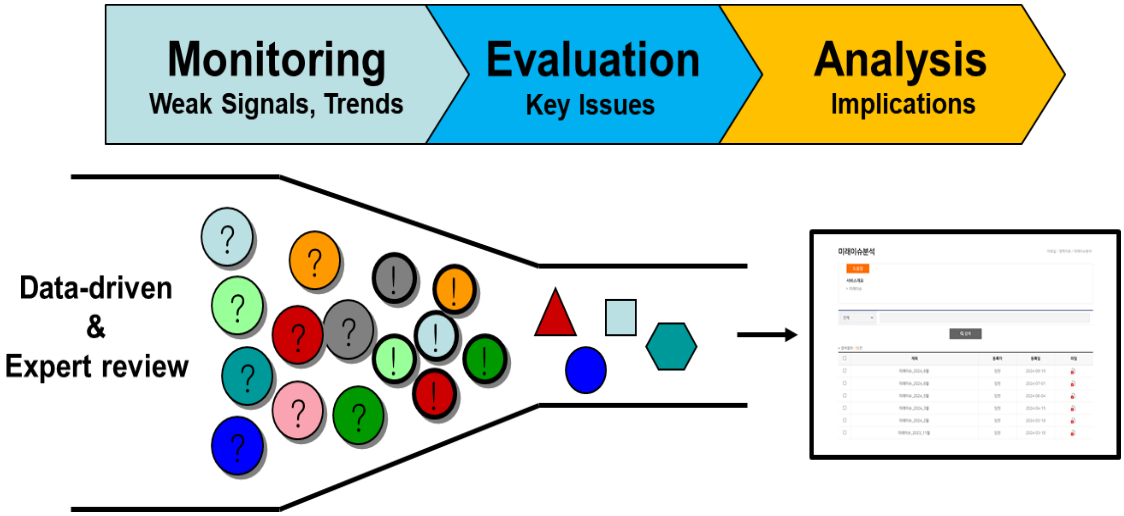
Role	Description	Example
Environmental Scanning	Monitor and analyze new issues that may be related with S&T	Technology-related issues are monitored and analyzed
Key Technologies	Identify technologies that have the potential to contribute to increasing national wealth and improving the quality of human life	Korean S&T Foresight Program for every 5 years KISTEP 10 emerging technologies
National Vision Building	Envision a desirable future for the country and provide long-term strategic direction	National S&T vision is established every 10 years
Support on Decision-Making	Support policymakers' decision-making with analytics on uncertain and complex issues	Foresight projects are carried out for specific issues Technology assessment is carried out every year

2.1 Environmental Scanning

In order to secure competitiveness in a rapidly changing and complex society, there is an increasing need for information on social, economic, environmental, technological, and policy-related issues in order to develop responses and plans at the corporate and national levels. Environmental scanning is a method of finding and analyzing new issues. It is the starting point for almost all technology foresight. In general, this method is adopted as the most basic one among various methods to grasp information and knowledge about the fields or subjects of technology foresight.

An environmental scan is typically conducted in three steps. First, in Step 1, quantitative analysis techniques are used to systematically analyze internet search data and documents such as reports, papers, and meeting minutes written by experts. Information about various social phenomena related to the topic is extracted, categorized, and organized. The analyzed issues and trends are then reviewed by experts. In the second step, key issues are selected from the collected issues. They are typically evaluated based on criteria such as impact and feasibility. Finally, key issues are further analyzed to understand their implications in the Korean context and the results are made publicly available on the KISTEP website.

[Figure 4-6] KISTEP’s three-step environmental scanning process

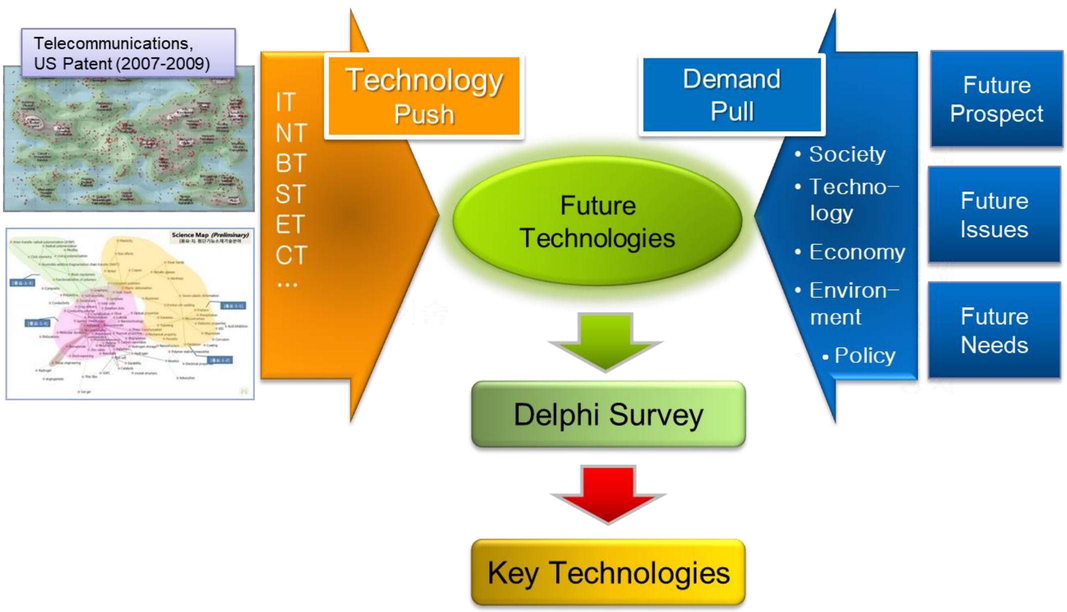


2.2 Identification of Key Technologies

2.2.1 Korea's science and technology foresight programs

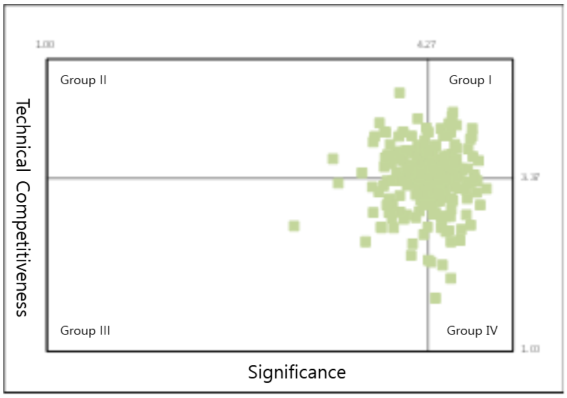
Korea's science and technology foresight programs are conducted every five years according to the Framework Act on Science and Technology. Since the first science and technology foresight program was launched in 1994, continuous improvements and developments have been made in methods, systems, and processes. In 2007, in order to secure linkage with the S&T master plan – the top-level national plan for science and technology, the timing was adjusted and the contents were supplemented, with the results of the third foresight program re-announced. The science and technology foresight programs conducted since then have been recognized as the core basic data for top-level science and technology planning at the national level. The main objective of the S&T Foresight Program is to provide vision and strategic direction in the field of science and technology, specifically to identify future technologies that have the potential to contribute to increasing national wealth and improving the quality of human life. In the sixth program, focus was placed on efforts to improve policy utilization.

[Figure 4-7] Methods of the 6th S&T Foresight Program to derive key technologies



In addition to presenting various future technology challenges, new attempts were made to provide concrete answers to the question of which technologies should be chosen strategically to enhance national competitiveness for the 6th S&T foresight program. For example, two criteria were used to identify strategically important technologies. One is “Significance to future society” and the other is “Technical competitiveness”. Future technologies of Group I (Leading Investment) that are high both in significance and technical competitiveness include 71 technologies of the 241 future technologies, including “Big data and AI-powered smart airport systems,” and “Real-time controlled power grid high reliability and safety systems.” Future technologies of Group IV (Catch-up Investment) with high significance but low technical competitiveness include 65 future technologies, including “Medical services, new drugs, and new materials development using microgravity” and “Three-dimensional quantum device technology for maximizing quantum computing performance.”

[Figure 4-8] Distribution of 241 future technologies by significance and technical competitiveness criteria



Classification	Group Ⅰ	Group Ⅱ
No. of Technologies (Percentage)	71 (29.5)	60 (24.9)
Classification	Group Ⅲ	Group Ⅳ
No. of Technologies (Percentage)	45 (18.7)	65 (27.0)

2.2.2 KISTEP’s 10 Emerging Technologies

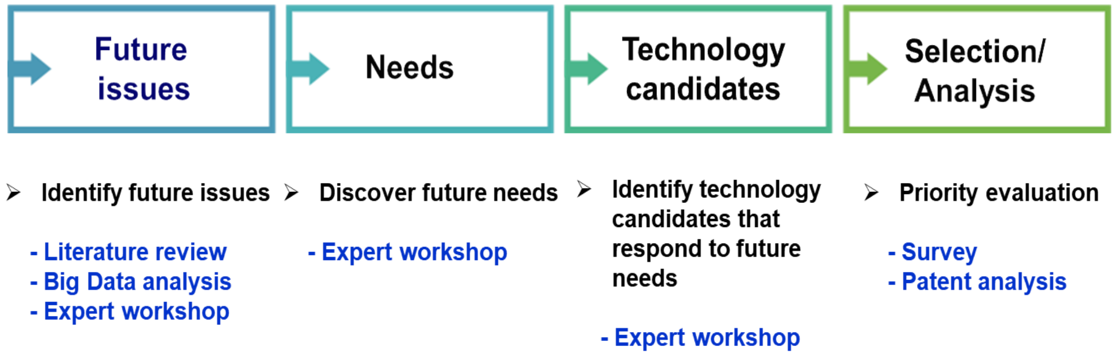
It is necessary to present the future direction of science and technology in response to the future society through the selection of emerging technologies. KISTEP has identified 10 emerging technologies each year, considering their economic and social impact in 5-10 years. Since 2013, KISTEP has been identifying emerging technologies that respond to specific topics.

[Table 4-2] List of KISTEP's 10 emerging technologies

	2021 (Contactless society)	2022 (Carbon neutral)	2023 (Next-generation security)
1	Non-Invasive Biometric Cardiovascular Disease Management Technology	Carbon Dioxide Capture and Conversion Technology	Infrastructure-integrated security technology for autonomous driverless vehicles
2	Level 4 autonomous vehicles for the transportation disadvantaged	Bio-based feedstock and product production technologies	Intelligent cybersecurity control and automated response technologies based on artificial intelligence
3	Personalized curation technology based on Learning eXperience Platform (LXP)	Carbon Reduction Blast Furnace and Electric Furnace Process Technologies	5G/6G network security technology
4	Self-driving last-mile delivery services	High-capacity, long-life secondary battery technology	Manufacturing (industrial) supply chain and system security vulnerability diagnosis automation technology
5	Intelligent Edge Computing	Clean Hydrogen Production Technology	Functional cryptography and application technologies such as homomorphic encryption for privacy-enhancing data safety utilization
6	Beyond Screen technology to break down interface barriers	Ammonia Generation Technology	User protection and security technologies in virtual environments such as metaverse
7	VR/Hologram-based real-time collaboration platform	Grid-connected systems	Quantum cryptography for absolute data security in the quantum era
8	AI security technology, cyber guardian in the hyper-connected era	High-efficiency solar cell technology	Cybercrime prevention and tracking technology that exploits digital new technologies
9	Non-face-to-face hyper-realistic media production and relay technology	Very large offshore wind systems	Cloud and edge security technologies for secure virtualization environments
10	Green packaging technology that reduces online shopping waste	Rare earth recovery technology	Cryptocurrency reliability assurance technology for safe digital economy utilization

In order to prepare for future issues on specific topics, the KISTEP’s 10 Emerging Technologies were selected through a four-step process. In step 1, future issues were identified by reviewing published literature and gathering expert opinions. The issues were then finalized through a survey of technology foresight experts and KISTEP policy audiences. In step 2, expert workshops were held to identify future needs to address the future issues identified in step 1. In step 3, we identified technology candidates related to the future needs. In step 4, we selected the 10 emerging technologies through discussion and written assessments at an expert workshop, and then then analyzed their characteristics, relationships to other emerging technologies, technical challenges, and policy recommendations.

[Figure 4-9] KISTEP's process for identifying 10 emerging technologies

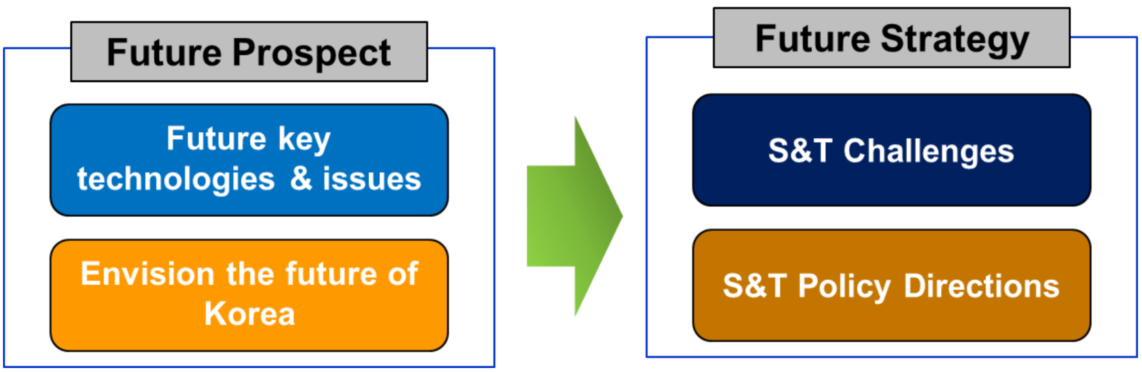


In Korea, major organizations such as KISTEP and KISTI identify and announce emerging technologies every year. According to the results of a survey of researchers on the utilization of emerging technologies announced by major domestic institutions, KISTEP's 10 emerging technologies are the most recognized and utilized (임현 외, 2019). The survey also found that the most common purpose for leveraging new technologies is to “generate ideas to plan new R&D projects and challenges” (80.3%).

2.3 Vision Building

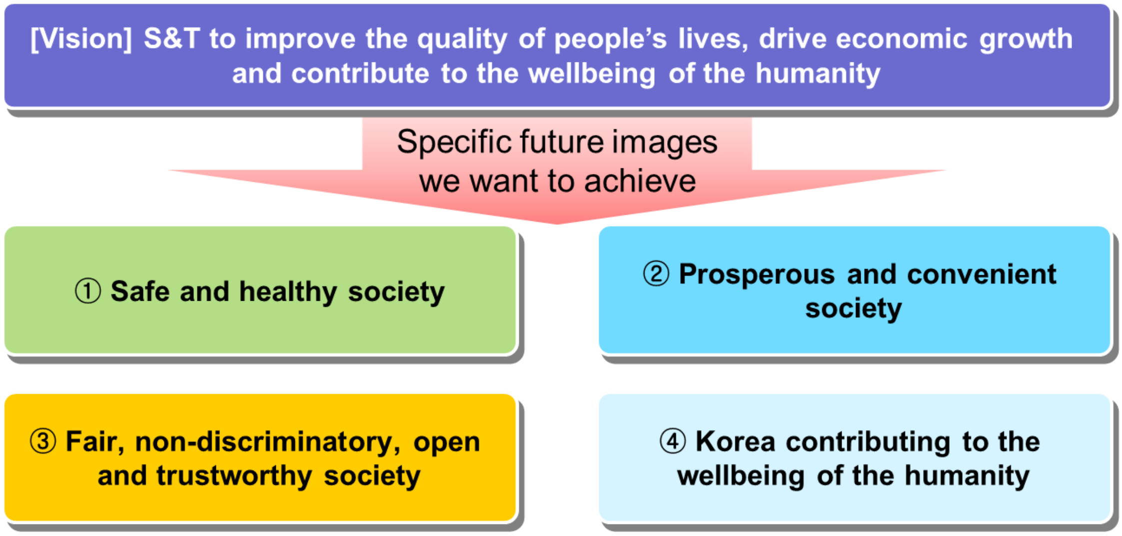
In Korea, S&T long-term visions have been established every ten years. Two long-term visions (① the long-term vision for S&T development by 2025 (in 1999), ② S&T future vision for 2040 (in 2010)) were established by KISTEP to contribute to the national development by predicting and responding to the fast-changing future in a preemptive manner. The recent S&T vision, "Korea's S&T future strategy for 2045", established by STEPI in 2019, laid out the scientific and technological vision and mid-to-long term policy direction in preparing for 2045. The establishment of the future vision for science and technology consists of two steps to establish future strategies based on the outlook for the future society.

[Figure 4-10] Two-step process for establishing a S&T vision



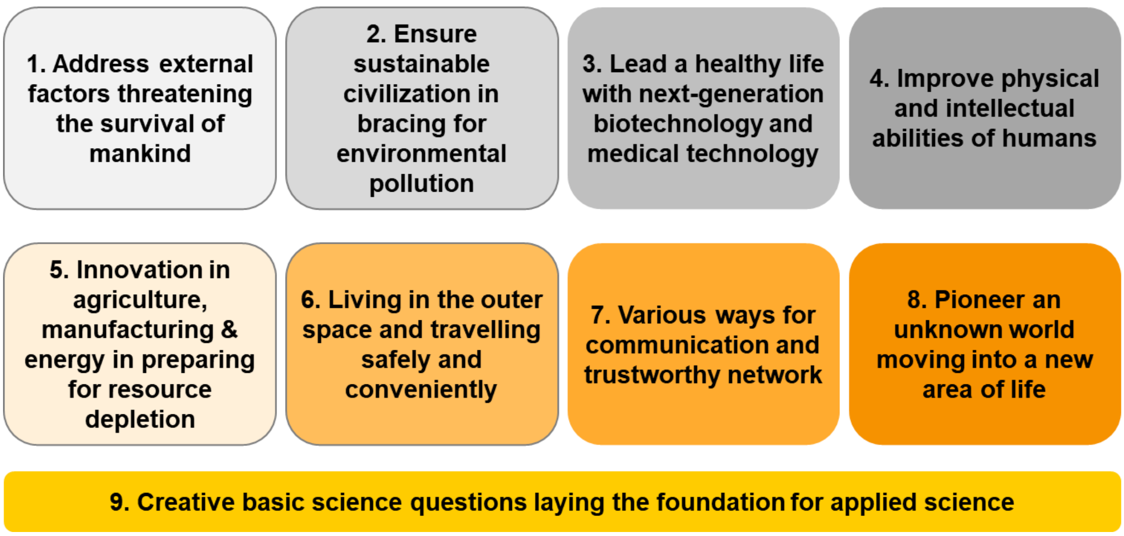
In the first step, the S&T vision described the future of Korea we hope for in 2045. The S&T vision was derived from past long-term strategies, public surveys, and megatrends analysis. This vision of the future was crystallized into four desirable futures: ① safe and healthy society, ② prosperous and convenient society, ③ fair, non-discriminatory, open and trustworthy society, ④ Korea contributing to the wellbeing of the humanity.

[Figure 4-11] S&T vision and specific future images we want to achieve in 2045



In the second step, technical challenges and policy directions were presented as specific strategies to achieve the goals. Regarding S&T challenges, we presented nine S&T challenges needed to realize the future we want and suggested directions for technology development. In particular, the ninth challenge offered creative challenges in basic science which laid the groundwork for eight major challenges (including 160 technologies) and applied science.

[Figure 4-12] S&T challenges to be resolved for realizing the future vision



In order to continuously secure innovative capabilities in science and technology, which are the basis for solving the nine S&T challenges presented above, eight S&T policy directions were proposed. Specifically, policy directions and detailed tasks were presented from a mid-to long-term perspective for each of the eight major S&T areas, including human resources, R&D, business and industry, social issues, regional, global, science advanced countries, and future directions. The S&T vision was expected to provide a long-term guideline for S&T-based short-to-mid-term strategies and plans established every five years including S&T foresight program (future technologies, 2021-2045), S&T master plan (policies, 2023-2027) and mid-to-long-term national R&D investment strategy (areas of investment).

[Figure 4-13] S&T policy directions for solving S&T challenges

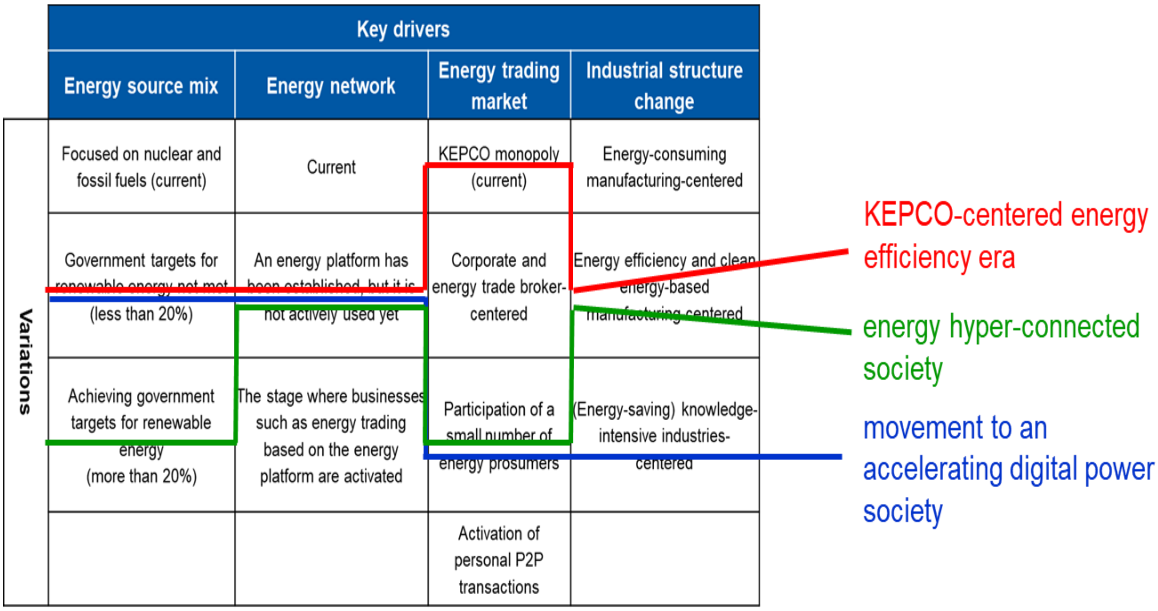
2.4 Support on Decision-Making

Governments face an increasingly complex and uncertain environment as they design public policies to address pressing issues. In addition, technological advances have both positive and negative impacts on society. Technology foresight using scenario planning techniques helps to understand these complex and uncertain issues, and technology assessment helps to prepare in advance for the negative impacts of technological advances.

2.4.1 Foresight study with scenario planning

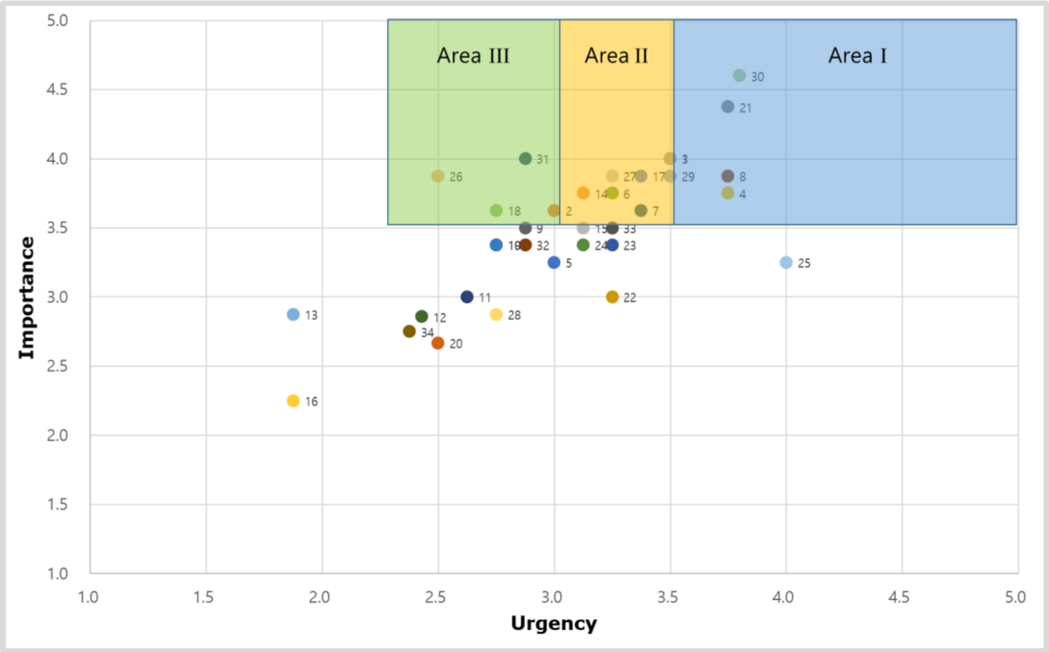
Technology Foresight Center in KISTEP conducts technology foresight study every year by selecting uncertain and complex issues related to science and technology. One of them is the foresight study on “Future of Energy Consumption by digital technology development in 2030” conducted in 2019. As Korean government moves towards the goal of carbon neutrality by 2050, the supply of renewable energy such as solar and wind power is becoming more important. In the energy sector, digital transformation is expected to bridge the gap between energy production and consumption in conjunction with the deployment of renewable energy and provide the ability to integrate the entire energy system. However, Korea still maintains a centralized energy system based on nuclear power plants and fossil fuels, and linkage and conversion to distributed power grids are being made slowly. In this regard, using the scenario methodology, we studied the 'future of energy consumption by digital technology development in 2030’ and identified countermeasures. After deriving various factors through STEEP analysis, four key factors, including energy source mix, energy network, energy trading market, and industrial structure change, were selected through evaluation. After identifying the spectrum of change for each key factor, the Bayesian model was used to identify three scenarios such as “KEPCO-centered energy efficiency era”, “movement to an accelerating digital power society”, “energy hyper-connected society”.

[Figure 4-14] Key drivers and three scenarios



The 34 policy measures derived from each scenario are expected to be utilized as a basis for establishing a long-term strategy for national energy policy. The portfolio analysis was conducted according to the importance and urgency of the policy measures, and the policy measures with high importance were categorized into three areas. Among them, Area I is an area of high importance and urgency, and includes measures such as "opening the electricity retail market (to break the KEPCO monopoly)," "improving the energy production and trading system," "strengthening energy demand management R&D," and "revitalizing microgrids." Area III is an area that has high importance but low urgency and should be promoted in the mid- to long-term. It includes policy measures such as "RE-100 corporate and consumer benefits," "establishment of policies to maintain energy supply stability," and "strengthening policies to diversify energy sources such as hydrogen energy."

[Figure 4-15] Policy measures divided into three areas

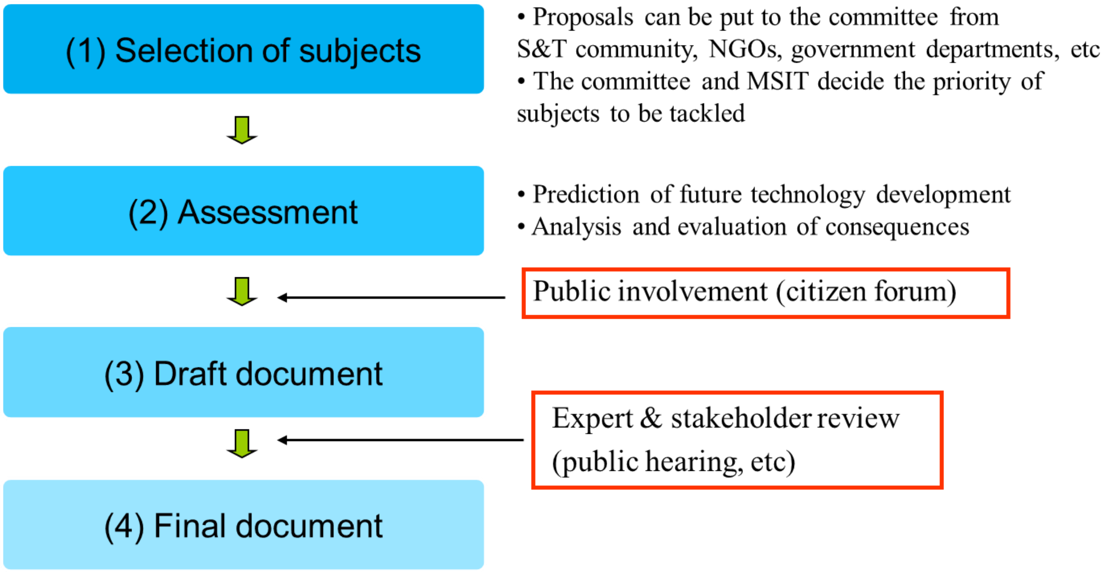


2.4.2 Technology Assessment

KISTEP evaluates the impact of the development of science and technology on the economy, society, culture, ethics, and environment based on Article 14 of the Framework Act on Science and Technology. The technology assessment is conducted through discussions among various stakeholders, including experts in various fields and citizens, and aims to derive policy recommendations to develop technology in a desirable direction. Based on this, it aims to maximize the positive impact of the technology and minimize the negative impact by guiding the technology to develop in a desirable direction from the beginning of policy formulation.

KISTEP is commissioned annually by the Ministry of Science and ICT to conduct technology assessments, and the process is as follows.

[Figure 4-16] Procedure for conducting a technology assessment



First, the technology selection committee establishes the criteria for selecting the technology subject, deliberates on the selection of the technology subject, and once the technology subject is confirmed, the technology subject is defined and scoped through preliminary analysis of the technology subject and the main evaluation topics are derived. Next, the technology assessment is conducted simultaneously through the Technology Assessment Committee, citizen forums, and online opinion gathering, and the assessment results are derived by comprehensively considering the opinions of these various assessment entities. Afterward, the final evaluation results which reflect additional opinions through public hearing are reported to the steering committee of the National Science and Technology Advisory Council, and the heads of relevant central administrative agencies are required to reflect the results in research and planning for national R&D projects in their respective fields or to take measures to minimize negative impacts.

3. Recommendations for Establishing a Technology Foresight Center in Peru

Concytec in Peru proposes to make its technology foresight center an internal organization rather than a separate entity. Initially, the Technology Foresight Center can be set up as an internal organization within Concytec, but as the size and role of the Technology Foresight Center grows, it should be considered as an independent organization. The human resources of Concytec's proposed Technology Foresight Center are shown in the table below.

[Table 4-3] Human resources of Concytec's proposed Technology Foresight Center

Human Resources	Roles
Chief Coordinator	• Responsible for the general direction of the center and the management of resources
Specialists (three positions)	• Carry out studies and analysis of technological foresight in their areas of specialization. • Identify emerging trends and disruptive technologies. • Collaborate with other specialists and the lead coordinator in the development of reports and strategies. • Participate in conferences and events related to technological foresight. • Assist in the training and dissemination of knowledge of technological foresight
Administrative Support	• Administrative Assistant: Assists in the management of documentation, coordination of events and internal communication • Head of Communication and Dissemination: Responsible for the dissemination of the results of the studies and the relationship with the media
External Advisory Committee	• Composed of experts in technology, economics, sociology and the environment • Advises the Chief Coordinator and specialists on the validation of studies and strategic recommendations

In recent years, as the amount of data on trends, future issues, and more has grown and the connections between data have become more complex, it has become increasingly important to use quantitative techniques like data analysis to monitor and analyze future issues rather than relying solely on experts. Horizon scanning, which systematically examines evidence for future trends rather than making predictions, helps governments analyze whether they are adequately prepared for potential opportunities and threats. This allows them to formulate policies that are resilient to a variety of future environments, which is why it's important that Concytec's proposed Technology Foresight Center also employs a specialist with expertise in data analysis. The use of Artificial Intelligence (AI) in Technology Foresight has been gaining momentum in recent years, as organizations and governments seek more sophisticated and data-driven approaches to anticipate and plan for the future. AI can process vast amounts of data from various sources, including social media, economic indicators, scientific research, and news articles. By analyzing these data, AI can identify patterns, trends, and weak signals that may indicate emerging changes or potential future developments. Therefore, it is also recommended to employ an AI data analyst.

The key functions of the proposed Technology Foresight Center are shown in the table below. The key functions are well presented, including detecting and analyzing S&T-related issues and trends that will affect Peru and conducting foresight studies using scenario methodology. Introducing government officials to the future and providing training to develop forecasting capabilities within government is also an important function of the Technology Foresight Center. However, given Peru's limited government R&D budget, it is necessary to add a function to identify key technologies. In addition, the Technology Foresight Center should play a guiding role in determining Peru's long-term S&T policy direction. Therefore, it is necessary to add S&T future visioning as one of the Technology Foresight Center's key functions.

[Table 4-4] Key functions of Concytec's proposed Technology Foresight Center

Key Functions	Description
Trend Monitoring and Analysis	• Conduct ongoing studies on technology trends and their potential impact on various sectors
Future Scenarios	• Develop future scenarios to foresee possible developments and prepare for different eventualities
Research and Publications	• Publish reports, articles, and studies that provide insights into the future of technology.
Training and Education	• Offer workshops, courses and seminars to train professionals and stakeholders on technological foresight
Public Policy Advisory	• Collaborate with government entities to develop policies that promote the responsible and beneficial adoption of new technologies
Networks and Collaborations	• Establish and maintain collaborative networks with other research centers, universities and international organizations

Environmental changes such as pandemics, climate change, energy crises, demographic shifts, and artificial intelligence have demonstrated how important long-term thinking and science and technology are in our society. Public policy focused on individual sectors or industries and short-term thinking driven by electoral considerations fail to see the profound intersectoral and interdisciplinary origins of disruption and breakthroughs. Technology foresight that provides a long-term, agile, and holistic vision is key to responding to these abrupt environmental changes. For technology foresight to be effective, it must be coordinated across administrative levels, agencies, and organizations that produce foresight knowledge. There must be an independent organization responsible for representing technology foresight knowledge that is disseminated across the administration, and that can leverage these insights for appropriate policy design and decision making. In the future, Concytec’s proposed Technology Foresight Cednter should be expected to evolve into an independent organization that provides comprehensive strategic direction within government.

CHAPTER 5

Establishment of Peru's R&D Planning and Technology Foresight Center: Guidelines for Policy Recommendations

S&T Planning and Technological Forecasting for Prioritized Sectors in Perú

Chapter 5. Establishment of Peru's R&D Planning and Technology Foresight Center: Guidelines for Policy Recommendations

1. Introduction[6]

The rapid rise of digital transformation and the advent of Industry 4.0 are reshaping economies and industrial structures worldwide. For emerging markets, this transformation presents an opportunity to leap forward by embracing new technologies and innovations. In this context, it is crucial for Peru to build a robust capability in technology foresight that enables the country to anticipate, prepare for, and effectively respond to future technological developments. The establishment of an R&D Planning and Technology Foresight Center is therefore essential to strengthen Peru's strategic positioning in the global technological landscape. This report explores the rationale behind the creation of such a center and outlines the methodology, guidelines, and policy recommendations necessary to achieve this vision.

Technological foresight has become an essential tool for guiding strategic decisions in rapidly evolving industries, particularly in emerging economies. By leveraging methods such as the Delphi survey, countries can identify key areas of technological growth, set investment priorities, and address potential challenges. This approach has been successfully used worldwide to build consensus on future directions and inform policy and R&D planning (Linstone & Turoff, 1975; Adelson & Aroni, 1975). To achieve this, the center will focus on the following key goals. First the center aims to develop a robust technology foresight framework that will support scenario planning, trend analysis, and policy advisory. This includes creating a systematic approach to monitoring technological trends, predicting

[6] <Chapter 5. Establishment of Peru's R&D Planning and Technology Foresight Center : Guideline for Policy Recommendations> is written by Ahreum Hong, Professor, Kyung Hee University.

potential disruptions, and advising on strategies to leverage these developments for national growth. Second, One of the fundamental components of the center is the establishment of a strong IT infrastructure that will serve as the backbone for supporting six key government ministries. This infrastructure will facilitate data sharing, collaboration, and integration across sectors, enabling seamless coordination for industry promotion, innovation initiatives, and cross-ministerial projects. Third, By leading the Latin American ISO group, the center will help position Peru as a hub for technology standards and innovation within the region. This involves fostering collaboration with international technology organizations, setting benchmarks for technological excellence, and driving the adoption of industry standards across Latin America.

For Peru, understanding the strengths and barriers across sectors like mining, agriculture, and manufacturing is crucial to developing sustainable and competitive technological solutions (Kim, 2001; Kim & Oh, 2000). Recent studies have emphasized the importance of government support in infrastructure and workforce development, as well as international collaboration, to overcome limitations and drive innovation (KISTEP, 1999; Ministry of Information and Communication, 2001). The first and second Delphi survey result aims to provide a strategic roadmap for Peru by offering insights into the country's technological capabilities and guiding future policy initiatives. In the face of accelerating technological change, it is essential for Peru to build a forward-looking institution that can foresee and prepare for future challenges and opportunities. The proposed R&D Planning and Technology Foresight Center will not only equip the nation with the tools to navigate the complexities of digital transformation but also empower it to lead regional efforts in innovation. By integrating R&D roadmapping and technology foresight research and developing a comprehensive IT infrastructure, the center will provide a solid foundation for Peru's sustained technological and economic growth, reinforcing its ambition to become a leader in technology foresight and innovation.

The main goal of this project is to enhance Peru's capacity for R&D planning and technology foresight through the establishment of a dedicated Technology Foresight Center. This initiative is part of a broader collaboration between South Korea's STEPI (Science and

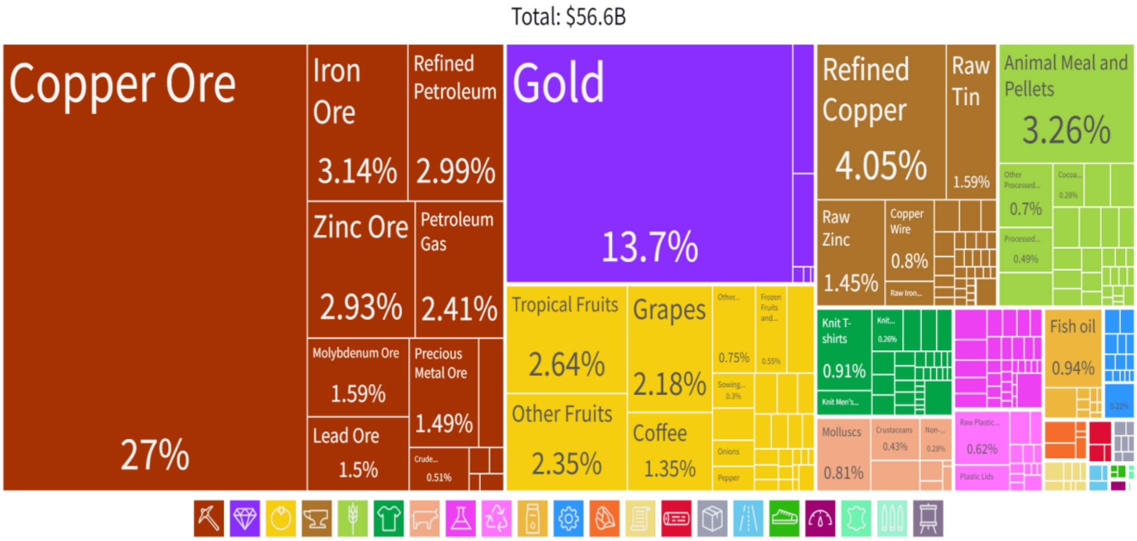
Technology Policy Institute) and Peru's CONCYTEC (National Council for Science, Technology, and Technological Innovation). The study focuses on creating a strategic roadmap for developing Peru's technological capabilities by leveraging methodologies like technology roadmapping and Delphi surveys, which identify key areas for technological growth, set investment priorities, and outline future development directions. Share Korea's experience and knowledge in R&D governance, strategic planning, and technology foresight with Peruvian stakeholders. This involves in-depth workshops and training sessions to provide practical insights into R&D program development and the establishment of a technology foresight framework. Strengthen the capabilities of Peruvian policymakers, research institutions, and industry stakeholders by building a robust IT infrastructure to support R&D and innovation projects. This also includes guidance on developing strategic partnerships and collaborative efforts across government, academia, and the private sector. Propose a structured plan for setting up a dedicated center in Peru that will lead the country's efforts in monitoring technological trends, conducting scenario analyses, and providing policy advisory services. The center aims to position Peru as a regional leader in technology innovation by driving the adoption of international technology standards and fostering collaboration with global technology organizations. Overall, the project seeks to lay the groundwork for sustainable and competitive technological advancement in Peru by offering a strategic blueprint for future R&D initiatives, thus reinforcing Peru's long-term economic growth and innovation potential.

2. R&D Requirements for Establishing the Technology Foresight Center

Peru's R&D and innovation landscape has shown steady growth over recent years, driven by efforts to diversify the economy and foster technological advancements. However, the nation still faces challenges, such as limited investment in research and development, insufficient infrastructure, and gaps in innovation capabilities across key industries. While there have been initiatives to support technology-driven growth, a more cohesive and strategic approach is needed to fully harness Peru's potential as a regional leader in innovation. To overcome these barriers, it is crucial to build a robust ecosystem that can foster collaboration between government bodies, academia, and the private sector.

Below Figure provides a detailed breakdown of Peru's export composition, illustrating the diversity of its export sectors. The leading export commodity is copper ore, which makes up a substantial 27% of the total exports, reflecting Peru's rich mineral resources and strong mining industry. This is followed by gold, accounting for 13.7%, further underscoring the country's reliance on mining. Other significant mineral exports include refined copper (4.05%), iron ore (3.14%), and zinc ore (2.93%). Apart from minerals, the chart shows exports of agricultural products such as tropical fruits (2.64%), grapes (2.18%), and coffee (2.35%), highlighting Peru's growing agricultural sector. There are also smaller but notable shares for animal meal and pellets (3.26%), refined petroleum (2.99%), and various processed foods. Additionally, textiles like knit T-shirts (0.91%) represent the country's manufacturing sector. Overall, the chart demonstrates Peru's strong emphasis on mining and agriculture, with smaller contributions from other sectors like energy, food processing, and manufacturing (STEPI, 2022).

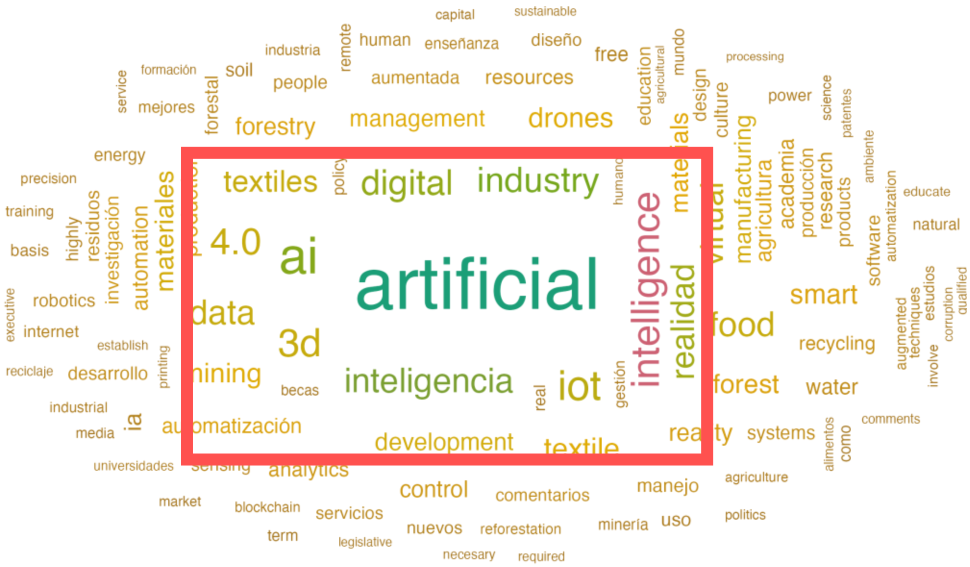
[Figure 5-1] Detailed breakdown of Peru's export composition (2023)



The rapidly evolving technological landscape underscores the importance of systematic technology foresight in shaping Peru's national R&D strategy. Technology foresight involves anticipating future technological trends, assessing their potential impact, and identifying opportunities for innovation. For Peru, a systematic foresight approach would enable policymakers and industry leaders to make informed decisions, prioritize investments, and develop strategies that align with global technological shifts. This is particularly essential for emerging economies, where strategic planning can be a powerful tool for achieving sustainable development and economic competitiveness.

Experts across the six strategic industries have recognised that there is much interest in and a great need for technologies related to industrial transformation, such as artificial intelligence, digital innovation, and Industry 4.0. A strategy must be put in place to foster Peru's internal ICT and data-related experts and connect them with strategic industries so as to promote the industrial structure (STEPI, 2022, 2023).

[Figure 5-2] Word cloud of the results of the Peruvian emerging technology expert consultation



Source: Author’s Own Work (2024)

The word cloud represents the key themes and focus areas identified through first Delphi survey responses related to Peru's technological and industrial future. The most prominent term is "artificial", indicating a strong emphasis on artificial intelligence (AI), which is also highlighted with related terms such as "intelligence," "ai," "iot" (Internet of Things), and "digital." This suggests that Peru is keen on advancing technologies that integrate AI and digital transformation. Other notable terms include "industry," "4.0," "data," and "automation," pointing towards a focus on Industry 4.0 concepts and the broader adoption of smart technologies across different sectors. Additionally, words like "management," "drones," "3d," "textiles," "food," "forestry," and "development" hint at specific applications of these technologies in diverse fields such as agriculture, manufacturing, energy, and resource management. The presence of terms like "recycling," "water," "sustainable," and "forest" also indicates an interest in sustainability and environmental management. The word cloud reflects a broad vision of technological integration across industries, aiming to boost innovation and efficiency (STEPI, 2022).

Across all six industries, a few recurring themes stand out. Artificial intelligence (AI) is a pivotal technology, driving automation, optimization, and innovative solutions across sectors. The emphasis on digital transformation—including technologies like big data, IoT, 3D printing, and smart systems—reflects a shift toward greater efficiency, connectivity, and data-driven insights. Additionally, sustainability and environmental management are critical focus areas, as seen in the increased attention on smart farming, circular economies, and ecosystem conservation. Overall, the integration of advanced technologies to foster innovation, sustainability, and operational efficiency is at the core of the strategic direction for these industries from 2022 to 2039.

[Table 5-1] Key Technologies & Trends of the results of the Peruvian 6 Industries

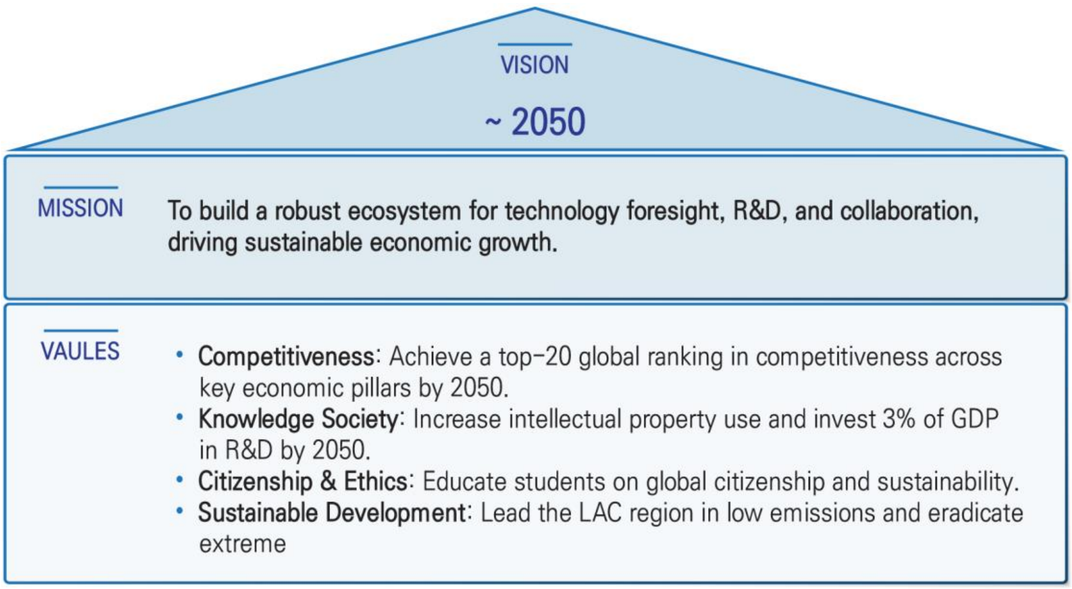
Industry	Key Technologies & Trends	Predicted Growth/Decline
Agribusiness	• AI, Big Data, Precision Agriculture, Blockchain, Pest Control, Supply Chain	• High emphasis on AI and precision agriculture technologies
Advanced Manufacturing	• AI, 3D Printing, Industry 4.0, IoT, Virtual/Augmented Reality, Cyber Physical Systems	• High growth in Robotics, development as a complementary industry
Mining and Materials	• AI, 3D Printing, Geotechnical Engineering, Clean Energy Mining, Material Processing	• Growth in Clean Energy Mining, Casting & Welding, stable Material Processing
Textiles and Apparel	• Smart Textiles, 3D Technology, Circular Economy, Sustainable Fibers, Digital Design	• Growth in Logistics, Textile Manufacturing; decline in Apparel Manufacturing, Textile Materials
Creative Industry	• AI, Virtual/Augmented Reality, 3D Technologies, Big Data Analytics, Media Services	• Growth in Media Services; decline in Financial Services over time
Forestry	• AI, Forest Management, Sensor Technology, Data Monitoring, Conservation Efforts	• Growth in Environmental Management, Precision Machineries; decline in Furniture Manufacturing

Source: Author's Own Work (2024)

Globally, several countries have successfully established technology foresight centers that play a pivotal role in guiding national innovation strategies. Countries like Japan, Germany, and South Korea have developed advanced foresight frameworks that integrate AI and data analytics to predict future technology trends, assess risks, and create long-term development plans. These centers not only provide valuable insights for government policy but also promote collaboration across industries, facilitating innovation and economic growth. By analyzing these successful models, Peru can identify best practices and adapt them to create a tailored framework that addresses its unique challenges and opportunities.

The establishment of a Technology Foresight Center in Peru will require a comprehensive framework that integrates several key components. First, a solid IT infrastructure must be developed to support data collection, processing, and analysis across various sectors. Second, the center should incorporate AI research capabilities to enhance technology monitoring and prediction. Additionally, it is essential to create a collaborative platform that brings together stakeholders from government, academia, and the private sector to foster innovation and knowledge sharing. The framework must also include mechanisms for continuous training and capacity building to ensure that the center remains adaptive to emerging technologies and market trends. By implementing these elements, the center will be well.

[Figure 5-3] Vision and Goals 2050



■ Vision:

- By 2050, Peru aims to become an industrialized country driven by R&D, capable of fully harnessing the power of the latest technological advancements. The country will have addressed and bridged various gaps, earning recognition as one of the most competitive nations in Latin America and the world. Consolidated efforts and achievements across economic, social, and environmental domains will enable Peru to be seen as a sustainably developed nation. This progress will transform Peru into a knowledge-based society where education, research, innovation, high-quality human capital, and technology drive continuous social improvement, backed by ethical behavior, respect for the law, and moral conduct.

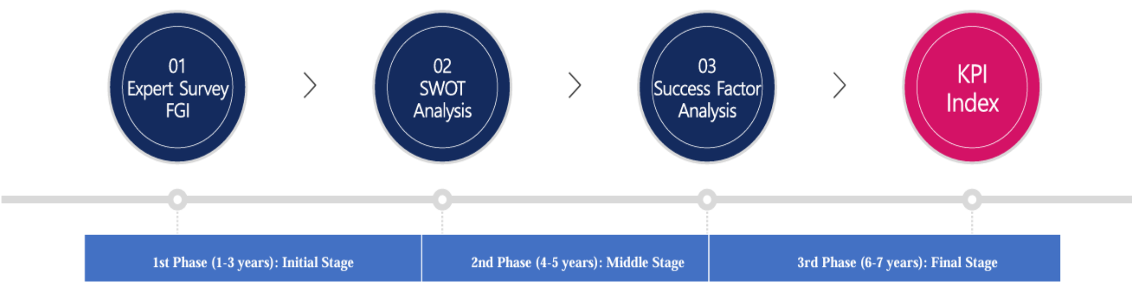
■ Goals and Values

1. **Competitiveness:** By 2050, Peru will rank among the top 20 most competitive countries, excelling across key pillars such as enabling environment, human capital, markets, innovation ecosystem, transformation readiness, disruption management, and resilience.
2. **Knowledge Society (Intellectual Property):** By 2050, Peru's economic sectors will see a significant increase in the use of intellectual property rights.
3. **Knowledge Society (R&D Investment):** By 2050, Peru will allocate 3% of its GDP to research and development.
4. **Citizenship and Ethical Behavior:** By 2050, a high percentage of Peruvian students will demonstrate a solid understanding of global citizenship and sustainability through education.
5. **Sustainable Development (Emissions):** By 2050, Peru will have the lowest fuel emissions in the Latin American and Caribbean (LAC) region.
6. **Sustainable Development (Poverty Reduction):** By 2050, Peru will have successfully eradicated extreme poverty.

3. Strategies for Applying Technology Foresight to Public R&D Planning

Effective technology foresight is essential for guiding public R&D planning, allowing governments to anticipate future technological trends and align national research efforts with long-term strategic goals. To achieve this, several proven methodologies can be employed, each offering unique insights and benefits. These methodologies include scenario analysis, Delphi method by Peruvian expert, and technology roadmapping. By leveraging these approaches, policymakers can better understand emerging opportunities and challenges, enabling them to make informed decisions that foster innovation and economic growth. Integrating these foresight tools into the R&D planning process ensures a systematic evaluation of future possibilities, helping to set priorities, allocate resources effectively, and promote collaboration across various sectors.

[Figure 5-4] Effective technology foresight Methodology



Source: Author's Own Work (2024)

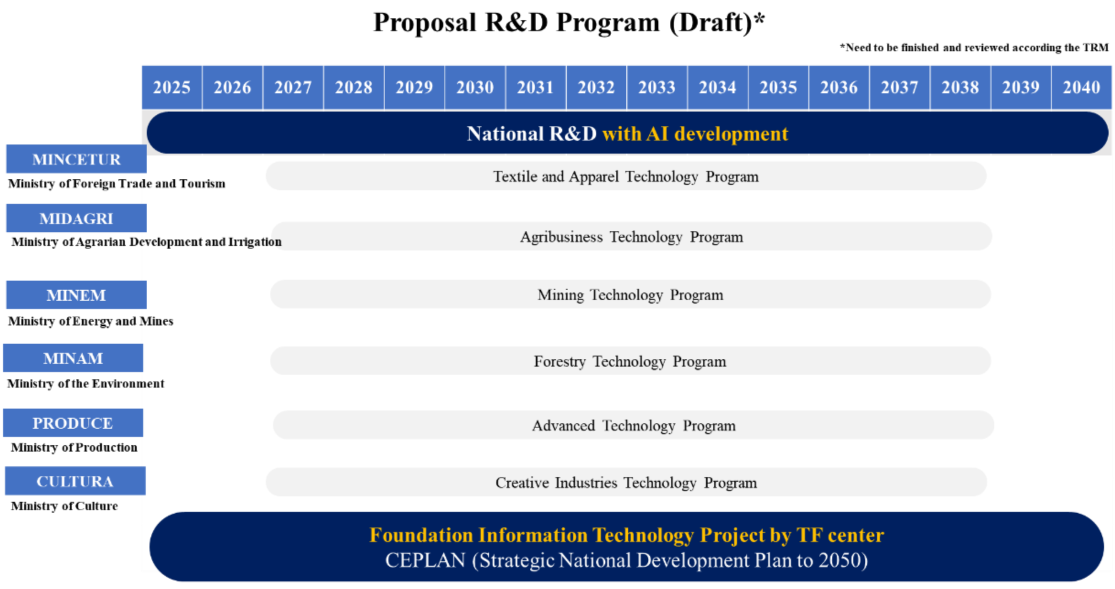
Figure 5-4 outlines a strategic process to assess and enhance project success through various stages, focusing on expert feedback, SWOT analysis, success factor identification, and performance monitoring.

1. **Expert Survey (FGI)**: The process starts with focused group interviews (FGI) with experts to gather qualitative data and insights.

2. **SWOT Analysis**: A SWOT analysis (Strengths, Weaknesses, Opportunities, and Threats) is used to identify and evaluate internal and external factors that could affect project success.
3. **Success Factor Analysis**: This step focuses on identifying crucial factors that influence project outcomes.
4. **KPI Index**: The final phase establishes Key Performance Indicators (KPIs) to measure the success of both short- and long-term goals, allowing for continuous strategy adjustments based on quantifiable data.

In 2025, Peru will establish its Technology Foresight Center (TFC) with a clear vision to strengthen AI and ICT infrastructure. The TFC will work closely with various government ministries to develop national R&D programs in industries such as textiles, agribusiness, mining, forestry, and creative industries. Through the TFC, Peru will enhance AI capabilities, bolster ICT infrastructure, and lead regional efforts in technology standards, positioning itself as a regional innovation hub. By 2050, the TFC aims to integrate AI into all key sectors, enhancing Peru's global competitiveness in research and innovation.

[Figure 5-5] Integration of technology foresight into national R&D strategies



Source: Modifed Author's Own Work (2024)

4. Short Term, Mid-term to Long-Term Strategic Recommendations for R&D Program Planning and Technology Foresight Center Establishment

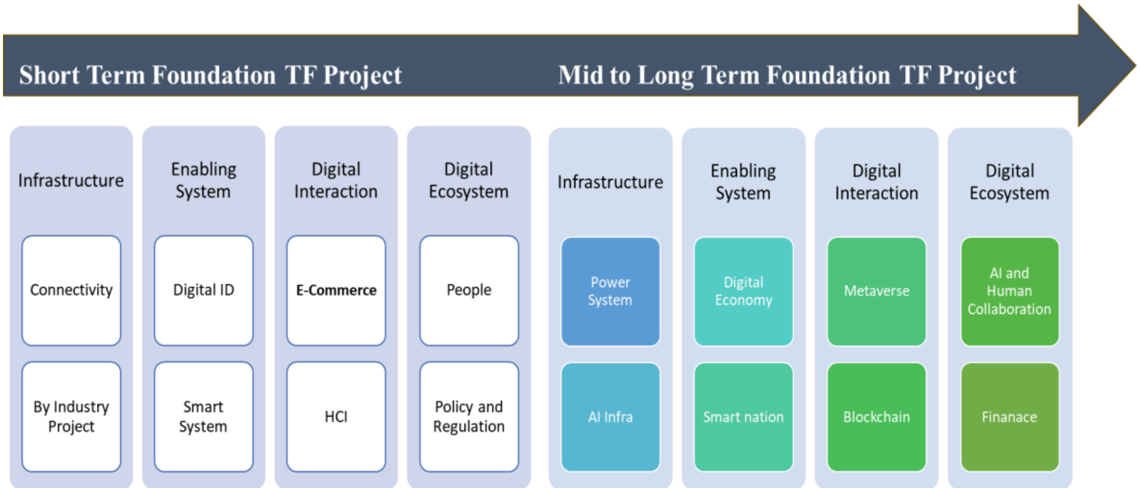
Peru’s technology foresight strategy aims to establish a solid foundation in ICT and AI to enhance its position as a leader in Latin America. The short-term goal focuses on strengthening foresight capabilities and creating collaborative networks. In the mid-term, Peru plans to build a robust national platform emphasizing automation, Industry 4.0, and innovation for sustainable growth. Long-term, the goal is to lead global technology foresight by leveraging AI and green technology to drive innovation and economic prosperity."

[Figure 5-6] Short Term, Mid-term to Long-Term Strategic Recommendations

Source: Author’s Own Work (2024)

The "Short Term Foundation TF Project" focuses on establishing critical digital infrastructure and enabling systems that will drive Peru's transition into a digitally connected economy. In the first phase, connectivity and smart systems are prioritized, including the introduction of Digital IDs and E-commerce platforms. As the project transitions to a mid- to long-term phase, the focus shifts towards more advanced technologies like AI infrastructure, the metaverse, and blockchain, ensuring collaboration between AI systems and human capital. This holistic approach will support Peru’s innovation and leadership in the digital economy.

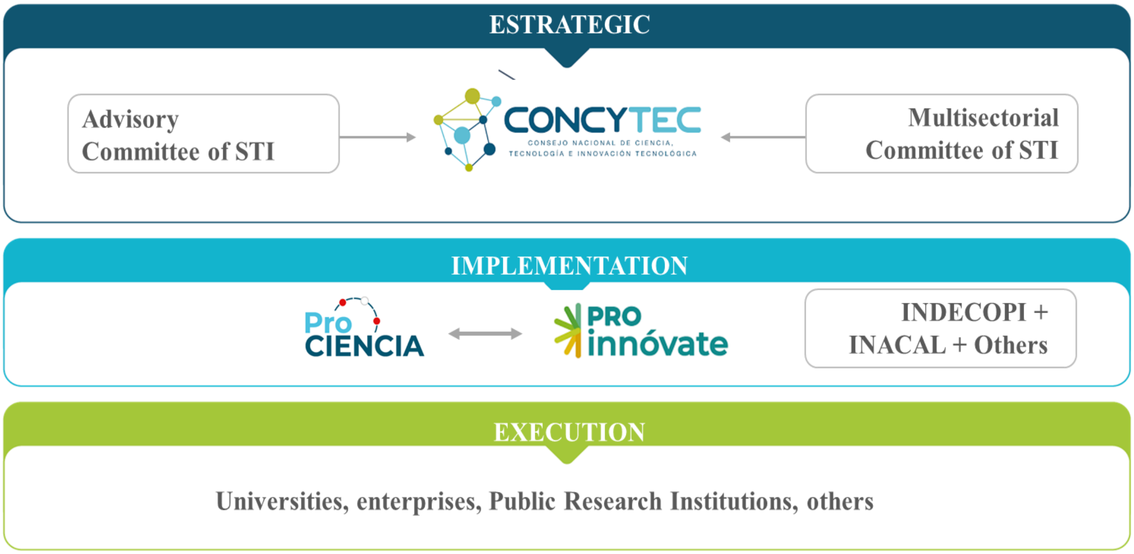
[Figure 5-7] Strategic planning for the establishment and operation of the center



Source: Author's Own Work (2024)

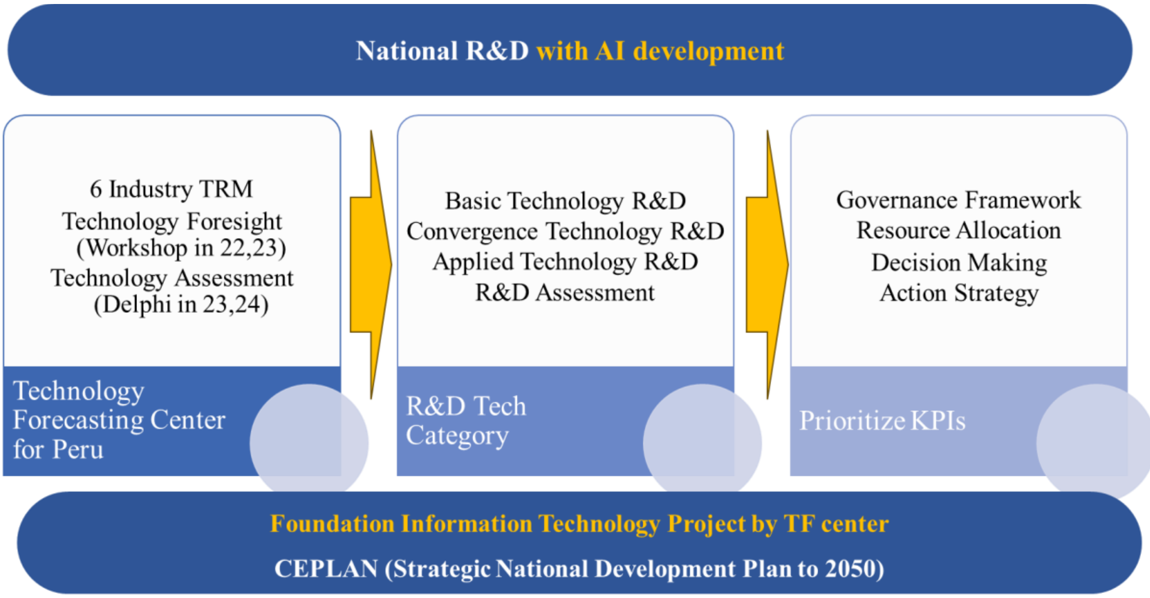
Figure 5-8 is the the capacity of CONCYTEC's practitioners and the stakeholders (government policymakers, research institutions, universities, the private sector, etc.) related to S&T planning and technological forecasting for prioritized sectors. This organizational structure highlights the integration of key agencies and committees in the Peruvian innovation ecosystem. At the strategic level, CONCYTEC (The National Council for Science, Technology, and Technological Innovation) leads with support from the Advisory and Multisectorial Committees of STI (Science, Technology, and Innovation). Implementation is carried out by agencies like ProCiencia and ProInnovate, which collaborate with other important institutions like INDECOPI and INACAL. The execution phase involves a wide range of public and private entities, including universities and research institutions, to drive innovation across the country

[Figure 5-8] STI Governance Structure



Peru's national R&D development, aligned with AI integration, aims to bolster innovation across critical industries. The roadmap emphasizes key milestones, starting with a Technology Forecasting Center focused on foresight workshops and assessments in the short term. The R&D phase will address basic, convergence, and applied technologies, driving advancements across sectors. Governance will guide resource allocation, decision-making, and strategy implementation. The foundational IT project by the Technology Forecasting Center aligns with the nation's Strategic National Development Plan to 2050, fostering a future-ready ecosystem."

[Figure 5-9] Perú's national R&D development



Source: Author's Own Work (2024)

Several key industries, such as agriculture, forestry, mining, and media, are outlined with specific goals to be achieved by 2035. Each sector's roadmap prioritizes technological advancements and development, focusing on domestic R&D, international cooperation, and strategic investment. Some of the priority areas include. By 2035, 30% of farms and enterprises should adopt smart technologies in Smart Farms. Goals include achieving 50% saw milling quality and reforesting 7.5 million hectares in Forestry Sector. A focus on geotechnical engineering improvements and clean energy mining, aiming for 75% of mining processes to use clean energy in Mining. Development of financial services, digital content, and media services, with a strong emphasis on international collaboration and domestic R&D in Media and Digital Content. These roadmaps emphasize the role of R&D investment in driving innovation and achieving long-term goals across multiple industries.

[Figure 5-10] Each sector's roadmap prioritizes technological advancements and development

•TRM Agribusiness:

Top Priority Product

TECHNOLOGIES	GOAL BY 2035	SHORT BEFORE 2026	UP TO 2030	UP TO 2035
SMART FARMS	30% of farms and enterprises reach smart level	Domestic R&D	Domestic R&D	
FOOD PROCESSING	50% of firms adopt modern technology	Int. Coop/Domestic R&D/Investment	Domestic R&D/Investment	
AGROMECHANICS	50% of farms adopt agromechanics components as irrigation systems	Domestic R&D/Investment	Domestic R&D/Investment	
AGROCHEMICALS	50% of farms use agrochemicals according to overseas rules	Int. Coop./Domestic R&D	Domestic R&D	
AGROBIOLOGY	50% of farms use nature fertilizers and agrobiology inputs	Domestic R&D	Domestic R&D	

Top Priority

Technology Roadmap for Forestry sector

Forestry		R&D Investment Roadmap		
Technologies	Goal by 2035	Short (~2026)	Mid (~2030)	Long (~2035)
Saw milling	50 % of saw milling produce with high quality		International cooperation	
			Domestic R&D	
Environmental management	7.5 million hectares reforested		Int. cooperation	
Precision machineries	30% of timber industry adopt precision machineries		International cooperation	
			Domestic R&D	
Surface treatment	70 % of timber industry adopt surface treatment for timber preservation		Domestic R&D	
			International cooperation	
Furniture manufacturing	Furniture manufacturing represent 30% of timber industry		Domestic R&D	
			International cooperation	

•TRM Creative Industries:

Top Priority

Introduction | Technology Foresight | Technology Assesment | Technology Roadmap

Technology Roadmap

	Goal by 2035	Short (~2026)	Mid (~2030)	Long (~2035)
Financial Services	60%	International Collaboration		
Marketing Services	100%	International Collaboration	Domestic R&D	International Collaboration
Media Services	70%	Domestic R&D	International Collaboration	
Digital Content	70%	Domestic R&D		International Collaboration
Copyright System	50%	Domestic R&D		International Collaboration

Top Priority

Mining and Materials		R&D Investment Roadmap		
Technologies	Goal by 2035	Short (~2026)	Mid (~2030)	Long (~2035)
Resource Exploration	50% of exploraionare made by local companies	Licensing	Domestic R&D	Intl. Collaboration
Material Processing	75% of mining processes are more clean	Training experts	Domestic R&D	Intl. Collaboration
Geotechnical Engineering	50% of geotechnical studies improve with modern equipments	Licensing	Domestic R&D	
Clean Energy Mining	75% of mining uses clean energy techniques	Domestic R&D		
Casting and Welding	50% productivity increase with modern infrastures	Training experts	Domestic R&D	Intl. Collaboration

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•TRM Advanced Manufacturing:

Top Priority Process in AI&Robot

Technology Roadmap – Adv. Manufacturing

	Goal (2035)	Short (~2026)	Mid (~2030)	Long (~2036)
Automation	50% of companies have SCADA systems	Licensing	Domestic R&D	
Clean Manufacturing	50% of companies have RECP plans	Int. Cooperation	Training experts	Domestic R&D
Digital Production	30% of manufacturers adopt digital design and modeling	Licensing	Domestic R&D	
Engineering Services	5-fold increase in the output of engineering services companies	Training experts	Domestic R&D	
Robotics	30% productivity increase with collaborative robots	Licensing	Domestic R&D	

Top Priority

Technology Roadmap Textile & Apparel

	Goal by 2035	Short (-2026)	Mid (-2030)	Long (-2035)
Textile Materials	90% of manufacturers include new technologies	Domestic R & D	Domestic R & D	Domestic R & D
Textile Manufacturing	80% of manufacturers adopt new technologies	Domestic R & D	Domestic R & D	Domestic R & D
Apparel Manufacturing	80% of manufacturers adopt advanced technology	Domestic R & D	Domestic R & D	Domestic R & D
Digital Design	80% of manufacturers adopt digital design tools	Domestic R & D	International Collaboration	Domestic R & D
Logistics	75% of manufacturers use sophisticated systems.	Domestic R & D / International Collaboration	Domestic R & D	Domestic R & D / International Collaboration

1. Agribusiness

- **Keywords**: *Artificial Intelligence, Big Data, Precision Agriculture, Machine Learning, Blockchain, Pest Control, Supply Chain*
- **Technology Roadmap**: Technologies like "Smart Farm" are expected to have a transformative impact, promoting precision agriculture and data-driven solutions. While traditional areas like "Food Processing" may decline, there is a focus on integrating AI and big data to optimize agricultural efficiency and sustainability.

2. Advanced Manufacturing

- **Keywords**: *Artificial Intelligence, 3D Printing, Industry 4.0, Cyber Physical Systems, Internet of Things (IoT), Augmented Reality, Virtual Reality*
- **Technology Roadmap**: "Robotics" and automation are highlighted for high growth, with advanced manufacturing acting as a supportive pillar to other sectors like mining and textiles. The adoption of IoT, 3D printing, and AI technologies will enable integration across different industries, enhancing productivity and innovation.

3. Mining and Materials

- **Keywords**: *Artificial Intelligence, 3D Printing, Manufacturing Technology*
- **Technology Roadmap**: Emphasis is placed on "Casting and Welding" and "Geotechnical Engineering" for future growth, with AI being central to improving mining operations. Technologies such as 3D printing are seen as tools to advance material processing and equipment manufacturing, paving the way for more efficient resource management.

4. Textiles and Apparel

- **Keywords**: *Smart Textiles, 3D Technology, Technology Development, Circular Economy, Sustainable Textiles*
- **Technology Roadmap**: Growth is anticipated in "Logistics" and "Textile Manufacturing," with a focus on integrating digital design and smart textile technologies. Sustainability is

a key trend, with the circular economy and sustainable fibers driving innovation, as traditional sectors like “Apparel Manufacturing” see a decline.

5. Creative Industry

- **Keywords**: *Artificial Intelligence, Virtual Reality, Augmented Reality, 3D Technology, Big Data*
- **Technology Roadmap**: The creative industry is set to expand through investments in "Media Services," driven by the adoption of AI, VR, and AR. These technologies will redefine content creation and offer new, immersive experiences. While "Financial Services" may experience short-term gains, the focus remains on digital and immersive technology development.

6. Forestry

- **Keywords**: *Artificial Intelligence, Forest Management, LiDAR, Sensor Technology, Ecosystem Conservation*
- **Technology Roadmap**: Technologies like AI and LiDAR are essential for advancing "Environmental Management" and "Precision Machineries," promoting sustainable practices in forestry. The focus is on using sensor technology for real-time monitoring, aiding conservation and efficient resource management.

The roadmap outlines the strategic goals for various industries in Peru, with a clear focus on leveraging research and development (R&D) for technological advancements. By 2035, sectors like agriculture, forestry, mining, and media aim to integrate smart technologies and international collaboration to enhance productivity, sustainability, and innovation. For instance, smart farming targets 30% adoption by 2035, while the forestry sector focuses on reforestation and precision machinery. This comprehensive plan highlights the importance of R&D investment and domestic efforts in fostering industrial growth and competitiveness. Technology roadmaps in key sectors like advanced manufacturing and textile and apparel, outlining goals for 2035 and identifying top priorities for short, mid, and long-term development.

For Advanced Manufacturing:

- **Automation:** 50% of companies are expected to implement SCADA systems by 2035, with steps like licensing, domestic R&D, and international collaboration.
- **Clean Manufacturing:** 50% of companies aim to adopt RECP plans through a mix of international cooperation and R&D.
- **Digital Production, Engineering Services, and Robotics:** These areas emphasize increased productivity, expert training, and advanced R&D efforts.

For Textile & Apparel:

- **Textile Materials & Manufacturing:** 80-90% of manufacturers are expected to adopt new technologies and advanced systems.
- **Digital Design & Logistics:** A focus on digital tools and systems will help enhance efficiency, supported by domestic and international collaboration.

This technology roadmap outlines key goals for advancing manufacturing and textile sectors by 2035. It highlights priorities such as automation, clean manufacturing, digital production, and the increased adoption of advanced systems in apparel manufacturing. The plan emphasizes a phased approach involving domestic research and development, international collaboration, and expert training, with a long-term goal of significantly improving productivity across these industries."

1. Expected Outcomes

The structure presented in the image outlines a strategic framework for various industries over three terms: short-term, mid-term, and long-term, each of which includes Key Performance Indicators (KPIs) and target metrics for technological development. The focus is on different sectors such as textile and apparel, agribusiness, mining technology, forestry technology, advanced technology, and creative industries. Each sector has specific objectives tied to national goals for economic growth and innovation.

Phase Vision (2024-2030) This outlines the long-term vision for each industry, aiming to drive innovation and sustainable development. The strategies include integrating AI and advanced technologies into various sectors. Analysis from Dr. Sehee Park

KPIs (Key Performance Indicators) These indicators measure the performance and progress of each sector in the short term (2025-2027), mid-term (2028-2032), and long-term (2033-2040). Each KPI is aligned with national objectives like competitiveness and technological advancement. Presented by Dr. Song Chi Woon

Action Strategy Each action strategy is aligned with the 10 principles of R&D programs, ensuring that each sector follows sustainable and innovative pathways to achieve its vision. Presented by Prof. JongIl Lee and Dr. Hyun Lim

To successfully apply technology foresight to public R&D planning, a strategic implementation approach is required. The first step is to establish a legal framework by 2025 that facilitates the integration of AI research with technology foresight activities. This will provide the necessary regulatory support to promote innovation while ensuring ethical and responsible use of AI technologies. Additionally, it is crucial to secure funding for robust IT infrastructure and develop comprehensive AI guidelines to standardize public R&D processes, enabling seamless collaboration and data sharing across sectors. Another key component involves strengthening international collaborations, positioning Peru to lead the Latin American Innovation group, which will foster cross-border knowledge exchange and enhance regional technological capabilities.

The expected outcomes of these initiatives are transformative. By integrating AI into technology foresight, Peru will achieve enhanced foresight and decision-making capabilities, enabling more accurate predictions of technological trends and better alignment of national R&D strategies. Furthermore, the adoption of generative AI applications will lead to increased efficiency in public R&D, streamlining processes, and expediting the development of new technologies. Ultimately, these efforts will solidify Peru's position as a global innovation hub, leading Latin America in standardization efforts and driving the adoption of best practices across industries.

2. Conclusion and Policy Recommendation

As Peru embarks on its journey towards becoming a technologically advanced and industrialized nation, strategic investments in R&D and innovation will be pivotal. The establishment of a dedicated Technology Foresight Center will act as a cornerstone in these efforts, providing vital support to various ministries in identifying emerging technologies and trends that will shape the next industrial revolution across key sectors such as agribusiness, mining, and advanced manufacturing. The following sections outline specific policy recommendations and strategic initiatives to ensure the successful realization of this vision.

The first and second Delphi survey provided insights into Peru's technological capabilities and investment priorities, serving as baseline data for future R&D programs and policy development. It emphasized the importance of human resources, infrastructure, and investment, highlighting the need for skilled professionals in the textile and agricultural sectors and infrastructure development in mining. Clean technology manufacturing was identified as a top investment priority, while the economic importance of technology was especially noted in mining and food processing *(Kim, 2001; Kim & Oh, 2000).*

The survey identified the United States and the European Union as leaders in technological advancement, particularly in mining and advanced manufacturing, with most technologies projected to emerge between 2028 and 2032. Respondents stressed the need for policy support in infrastructure development and workforce training, with active government investment being crucial for sectors like mining and agriculture *(KISTEP, 1999; Ministry of Information and Communication, 2001)*. International collaboration was seen as a key method for technology acquisition across most industries, while internal R&D was effective in textiles, apparel, and food processing *(Adelson & Aroni, 1975; Linstone & Turoff, 1975).*

Key barriers included infrastructure shortages and a lack of skilled professionals, particularly in agriculture and mining. Investment priorities were divided into short-term (clean tech, food processing, textiles), mid-term (materials processing, digital design, agricultural biotechnology), and long-term (smart agriculture, robotics, digital design). These findings will help guide future R&D efforts and policy initiatives, outlining a strategic roadmap for Peru's scientific and technological advancement.

Implement AI algorithms that can process vast amounts of data to identify patterns and predict technological advancements across industries. This will help in creating comprehensive technology maps and scenarios that guide national R&D priorities. Using AI-generated forecasts to advise policymakers on strategic areas of investment, ensuring that public R&D efforts are aligned with global technological shifts.

A robust IT infrastructure is the backbone of any technology foresight initiative. For the Technology Foresight Center to function effectively, there must be a comprehensive IT framework that facilitates data sharing, analysis, and collaboration among various governmental bodies. The following actions are recommended. Develop a central digital platform that links the six key ministries : *MINCETUR (Tourism and Trade), MIDAGRI (Agriculture), MINEM (Energy and Mining), MINAM (Environment), PRODUCE (Production), and CEPLAN (National Strategic Planning Center).* This platform should enable seamless data exchange, project management, and real-time communication to ensure efficient coordination across sectors. Allocate funding to upgrade existing IT systems within the ministries to support the new platform. This includes investing in high-performance computing systems, cloud storage, and secure data exchange protocols. Define clear standards for data formats, security, and privacy to ensure that information shared across the platform can be effectively utilized by all stakeholders. This will facilitate better integration and analysis of data, leading to more informed decision-making.

Peru can enhance its capacity to develop comprehensive technology roadmaps, streamline planning processes, and improve the efficiency of R&D initiatives. Key applications include that Generative AI can assist in creating detailed roadmaps that outline the development trajectories of specific technologies. By analyzing historical data, market trends, and research publications, AI can generate models that predict the future evolution of technologies, enabling better long-term planning. AI tools can evaluate multiple scenarios, assessing the potential risks and benefits associated with different technological investments. This will allow policymakers to make data-driven decisions that maximize the impact of public R&D funding. Strong IT infra and governance can automate parts of the research process, such as literature reviews, patent analysis, and data collection. This reduces the time required for planning and allows researchers to focus on more strategic activities.

Peru has the opportunity to establish itself as a leader in digital innovation within Latin America. By leveraging its Technology Foresight Center, the country can drive regional collaboration, set standards for emerging technologies, and promote best practices across the continent. The following initiatives are recommended. Peru should actively engage with other Latin American countries to form a digital innovation consortium. This group can work together on joint research projects, share knowledge, and advocate for the adoption of new technologies across the region. As a leader, Peru can take the initiative to develop frameworks and standards that ensure the responsible use of AI, IoT, and other digital technologies. This includes creating guidelines for data privacy, cybersecurity, and ethical AI practices that can be adopted by neighboring countries. Encourage joint ventures and collaborations between Peruvian research institutions and international partners. This will help attract foreign investment, promote knowledge exchange, and drive innovation in critical sectors like agriculture, energy, and manufacturing.

The successful implementation of these strategies will yield several significant outcomes that will benefit Peru's R&D ecosystem and overall economic development. By integrating AI into the Technology Foresight Center, Peru will be better equipped to anticipate technological trends, allowing for more informed and strategic R&D planning. Generative AI applications will streamline planning and evaluation processes, enabling faster project execution and reducing the resources needed for research activities. This will lead to more effective utilization of public funds. With a robust IT infrastructure and strategic collaborations, Peru will be well-positioned to lead the Latin American Digital Innovation group. This will not only drive technological advancement within the country but also set Peru apart as a pioneer in digital innovation, attracting international investments and partnerships. Establishing Peru as a leader in technology foresight and digital innovation will promote collaboration across Latin America, leading to a more integrated and competitive regional economy.

In conclusion, Peru's vision of becoming a technologically advanced nation by 2050 can be achieved through strategic investments in AI and digital infrastructure. By expanding the Technology Foresight Center to include AI research capabilities, developing a robust IT framework, and promoting regional leadership, Peru can effectively position itself as a key player in the next wave of technological innovation. These initiatives will not only boost the country's competitiveness on the global stage but also foster a culture of innovation that will drive long-term sustainable growth. The proposed policy recommendations provide a roadmap for Peru to navigate the complex technological landscape, ensuring that the nation remains at the forefront of digital transformation and innovation.

[Table 5-2] R&D Profcess with Six Industries

Industry/ R&D Process	Planning	RFPs	R&D	Feedbacks	Evaluation
Textile and Apparel	Define R&D goals for textile innovation, focusing on sustainable production methods and materials.	Government releases RFPs for textile manufacturing technologies and logistics optimization.	Collaboration between agencies, players, and experts to pilot new textile manufacturing techniques and logistics solutions.	Refine textile production and logistics strategies based on market performance and industry feedback.	Measure success through FDI growth, technology adoption rates, and sustainability metrics.
Agribusiness	Identify critical areas for agricultural innovation, incluing sustainable farming techniques and food processing.	Government and agencies issue RFPs for sustainable farming technologies and agribusiness innovations.	Develop and test new farming techniques and agro-biological innovations through collaboration between industry players and experts.	Refine farming techniques and agribusiness strategies based on pilot results and stakeholder input.	Evaluate productivity improvements, sustainability impact, and industry adoption rates.
Mining Technology	Develop goals for clean energy mining and resource exploration technologies, with a facus on sustainability.	RFPs issued for clean energy mining technologies, resource exploration, and environmental impact solutions.	Collaborative efforts between government, agencies, and private mining firms to develop and pilot new mining technologies.	Adjust mining R&D goals based on environmental and performance outcomes from initial projects.	Assess success through sustainability outcomes, international collaborations, and mining efficiency improvements.

Industry/ R&D Process	Planning	RFPs	R&D	Feedbacks	Evaluation
Forestry Technology	Define innovation goals for sustainable forestry management and ecosystem conservation technologies.	RFPs issued by government agencies for precision forestry and sustainable forest management technologies.	Focus on developing and implementing sustainable forestry practices, including precision machinery and environmental management systems.	Adjust forest management strategies based on conservation outcomes and feedback from key stakeholders.	Evaluate improvements in conservation practices, ecosystem management, and sustainability measures.
Advanced Technology	Establish R&D objectives for automation, robotics, and advanced manufacturing solutions, focusing on efficiency and clean manufacturing.	Government issues RFPs for robotics and automation solutins in key manufacturing sectors.	Collaboration between agencies, manufacutrers, and experts to develop and implement automation technologies.	Refine automation and robotics strategies based on industry performance and technology adoption.	Evaluate manufacturing efficiency improvements, cost savings, and sector growth driven by automation and robotics.
Creative Industries	Idenfity areas for innovation in content creation, marketing technologies, and global digital distribution.	RFPs issued for digital content creation tools, marketing systems, and creative innovation technologies.	Collaboration between industry players and agencies to develop new creative content production technologies and marketing automation solutions.	Refine creative content and marketing strategies based on feedback from global markets and industry trends.	Assess success through global collaboration rates, content production efficiency, and marketing system performance.

[Table 5-3] Short Term, Mid Term and Long Term KPIs with Six Industries

Industry	Phase Vision (2024-2030)	Short-Term KPIs (2025-2027)	Mid-Term KPIs (2028-2032)	Long-Term KPIs (2033-2040)	Action Strategy (Aligned with the 10 Principles of R&D Programs)
Textile and Apparel (MINCETUR)	**Design Phase** • Design of Technologi Foresight Center • Legal infrastructure for TTFH **Implementation Phase** • Provision of Technologi Foresight services • Implementation of Technology Foresight Report (5-year)	• Number of new textile technologies identified and piloted. • Increase in industry collaboration. **Investment Priority:** Textile manufacturing > Apparel manufacuturing	• Growth in FDI for textile sector. • Establishment of innovation hubs for textile production **Investment Priority:** Textile materials > Apparel manufacuturing	• Global leadership in sustainable and efficient textile production. **Investment Priority:** Textile design > Advanced materials	**Action Strategy** **Operational Strategy**
Agribusiness (MIDAGRI)		• Launch of pilot projects for sustainable farming. • Development of new agricultural technologies. **Investment Priority:** Food processing > Agro-biological innovation	• Expansion of sustainable farming practices. • Growth in agritech investments. **Investment Priority:** Food processing > Smart farming	• Global leadership in innovative agribusiness practices. **Investment Priority:** Food processing > Agro-biology	

Industry	Phase Vision (2024-2030)	Short-Term KPIs (2025-2027)	Mid-Term KPIs (2028-2032)	Long-Term KPIs (2033-2040)	Action Strategy (Aligned with the 10 Principles of R&D Programs)
Mining Technology (MINEM)	**Design Phase** • Design of Technologi Foresight Center • Legal infrastructure for TTFH **Implementation Phase** • Provision of Technologi Foresight services • Implementation of Technology Foresight Report (5-year)	• Identification of sustainable mining technologies. • Resource exploration improvements. **Investment Priority:** Clean energy mining > Resource exploration	• Increased use of clean energy mining technologies. • Stronger international partnerships in the mining sector. **Investment Priority:** Sustainable mining technologies	• Global leadership in clean energy mining and resource management. **Investment Priority:** Clean energy > Resource exploration	**Action Strategy** **Operational Strategy**
Forestry Technology (MINAM)		• Introduction of sustainable forestry practices. • Improved forest management strategies. **Investment Priority:** Environmental management > Precision forestry	• Expansion of sustainable forest management techniques. • Growth in partnership for forest conservation technologies. **Investment Priority:** Sustainable forest technologies	• Long-term ecosystem management and conservation leadership. **Investment Priority:** Environmental management > Sustainable technologies	

Industry	Phase Vision (2024-2030)	Short-Term KPIs (2025-2027)	Mid-Term KPIs (2028-2032)	Long-Term KPIs (2033-2040)	Action Strategy (Aligned with the 10 Principles of R&D Programs)
Advanced Technology (PRODUCE)	**Design Phase** • Design of Technologi Foresight Center • Legal infrastructure for TTFH **Implementation Phase** • Provision of Technologi Foresight services • Implementation of Technology Foresight Report (5-year)	• Number of automation and robotics projects identified and piloted. • Growth in automation investments. **Investment Priority:** Automation > Clean manufacuturing	• Widespread adoption of automation and robotics technologies in key sectors. **Investment Priority:** Robotics > Advanced manufacuturing	• Leadership in automation and robotics for manufacturing. **Investment Priority:** Clean manufacturing > Automation	**Action Strategy** **Operational Strategy**
Creative Industries (CULTURA)		• Number of digital content and creative marketing projects identified and launched. **Investment Priority:** Digital contents > Marketing systems	• Expansion of global collaborations for creative content production. **Investment Priority:** Marketing technologies > Digital content	• Peru as a global hub for creative industries. **Investment Priority:** Digital content > Creative marketing	

[Table 5-4] KPIs for Technology Foresight Center Operation

Industry / Program	Phase Vision (2024-2030)	Short-Term KPIs (2025-2027)	Mid-Term KPIs (2028-2032)	Long-Term KPIs (2033-2040)	Action Strategy (Aligned with the 10 Principles of R&D Programs)
Textile and Apparel (MINCETUR)	**Design Phase (2024-2026):** Establish a foresight strategy for innovation in textile and apparel manufacturing. **Implementation Phase (2027-2030):** Implement industry-wide improvements in production technologies.		**(2024-2025)** **To Be Planned By Product and Servie By Technology**		**1. Goal**: Improve production efficiency and global competitiveness. **2. Legal infrastructure**: Develop regulations for innovation in textile production. **4. Agencies**: MINCETUR to coordinate implementation. **9. Dissemination**: Promote foresight findings to gloval stakeholders in textile manufacturing.
Agribusiness (MIDAGRI)	**Design Phase (2024-2026):** Establish a foresight strategy for agribusiness technologies. **Implementation Phase (2027-2030):** Promote the adoption of innovative technologies.				**1. Goal**: Improve productivity and sustainability. **6. Demand**: Identify market-driven needs for innovation in agribusiness. **4. Agencies**: MINCETUR to lead planning. **9. Dissemination**: Share foresight results with the global agritech community.

Industry / Program	Phase Vision (2024-2030)	Short-Term KPIs (2025-2027)	Mid-Term KPIs (2028-2032)	Long-Term KPIs (2033-2040)	Action Strategy (Aligned with the 10 Principles of R&D Programs)
Mining Technology (MINEM)	**Design Phase (2024-2026):** Develop a foresight strategy for resource exploration and sustainable mining technologies. **Implementation Phase (2027-2030):** Implement innovations in clean energy mining.	**(2024-2025)** **To Be Planned** **By Product and Servie** **By Technology**			**1. Goal**: Develop sustainable and efficient mining practices. **7. Intermediaries**: Engage with mining companies and international partners. **2. Legal infrastructure**: Establish environmental regulations for sustainable mining.. **8. Evaluation**: Regular performance assessments of ming innovations.
Forestry Technology (MINAM)	**Design Phase (2024-2026):** Develop foresight strategies for sustainable forest management. **Implementation Phase (2027-2030):** Promote the use of technologies for forestry conservation.				**1. Goal**: Improve long-term sustainability in forestry practices. **3. Duration**: Establish long-term forestry conservation programs. **9. Dissemination**: Share findings on sustainable forestry management globally. **6. Demand**: Identify key stakeholders for sustainable forestry practices.

Industry / Program	Phase Vision (2024-2030)	Short-Term KPIs (2025-2027)	Mid-Term KPIs (2028-2032)	Long-Term KPIs (2033-2040)	Action Strategy (Aligned with the 10 Principles of R&D Programs)
Advanced Technology (PRODUCE)	**Design Phase (2024-2026):** Develop foresight strategies for automation and robotics in manufacturing. **Implementation Phase (2027-2030):** Scale new technologies in advanced manufacturing.		**(2024-2025)** **To Be Planned** **By Product and Servie** **By Technology**		**1. Goal**: Improve efficiency and competitiveness in manufacturing. **4. Agencies**: Produce to coordinate technology foresight implementation. **8. Evaluation**: Monitor performance of automation technologies. **5. Finalcial resources**: Secure funding for new manufacturing technologies.
Creative Industries (CULTURA)	**Design Phase (2024-2026):** Establish foresight strategy for innovation in creative content production and marketing. **Implementation Phase (2027-2030):** Promote the use of technology in creative industries.				**1. Goal**: Improve global competitiveness in creative content production. **4. Agencies**: CULTURA to lead the foresight process. **9. Dissemination**: Share creative innovation findings globally. **5. Finalcial resources**:Secure funding for creative innovation projects.

[Table 5-5] Outline of Vision & KPIs for Technology Foresight Center Operational Model

<table>
<tr><th>Long-term Goal: Establish a competitive edge in global markets
strategies</th><th>1st Phase (1-3 years): Initial Stage</th><th>2nd Phase (4-5 years): Middle Stage</th><th>3rd Phase (6-7 years): Final Stage</th></tr>
<tr><td></td><td colspan="3">To Be Planned</td></tr>
<tr><td>Law/Policy Aspects
To Be Planned</td><td colspan="2" rowspan="4"><table><tr><th>KPI</th><th>Description</th></tr><tr><td colspan="2">To Be Planned</td></tr></table></td><td rowspan="4">To Be Planned
1. Governance Model
2. Budget Allocation
3. Human Resource Development

R&D 10 Principle
Action Strategy
TFC Operational Strategy Assessment</td></tr>
<tr><td>Organization Aspects
To Be Planned</td></tr>
<tr><td>Networks Aspects
To Be Planned</td></tr>
<tr><td>Supporting Agency Aspects
To Be Planned</td></tr>
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CHAPTER 6

Peru's R&D Landscape and Strategic Planning

S&T Planning and Technological Forecasting for Prioritized Sectors in Perú

Chapter 6. Peru’s R&D Landscape and Strategic Planning

1. Peru’s R&D Status and Governance[7]

1.1 Overview of STI Governance

Peru’s National System for Science, Technology, and Innovation (SINACTI) has been restructured to address issues of fragmentation and improve coordination among policies and funding for STI. It operates across three levels:

- Strategic Definition: Led by CONCYTEC and key stakeholders for policy design
- Implementation: Ensures resources for executing national STI policy
- Execution: Engages universities, businesses, and research centers in innovation

[7] <Chapter 6. Peru’s R&D Landscape and Strategic Planning> is written by Agnes Franco and Victor Izaguirre, Directors; and Felipe Valentín and John Collantes, Senior Specialists, all at CONCYTEC.

1.2 Policy Framework

Peru's policy framework for Science, Technology, and Innovation (STI) is structured to promote a cohesive system that integrates various stakeholders across the public and private sectors, guided by key legislative measures. The General Law of Science, Technology, and Technological Innovation (Law No. 31250) forms the backbone of this framework by establishing SINACTI (National System of Science, Technology, and Technological Innovation) under the leadership of CONCYTEC (Gobierno del Perú, 2024). This law outlines CONCYTEC's role in coordinating STI activities, defining it as a national authority responsible for formulating, directing, and supervising policies that drive scientific and technological development across sectors. Through this system, Peru seeks to strengthen STI capacity and align its scientific endeavors with economic and social needs.

A central element of Law No. 31250 is its focus on fostering collaboration among research institutions, universities, and industry, with a clear emphasis on prioritizing sectors that advance national competitiveness and productivity. The law encourages the integration of public research centers and higher education institutions within SINACTI to stimulate innovation and support strategic technological advancements. This collaboration is intended to create a productive research environment that aligns with global standards, while simultaneously addressing Peru's specific development challenges, particularly in areas like sustainable development, environmental management, and technology adaptation.

The STI policy framework is further enriched by regulations aimed at enhancing the legal and financial infrastructure for innovation. Law No. 31250 highlights the need for strategic funding mechanisms to support both basic and applied research, advocating for public-private partnerships as a means to sustain R&D investments. The law provides guidelines for leveraging external sources, such as international funds and Official Development Assistance (ODA), to expand the financial base available for scientific research and technological projects. Moreover, the law encourages the creation of innovation-friendly environments, such as technoparks and innovation clusters, which provide spaces for knowledge exchange, entrepreneurship, and commercialization of research outcomes.

Additionally, the framework underscores the importance of universities as essential contributors to the national STI agenda, with the University Law (Law No. 30220) specifying their role in training skilled professionals and conducting research. This law mandates universities to integrate STI priorities into their curricula, fostering a culture of research and critical thinking that contributes to the broader goals of SINACTI. Together, these policies create a structured yet flexible environment that promotes interdisciplinary collaboration, the efficient allocation of resources, and the strategic prioritization of technologies and research fields deemed essential for Peru's development.

In essence, Peru's STI framework, as laid out in these legislative foundations, strives to build a sustainable and responsive innovation ecosystem. By encouraging collaboration across sectors, ensuring effective governance, and fostering a robust legal and financial base, the framework aims to position Peru as an active participant in global scientific and technological advancement. This cohesive approach not only supports current development needs but also prepares the country to address future technological and economic challenges through strategic, knowledge-driven growth.

1.3 Funding Mechanisms

The ProCiencia program, the implementing arm of Peru's National Council for Science, Technology, and Technological Innovation (CONCYTEC), alongside the National Program for Technological Development and Innovation (ProInnóvate) under the Ministry of Production, has launched two major initiatives: the STI Ecosystem Improvement Project with a fund of $125 million, and the Innovation, Technological Modernization, and Entrepreneurship Program with a $140 million budget (Gobierno del Perú, 2023).

The Strengthening Peru's National Science, Technology, and Innovation System Project, led by ProCiencia, aims to enhance service quality to boost competitiveness and sustainable development, with a regional focus. This project, spanning from 2022 to 2027, is funded through a $100 million World Bank loan, supplemented by an additional $25 million from the public treasury. It focuses on three strategic areas: low-carbon and climate-resilient economy, health, and digital economy/ICT (Gobierno del Perú, 2023).

In parallel, the Innovation, Technological Modernization, and Entrepreneurship Program, managed by ProInnóvate, offers incentives for private innovation investment, early-stage funding for venture capital, technological modernization for MSMEs, and institutional capacity-building for the agency. This program is supported by $100 million from the Inter-American Development Bank (IDB) and $40 million from the public treasury. ProInnóvate will oversee these funds until 2026, aiming to foster growth and development among innovative companies and startups in Peru (Gobierno del Perú, 2023).

1.4 Proposal for the STI Planning Process

The budget planning process for R&D in Peru has historically been conducted independently by each institution, with submissions directly to the Ministry of Economy and Finance (MEF) without specific coordination with the Governing Body of Science, Technology, and Innovation. This fragmented approach limited the alignment of R&D plans and budgets with national priorities. However, with the enactment of Law 31250, which establishes the National System of Science, Technology, and Innovation (SINACTI), a coordinated budget planning framework is being implemented. This framework involves various institutions regulated under the law and aims to create a unified process to enhance the effectiveness and strategic alignment of R&D investments.

Under this new framework, the MEF plays a central role by setting budgetary limits and overseeing the allocation of resources across central, regional, and local government entities. As the top-level institution, the MEF ensures that funds for R&D are distributed efficiently and in accordance with national financial policies. Supporting this, the Multisectoral Commission on Science, Technology, and Innovation (CMCTI), established by Supreme Decree 025-2021-PCM, is tasked with monitoring adherence to the National Policy on Science, Technology, and Innovation (POLCTI). The CMCTI provides technical reports to guide policy updates and advances a strategic vision for STI. Complementing this, the Advisory Commission on Science, Technology, and Innovation (CCCTI) advises both the

CMCTI and CONCYTEC by identifying policy options, initiatives, and interventions that drive STI development.

CONCYTEC, as the specialized technical body for STI, holds a critical role in promoting and supervising science, technology, and innovation activities nationwide. With scientific, administrative, economic, and financial autonomy, CONCYTEC acts as a budgetary entity and ensures that R&D activities align with Peru's broader development goals. Additionally, ministries within the central government are responsible for allocating funds from their budgets to support R&D projects. These ministries, through their financial contributions, serve as key enablers in initiating the annual R&D budget planning process.

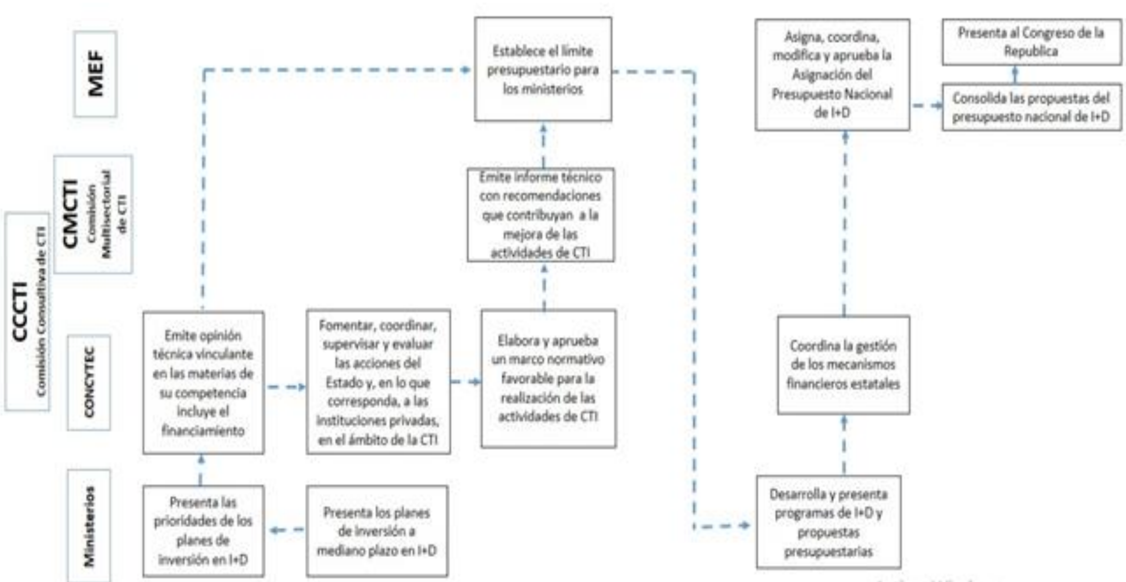
The coordination process for R&D funding begins with the ministries submitting their medium-term investment plans and priorities, highlighting areas crucial for STI development. These plans are analyzed by CONCYTEC, which provides technical opinions and specific recommendations for investment, in accordance with its role as the governing body under Law 31250. Subsequently, the CCCTI and CONCYTEC collaborate to establish a regulatory framework that defines the allocation of resources and the development of R&D projects in both public and private sectors. This collaboration ensures a coherent approach to funding and project planning.

Once reviewed, the CMCTI evaluates these plans and issues a comprehensive technical report with recommendations that align STI activities with the goals outlined in POLCTI. The MEF then uses these reports to establish budgetary caps for the ministries, ensuring resource allocation reflects strategic national priorities. This consolidated process ensures that R&D proposals and budgets from various ministries are integrated into a unified national budget for STI.

The final consolidated budget is presented to Congress for evaluation and approval. This step ensures transparency and accountability in resource allocation for R&D activities. Throughout this process, the CMCTI continuously monitors the implementation of POLCTI, issuing technical recommendations to adjust policies in response to emerging trends and challenges in the STI landscape. Simultaneously, the CCCTI provides strategic advice to the

CMCTI and CONCYTEC, ensuring that policy decisions and investment projects remain aligned with Peru's long-term development objectives.

This coordinated budget planning framework creates a robust foundation for STI in Peru, enabling the country to move toward a more strategic and sustainable vision for science, technology, and innovation. By integrating technical guidance, regulatory oversight, and institutional collaboration, this process ensures that R&D investments effectively contribute to national priorities and long-term development goals.



2. Peru's R&D Expenditure & Resource

2.1 Analysis of National R&D Expenditure

In 2023, Peru allocated a total of 1.23 billion soles to science, technology, and innovation (STI) through various programs and institutions, reflecting a significant commitment to fostering research and technological development across sectors. Of this total, CONCYTEC received 180 million soles, with 25.5 million directly allocated to its operations and 155 million directed to Prociencia. Funding sources for these allocations included 126.6 million in ordinary resources, 50.7 million from official credit operations (8.6 million from external debt and 42.1 million from the International Bank for Reconstruction and Development), and 3.2 million from donations and transfers (Ministerio de Economía y Finanzas del Perú, 2024).

In addition to CONCYTEC, other key recipients of STI funding included ProInnóvate, which received 140.9 million soles, and the 13 Public Research Institutes (IPIs), which together were allocated 263 million soles. Public universities received 106 million soles, while regional and local governments were allocated 150 million and 309 million soles, respectively. Furthermore, 81 million soles were dedicated to other STI-related activities, such as scientific campaigns in Antarctica, hospital-based research, and specialized health institutions, showcasing the diversity of investment areas.

Notably, 82% of the total funding for STI in 2023 came from two primary sources: ordinary resources, which accounted for 43.6%, and determined resources, contributing 38.3%. This strategic distribution underscores Peru's efforts to support research and innovation through a combination of national and external funding sources, ensuring broad-based development across multiple sectors and regions.

[Table 6-1] Breakdown of Funding Sources for STI in Peru (2023)

Source of Financing	Amount Accrued (Soles)	Percentage
ORDINARY RESOURCES	536,040,301	43.6%
DIRECTLY RAISED RESOURCES	11,900,854	1.0%
RESOURCES FROM OFFICIAL CREDIT OPERATIONS	118,914,717	9.7%
DONATIONS AND TRANSFERS	92,093,510	7.5%
DETERMINED RESOURCES	471,071,777	38.3%
TOTAL	1,230,021,159	100.0%

Source: Ministerio de Economía y Finanzas del Perú (MEF)

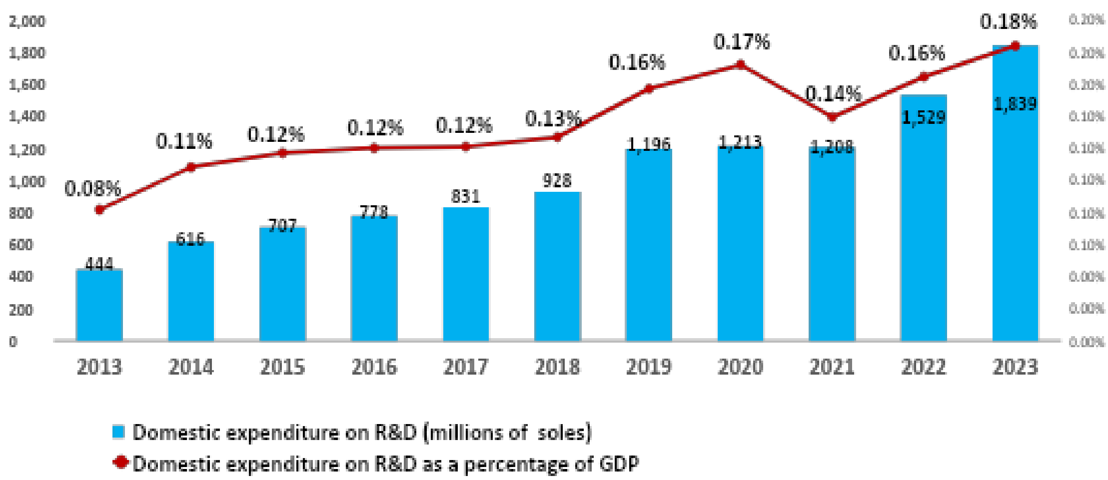
In 2013, R&D spending was 444 million soles, equivalent to 0.08% of GDP. Since then, this value has increased steadily, reaching 1,839 million soles in 2023, representing 0.18% of GDP. Between 2018 and 2019, significant growth was observed, going from 928 to 1,196 million soles. This increase in percentage terms was gradual, with peaks in 2019, with 0.16% of GDP, and in 2020, with 0.17%. Although R&D spending continued to grow in the following years, the pace was more moderate between 2019 and 2021, with a slight decrease in 2021, when spending stood at 1,208 million soles. However, between 2022 and 2023, R&D spending again showed a considerable increase, going from 1,529 to 1,839 million soles. In percentage terms, there was a slight decrease in 2021 (0.14%), but it recovered and exceeded previous values in 2023, reaching 0.18% of GDP.

In absolute terms, R&D expenditure in Peru has shown a sustained growth trend, with a notable increase in the last two years. This suggests a growing commitment by the government and other entities to promoting research and development. In relation to GDP, the growth of the percentage dedicated to R&D is slow but constant, a relevant figure for a developing country. However, the 0.18% recorded in 2023 is still low compared to developed countries, where this percentage usually exceeds 1%.

Despite recent growth in R&D investment, the percentage of Peru's GDP allocated to this sector remains insufficient to create a transformative impact on the economy and technological innovation. Fluctuations in the allocation, such as the decline in 2021, suggest challenges in resource distribution and program implementation, which may stem from economic constraints or shifts in public policies. These inconsistencies highlight the need for a more stable and strategic approach to funding science, technology, and innovation (STI) to ensure sustainable progress.

To address this, the National Policy on Science, Technology, and Innovation (POLCTI) has set a goal of increasing R&D investment to at least 1% of GDP by 2030, aligning with international recommendations. Achieving this target would enhance Peru’s competitiveness and productivity, while enabling the country to tackle critical social and economic challenges through sector-specific innovations. Although the current trajectory shows gradual progress, the low GDP percentage dedicated to R&D underscores a dual reality: an urgent need to strengthen funding policies and a unique opportunity to establish STI as a cornerstone of Peru's long-term development strategy.

[Figure 6-1] Peru Domestic expenditure on R&D and Domestic expenditure on R&D as a percentage of GDP (millions of soles)



Source: MEF, INEI

2.2 Allocation of R&D Resources

Investment in Science, Technology and Innovation (STI) in Peru has shown a marked geographical heterogeneity for a long time, concentrating mainly in the Lima region. In 2023, Lima received 457.3 million soles, representing 37.2% of the total. In contrast, the other regions of the country together received 772.7 million soles, equivalent to 62.8% of the total investment in STI.

[Figure 6-2] Regional Distribution of STI Expenditure in Peru

Source: MEF - Friendly Consultation

In 2023, Lima received an investment of 457.3 million soles, representing 37.2% of the total allocated to Science, Technology and Innovation (STI). The Cusco region follows with 270.2 million soles, equivalent to 22% of the total. The regions of Arequipa and Puno also stand out, receiving 57.3 and 58.2 million soles respectively, representing 4.7% of the total investment in STI for each. This distribution reflects a notable concentration of resources in Lima, although other regions are also beginning to receive greater attention and financial support.

On the other hand, the regions that have received the least budget in 2023 are Ica, Lambayeque, Tumbes, Huancavelica, Moquegua, Madre de Dios, Piura and Ancash. Each of these regions has an investment of less than 1% of the total allocated to Science, Technology and Innovation (STI), reflecting an unequal distribution that leaves these areas with limited resources for development in these crucial fields.

[Table 6-2] Regional Allocation of STI Expenditure in Peru

Nº	Departamento	Total general	Porcentaje(%)
1	AMAZONAS	17,261,087.17	1.4
2	ANCASH	10,593,788.62	0.9
3	APURIMAC	37,605,152.32	3.1
4	AREQUIPA	57,282,200.50	4.7
5	AYACUCHO	18,508,984.74	1.5
6	CAJAMARCA	32,935,699.15	2.7
7	CUSCO	270,188,024.44	22.0
8	HUANCAVELICA	7,419,912.75	0.6
9	HUANUCO	19,575,880.49	1.6
10	ICA	6,572,869.70	0.5
11	JUNIN	27,848,756.87	2.3
12	LA LIBERTAD	17,460,371.34	1.4
13	LAMBAYEQUE	6,676,297.96	0.5
14	LIMA	456,850,961.25	37.2
15	LORETO	39,841,387.54	3.2
16	MADRE DE DIOS	7,301,164.45	0.6
17	MOQUEGUA	8,898,891.65	0.7
18	PASCO	12,606,223.58	1.0
19	PIURA	9,229,345.61	0.8
20	CALLAO	42,850,321.41	3.5
21	PUNO	58,152,615.52	4.7
22	SAN MARTIN	15,518,930.28	1.3
23	TACNA	24,304,357.42	2.0
24	TUMBES	5,852,962.23	0.5
25	UCAYALI	18,085,456.20	1.5
TOTAL		1,229,421,643.19	100.0

Source: MEF Friendly Consultation

3. Peru's STI and STI Projects and Grants

3.1 Current STI Projects in Key Sectors

In 2013, FONDECYT commenced its operations as an Executing Unit following the approval of its Operational Manual, marking a significant step in supporting research and innovation in Peru. By 2021, PROCIENCIA began functioning as a key promoter of Science, Technology, and Innovation (STI) under the leadership of CONCYTEC. Over its 11 years of activity as an Executing Unit, FONDECYT funded a total of 3,536 projects, showcasing its impact on advancing STI in the country. Notably, 2015 was a standout year with 704 projects financed, followed by 2019 and 2018, with 473 and 447 projects funded, respectively, reflecting the program's growing momentum (PROCIENCIA, 2024).

The importance of regional grant distribution gained prominence in 2018 with the launch of the Project for the Improvement and Expansion of Services of the National System of Science, Technology, and Technological Innovation (SINACYT). Funded by a World Bank loan, this initiative aimed to decentralize research funding through the innovative "seed" modality. This approach specifically targeted entities outside the Lima Metropolitan Area and Callao, introducing simplified requirements for principal investigators to promote equitable access. As a result, 31 projects were successfully funded in regions beyond the capital, marking a milestone in fostering regional research capacity and addressing disparities in STI development.

■ Ordinary Resources

Ordinary resources come from tax collection and other sources, minus banking fees and services. These freely allocated funds are assigned through the Budget Law to central, regional, and local government agencies. During the 11 years of operation, ordinary resources allocated to PROCIENCIA accounted for, on average, 80% of total funding, covering 45% of the awarded projects, mainly in key instruments such as mobility programs and basic, applied, and technological development research projects.

■ Framework Fund for Innovation, Science, and Technology (FOMITEC)

Established in 2013 under Law No. 29951, FOMITEC was designed to promote STI development through targeted economic and financial instruments. Among its flagship programs were STARTUP PERU, led by the Ministry of Production, and CIENCIA ACTIVA, managed by CONCYTEC, which focused on funding human capital development, strategic research, and the scaling of technological products. Between 2014 and 2017, FOMITEC financed 222 projects, accounting for 11% of CONCYTEC's total STI investment. Additionally, it provided supplemental credit of S/ 300 million to support grants across several key initiatives: Centers of Excellence fostering national and international R&D collaboration, Research Circles strengthening multidisciplinary regional teams, Bold Ideas supporting innovative projects in health, agriculture, and the environment, and Advanced Human Capital Development promoting Ph.D. programs in priority areas both nationally and internationally.

■ SINACYT Project (2017-2022)

With a total investment of $100 million—$55 million from the Peruvian government and $45 million from the World Bank—this project aimed to enhance the National STI System, diversify the economy, and boost competitiveness. The initiative focused on six strategic and seven general priority areas, structured around three key components: strengthening SINACYT through regulatory development and the consolidation of STI data, promoting value chains in sectors such as wood, mining, and tourism to identify priorities and build capacity, and advancing SINACYT’s development by funding scholarships, STI projects, and scientific equipment to foster innovation and research excellence.

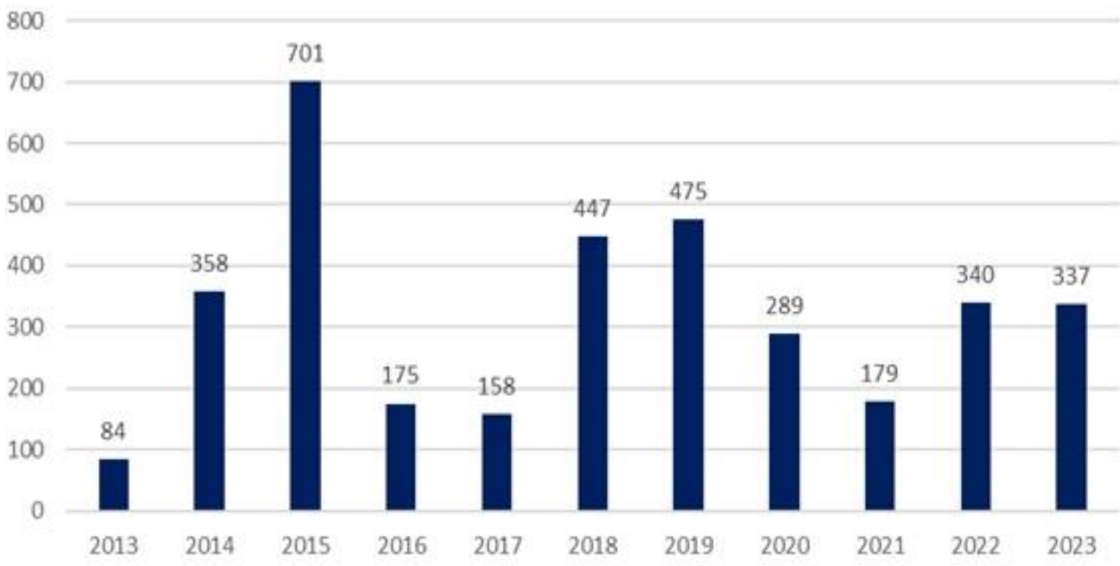
■ SINACTI II Project (2022-2027)

This new project, backed by a $125 million fund ($100 million from the World Bank and $25 million from the Peruvian government), focuses on advancing a low-carbon economy, improving health outcomes, and driving growth in the digital economy. It is structured around three key components: strengthening governance by enhancing STI institutions and services, generating knowledge in prioritized sectors through funding for Ph.D. programs, scholarships, and strategic projects, and fostering academia-industry linkages by developing technologies and reinforcing value chains in areas like mango, grape, and cotton textiles. Together, these initiatives represent approximately 14% of the allocated grants, underscoring their significant contribution to consolidating Peru's STI ecosystem.

3.2 Analysis of Grants and Funding Mechanisms

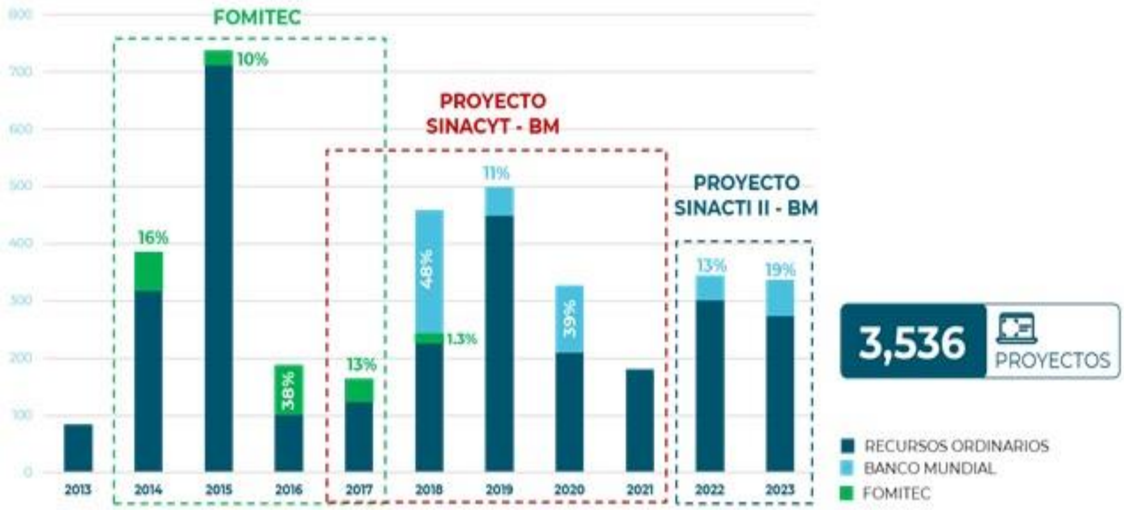
Between 2013 and 2023, a total of 3,536 grants were awarded, with the annual distribution detailed as in the Figure 6-3.

[Figure 6-3] Number of Grants by Year



The Figure 6-4 illustrates the number of grants funded by type of financial resource. While ordinary resources finance the majority (80%) of the grants, it is observed that when loan projects with the World Bank are implemented, the number of grants shows a more sustainable trend over time.

[Figure 6-4] Grants Funded by Type of Financial Resource



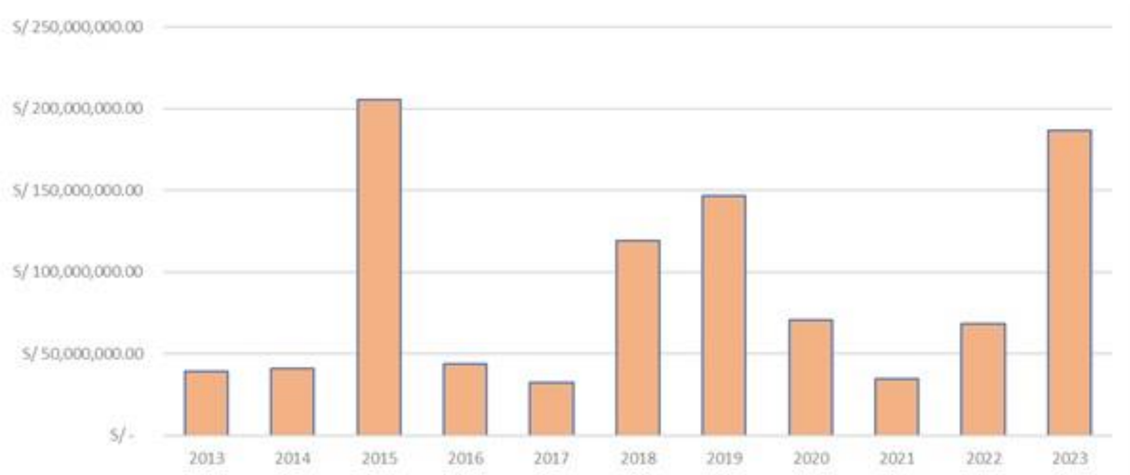
The data reveals a significant peak in grant awards around 2015, driven by increased resource availability, followed by a sharp decline in 2016 due to reduced funding and changes in distribution policies. In the years that followed, grant numbers exhibited moderate fluctuations, with another notable increase in 2018. These patterns highlight a strong correlation between the availability of external financing and the number of grants awarded, with periods of expansion often followed by contractions caused by financial adjustments or budgetary constraints.

■ Investment Analysis by Year

Between 2013 and 2023, Peru's central government invested a total of S/ 1,050,311,759.66 in Science, Technology, and Innovation (STI) nationwide through PROCIENCIA. The funds originated from ordinary resources (allocated by the Ministry of Economy and Finance, additional demand allocations, national and international agreements), FOMITEC, and public debt (World Bank contracts).

The years 2015 and 2023 stand out for their highest levels of investment, driven by substantial allocations: ordinary resources supported doctoral scholarships in 2015, while funding from loans contributed significantly in 2023. Over the period, the average annual investment across all funding sources amounted to S/ 95.5 million. However, PROCIENCIA's direct contribution to STI activities through grants funded by ordinary resources typically averaged around S/ 42.7 million per year, reflecting its consistent yet modest role in supporting innovation initiatives.

[Figure 6-5] Prociencia investment in STI



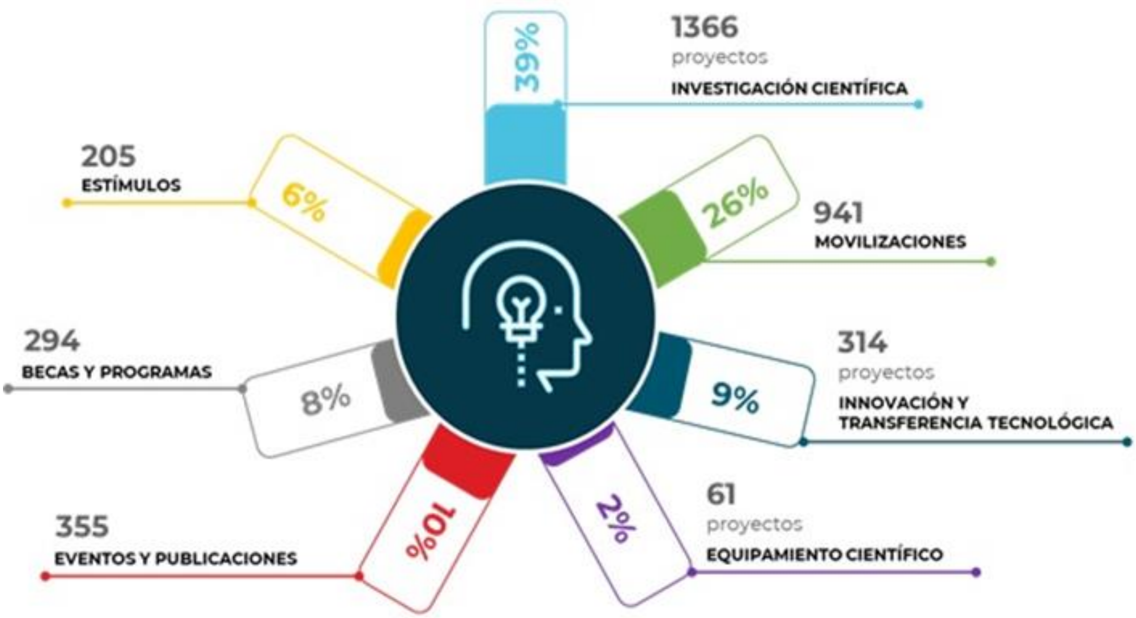
■ Types of Grants

The grants provided by CONCYTEC/PROCIENCIA are structured through financial instruments that define clear intervention objectives, which are further refined into financial schemes to diversify target audiences, expected outcomes, and eligible expenses.

These instruments are specifically designed to address the public good nature of scientific research by fostering knowledge generation, driving innovation through the development of new technologies, strengthening human resources and research infrastructure, and promoting the dissemination of STI. At their core, these financial instruments aim to establish a grant system that enhances productivity, boosts competitiveness, and contributes to economic development in both the medium and long term.

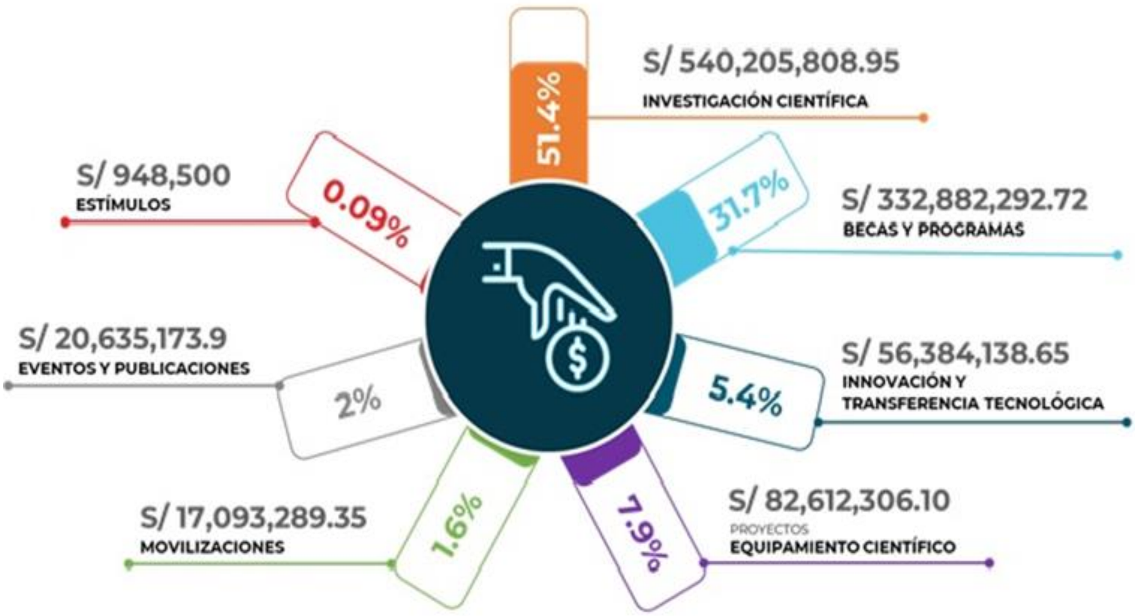
CONCYTEC/PROCIENCIA administers seven financial instruments to award grants across key areas of STI development. These include Human Capital Development, supported by scholarships, mobility programs, and incentive instruments; Knowledge Generation through research instruments; Strengthening Research Infrastructure via equipment funding; Innovation through technology transfer and innovation instruments; and STI Promotion and Dissemination through event and publication support. Collectively, these efforts have resulted in the funding of 294 scholarships and programs, 61 equipment acquisitions, 205 incentives, 355 events and publications, 314 innovation and technology transfer projects, 941 mobilities, and 1,366 scientific research projects, underscoring their impact on advancing Peru's STI ecosystem.

[Figure 6-6] Grant Distribution by Type of Financial Instrument



In terms of investment, 54.1% of the resources were dedicated to funding scientific research instruments, while 31.7% were allocated to scholarships and programs, establishing these as the most prominent and impactful instruments within PROCIENCIA/CONCYTEC's portfolio.

[Figure 6-7] Resource Distribution by Type of Financial Instrument



■ Geographic Distribution of Projects

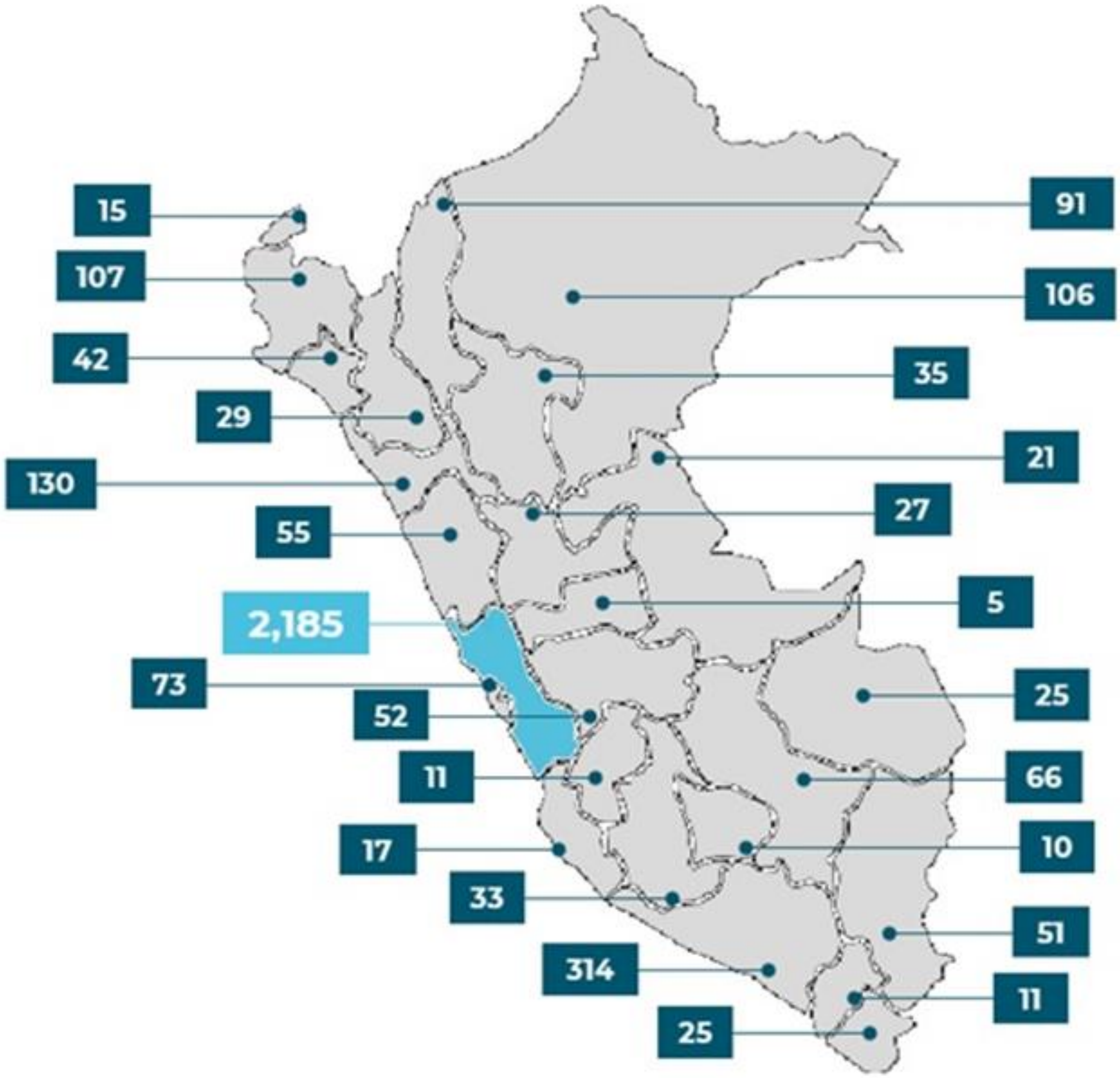
The distribution of funded projects across Peru reveals a significant concentration in Lima, which, as the country's capital and economic, political, and financial hub, benefits from advanced research infrastructure and a higher availability of highly qualified human resources. Despite Lima's dominance, other regions such as Arequipa, Amazonas, Piura, and La Libertad have also received funding, particularly due to their strong regional economies in sectors like mining and agriculture. However, at the start of the analysis period in 2013, the distribution of grants was notably unequal, with only nine regions outside Lima Metropolitan Area receiving PROCIENCIA/CONCYTEC funding. This imbalance stemmed from initial efforts to promote financial instruments, limited applications from regions outside the capital, and intense competition for a limited STI budget.

Recognizing these disparities, policies introduced in 2017 sought to diversify and decentralize STI funding, particularly through the "seed modality," which targeted regions outside Lima and Callao. This approach simplified application requirements for principal investigators and prioritized basic and applied research projects, the most in-demand offerings in the portfolio. By 2018, the impact of these policies was evident, as only Madre de Dios remained without a PROCIENCIA/CONCYTEC grant. The decentralization effort significantly increased participation from northern regions such as Piura, Lambayeque, La Libertad, and Ancash, as well as southern regions like Arequipa, Cusco, and Puno.

These decentralization policies have progressively balanced the distribution of STI projects, fostering greater knowledge generation, innovation, and researcher development across the country. While progress has been gradual due to budgetary constraints and regional management capacities, there has been a notable shift in equity. In 2013, Lima accounted for 77% of the grants, but by 2023, this share had decreased to 55%, reflecting considerable progress in resource allocation to other regions. Furthermore, this shift has expanded focus to the Amazon region, including Loreto, Amazonas, and San Martín, alongside the already active northern and southern regions.

Although Lima continues to lead in grant concentration, experiencing approximately 2,000% growth in funded projects between 2013 and 2023, other regions have shown even more dramatic growth, around 7 % during the same period. These highlights increased regional participation in PROCIENCIA/CONCYTEC's financial instruments over the past decade, underscoring substantial advancements in knowledge generation, human capital development, and the establishment of research groups and collaboration networks. As of December 2023, the nationwide distribution of grants reflects these significant strides toward a more inclusive and balanced STI ecosystem.

[Figure 6-8] Distribution of Grants Awarded Nationwide



4. Regional R&D Capabilities

4.1 Mapping of Regional R&D Resources and Capacities

The distribution of R&D resources and capacities in Peru shows significant regional inequalities. Lima concentrates 37.2%(MEF-SIAF-2023) of the national STI budget. Regions such as Cusco, Arequipa, and Puno have begun to stand out due to investments in sectors like agriculture, tourism, and renewable energy, while northern and Amazonian regions have significant potential in biodiversity and sustainable resource management. However, many regions lack advanced infrastructure, specialized talent, and adequate financial resources.

The weaknesses of the regional system include fragmented efforts to integrate local initiatives and a limited presence of coordination mechanisms in less developed areas. Regions such as Madre de Dios and Huancavelica receive less than 1% of the total STI budget, reflecting unequal distribution. The lack of research infrastructure restricts opportunities for innovation. Additionally, the shortage of skilled human capital in key specialties reinforces regional disparities, affecting these areas' ability to compete and contribute to the national STI ecosystem.

To address these inequalities, it is essential to strengthen research infrastructure in regions with greater economic potential, promote advanced training programs and researcher mobility, and implement balanced budget allocations that support regional innovation. Furthermore, it is recommended to reinforce collaborative networks among academia, industry, and government, using different mechanics to align regional projects with national priorities. These actions will foster more equitable development and leverage local strengths to contribute to Peru’s scientific and technological progress.

4.2 Strategic Recommendations for Regional Development

To achieve a more balanced distribution of STI funding and foster equitable regional development in Peru, it is crucial to adopt targeted strategies that address the concentration of resources in Lima while strengthening regional capabilities. The following recommendations focus on reducing disparities, expanding infrastructure, and fostering collaborative innovation.

Decentralize STI Funding and Strengthen Regional Infrastructure: Allocate a larger proportion of STI funding to regions outside Lima, prioritizing areas with high economic potential and pressing local needs. Establish new R&D facilities and technology hubs in underfunded regions, focusing on sectors such as agriculture, renewable energy, and biodiversity. These facilities should be equipped to support advanced research and innovation tailored to regional strengths. Moreover, capacity-building initiatives, including investments in equipment and digital infrastructure, are critical to enhancing local research capabilities.

Promote Training and Talent Development: Expand training programs to develop skilled human capital across all regions. Initiatives should include scholarships, mobility programs for researchers, and partnerships with local universities to ensure a pipeline of qualified professionals. Efforts should prioritize regions with limited access to educational and training resources, enabling them to attract and retain talent essential for STI development.

Align Policies with Local Needs: Tailor STI policies to address specific challenges and opportunities unique to each region. Engage regional stakeholders in the planning and decision-making process to ensure alignment with local priorities. Mechanisms such as Regional Development Councils or SINACYT should integrate regional perspectives into national STI strategies, ensuring that projects reflect the diverse needs and strengths of the country.

Enhance Collaboration through Public-Private Partnerships: Foster networks that connect academia, industry, and government to create innovation ecosystems in the regions.

Encourage partnerships through incentives such as tax benefits, grants, or co-financing mechanisms to stimulate joint R&D projects. Establishing regional innovation clusters and leveraging existing mechanisms like DER (ProInnovate) and CITE will enhance knowledge transfer and promote sustainable economic growth.

Monitor and Evaluate Impact: Develop a robust system to monitor and evaluate the outcomes of regional STI investments. Metrics should assess infrastructure utilization, talent retention, and the economic and social impact of funded projects. Regular reviews will help identify gaps and opportunities for improvement, ensuring continuous progress toward equitable regional development.

5. Public-Private Collaboration

5.1 Current Public-Private Collaboration:

The IVAI (Linkage Initiatives to Accelerate Innovation) program is an initiative promoted by CONCYTEC and implemented in collaboration with key public institutions such as the Ministry of Production, the Ministry of Foreign Trade and Tourism, ProInnóvate, PromPerú, and the ITP, with support from the World Bank. Its primary objective is to foster innovation and competitiveness by leveraging Peru's rich biological, geographic, climatic, and cultural diversity to develop high-value products and services that appeal to sophisticated and demanding markets willing to pay a premium for unique offerings.

This initiative focuses on driving transformative changes in eight strategic sectors across eight regions of the country. By strengthening value chains, the program aims to boost productivity, facilitate entry into more competitive markets, and enhance overall economic competitiveness. Each IVAI is designed to provide a strategic roadmap for its respective value chain, emphasizing collaboration through structured public-private dialogues involving academia, businesses, and government institutions. These dialogues play a critical role in identifying opportunities and aligning efforts to achieve sustainable growth.

The eight IVAIs currently under development span a diverse range of economic sectors that reflect Peru's unique strengths. These include tourism, aquaculture, agriculture (with a focus on Andean grains and tropical and subtropical superfruits), agroindustry (premium distilled beverages), manufacturing (wood, textiles, and apparel), and mining, including its supply chains. Together, these initiatives aim to harness Peru's regional potential, fostering innovation and driving economic transformation at both the local and national levels.

The DER (Boosting Regional Innovation and Entrepreneurship Ecosystems) program, managed by ProInnovate under the Ministry of Production, is a strategic initiative focused on decentralizing innovation in Peru. Its primary aim is to strengthen regional innovation ecosystems by promoting entrepreneurship, technological modernization, and capacity building in areas outside Lima. By aligning with regional comparative advantages, DER fosters sustainable growth in sectors such as agritech, energy, and manufacturing, helping to balance the geographic distribution of STI resources and benefits.

A key mechanism of DER is the InnovaSuyu platform, which provides a collaborative space for academia, industry, and local governments to connect and co-create. This platform has facilitated the establishment of regional clusters and supported SMEs in accessing funding and technical assistance for modernization projects. Furthermore, DER emphasizes capacity building through workshops, technical training, and funding for regionally relevant projects.

5.2 Strategic Initiatives to Enhance Collaboration:

To strengthen public-private partnerships (PPPs) and align STI initiatives with national priorities, it is essential to implement targeted incentives and streamlined processes that foster collaboration and innovation. These initiatives should address current gaps, promote shared goals, and ensure efficient project execution to achieve sustainable outcomes.

Proposed Incentives:

- **Tax Credits for R&D Investments**: Introduce enhanced tax deductions for companies investing in collaborative R&D projects with public research institutions. For example, a 200% tax deduction on eligible R&D expenditures can incentivize businesses to participate actively in innovation ecosystems. Expand eligibility criteria to include SMEs and startups, particularly those focused on regional development or aligned with priority sectors such as agritech, renewable energy, and digital transformation.
- **Innovation Grants**: Establish competitive grants co-financed by the public and private sectors, with priority given to projects that address national challenges like climate resilience, health innovations, or sustainable industrial practices. Provide specialized grant schemes for regions outside Lima to promote decentralized innovation and address local needs, leveraging programs like ProInnovate as a model.
- **Collaborative R&D Programs**: Create sector-specific R&D clusters that unite academia, government, and industry to co-develop innovative solutions. These clusters could focus on areas like advanced materials, biotechnology, or sustainable mining technologies. Encourage cross-sectoral collaboration through matchmaking platforms that connect researchers and businesses with complementary expertise.

CHAPTER 7

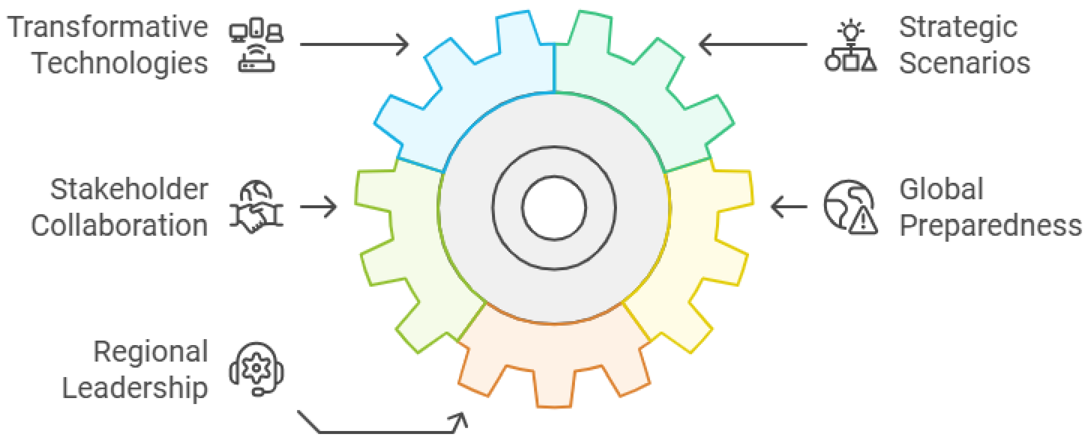
Establishment Draft of Peru’s Technology Foresight Center

S&T Planning and Technological Forecasting for Prioritized Sectors in Perú

Chapter 7. Establishment Draft of Peru's Technology Foresight Center

1. Purpose of the Foresight Center[8]

The Peru Center for Technology Foresight Center (PCTP) aims to anticipate and shape Peru's technological future by identifying transformative technologies and developing strategic scenarios that guide sustainable development policies. Serving as a catalyst for innovation and growth, CPTP collaborates with national and international stakeholders to strengthen Peru's science, technology and innovation ecosystem. The center promotes preparedness for global challenges, positioning Peru as a proactive contributor to regional foresight and technological progress.



[8] <Chapter 7. Establishment Draft of Peru's Technology Foresight Center> is written by Agnes Franco and Victor Izaguirre, Directors; and Felipe Valentín and John Collantes, Senior Specialists, all at CONCYTEC.

2. Proposed Structure and Functions

The functions of the Peru Center for Technology Foresight (CTF) include the following:

- Trend Monitoring and Analysis: conduct ongoing studies to monitor technology trends and assess their potential impact on critical sectors in Peru, such as agriculture, healthcare, energy, and education. This will enable the center to provide timely insights and identify early signals of disruptive changes.
- Identification of Key Technologies: through regular technology foresight exercises, such as a comprehensive Peruvian Technology Foresight exercise every five years, identify key technologies that could be vital for Peru's future growth and competitiveness.
- Future Scenarios Development: create diverse future scenarios to anticipate potential technological developments and prepare for various possible outcomes. This scenario-building approach will guide Peru's strategic planning and help policymakers envision different future pathways.
- Vision Building: establish a long-term vision for Peru’s technological and socio-economic direction. This vision will reflect Peru's strategic objectives and offer a roadmap for integrating emerging technologies into national priorities.
- Research and Publications: produce reports, articles, and studies that provide insights into technological advancements and future possibilities. These publications will serve as a resource for decision-makers, researchers, and the public.
- Training and Education: organize workshops, courses, and seminars to train professionals, policymakers, and other stakeholders in technological foresight, scenario planning, and innovation management.
- Public Policy Advisory: collaborate closely with government agencies to develop policies that encourage the responsible and beneficial adoption of new technologies. This advisory function will ensure that technology adoption aligns with Peru’s social and economic goals.

- Networks and Collaborations: build and maintain networks with local and international research centers, universities, and organizations. These collaborations will facilitate knowledge exchange and foster a strong ecosystem for foresight and innovation.

The proposed structure for the Peru Center for Technology Foresight is as follows:

■ **General Directorate**

- Oversees the entire operation of the Technology Foresight Center, making strategic decisions, setting priorities, and ensuring alignment with national and organizational goals.
- Manages daily operations and coordinates activities across different teams within the center, ensuring that projects and tasks are completed efficiently.

■ **External Advisory Committee**

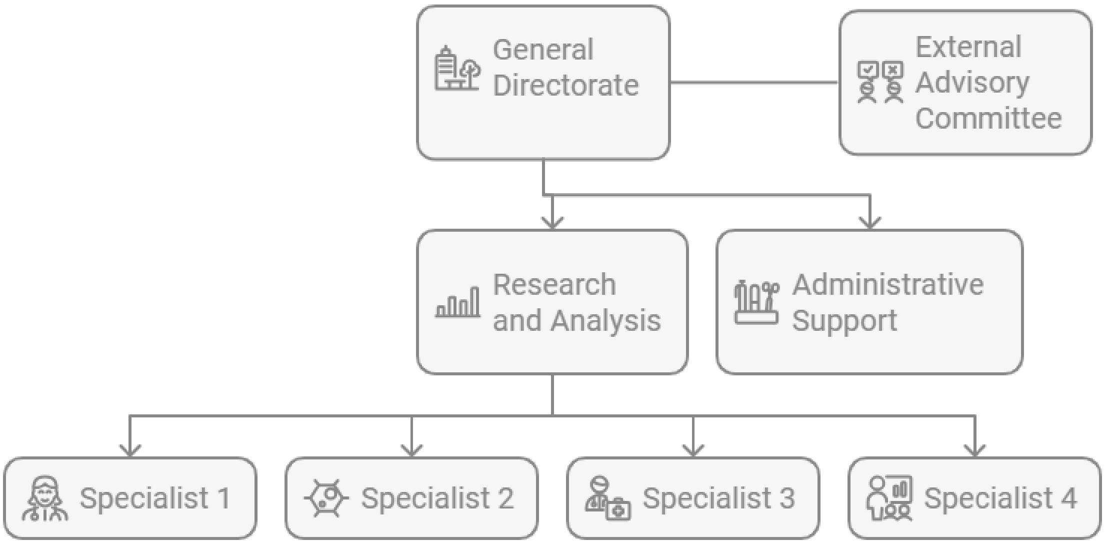
- Composed of external experts and stakeholders, this committee provides guidance, feedback, and strategic insights, helping the foresight center stay aligned with international best practices and emerging global trends.

Research and Analysis Coordinador: Responsible for conducting studies, gathering data, and analyzing trends related to technological foresight. This team identifies emerging technologies and their potential impacts on different sectors.

- Technology Foresight Specialist 1: Focuses on a specific area of research within the foresight center, providing expertise and insights into particular sectors or technologies relevant to Peru's development.
- Technology Foresight Specialist 2: Contributes to research projects by analyzing technological trends and creating reports. This specialist may focus on a different area of foresight, complementing the skills of the other specialists.
- Technology Foresight Specialist 3: Engages in scenario planning and helps develop long-term technological projections. This specialist may also support cross-disciplinary projects.

- Technology Foresight Specialist 4 (with expertise in data analysis): Specializes in data analysis, interpreting large datasets to uncover trends and insights that guide the foresight center’s research. This role is critical for validating trends and supporting evidence-based recommendations.

Administrative Support: Provides logistical and clerical assistance, managing schedules, communications, and documentation. Ensures that operational processes run smoothly and efficiently.



3. Roadmap for Implementation

The following information outlines the roadmap for implementing the Peru Center for Technology Foresight Center (PCTP), detailing the phases and planned actions toward full operation.

1. Initial Planning and International Collaboration (2024)

- Collaboration with international partners, such as the Science and Technology Policy Institute (STEPI) of South Korea, to gain insights and technical support
- Define the center's purpose, objectives, and main functions in alignment with Peru's science, technology, and innovation goals.
- Conduct stakeholder consultations to gather input and ensure the center's relevance to national and regional priorities.

2. Approval of Implementation Framework (2025)

- Develop an implementation framework detailing the proposed budget, organizational structure, and functional areas.
- Obtain approvals for budget allocation and official authorization to operate under CONCYTEC's oversight.
- Define and formalize roles and functions within the center, ensuring alignment with Peru's national strategies.

3. Phase 1: Initial Operations as an Internal Unit within CONCYTEC (2026)

- Establish PCTP as an internal unit within CONCYTEC with a small team of 2 - 3 technology foresight specialists.
- Begin basic foresight activities, such as environmental scanning and initial technology assessments, to demonstrate early value

- Establish foundational partnerships with local universities and research institutions.

4. Phase 2: Semi-Autonomous Phase (2027 - 2028)

- Expand the team to include 3 - 4 specialists, with new roles for scenario planning, policy advisory, and sector-specific technology analysis
- Conduct more comprehensive foresight studies in collaboration with government ministries and public institutions.
- Seek additional funding from international partnerships to support expanded operations.

5. Phase 3: Full Operational Autonomy (2029)

- Transition to an autonomous entity with a dedicated team of more than 5 professionals, including specialists in advanced technology, scenario analysis, and policy development
- Secure operational and financial independence, allowing the center to manage its projects and funds.
- Position the PCTP as a key player in regional foresight, establishing partnerships with other technology foresight centers in Latin America and internationally.

CHAPTER 8

Capacity Development Process

S&T Planning and Technological Forecasting for Prioritized Sectors in Perú

Chapter 8. Capacity Development Process

1. Kickoff Meeting (February 13, 2024)

■ Kick-off Meeting Overview

- Date: February 12, 2024 08:00-09:30 pm (@Peru) / February 13, 2024 10:00-11:30 am (@Korea)
- Working Language: English
- Venue: Virtual meeting via Zoom

■ Objectives and Activities

- Review the 2023 work progress and outline the 2024 work plan.
- Define specific activities and directions for the 2024 project.
- Prepare for the Capacity Building Training Workshop in Korea and the Dissemination Workshop in Peru.
- Discuss the role and scope of activities for the local consultant in 2024.
- Conduct a more in-depth analysis of the Delphi Survey.

■ Program Schedule

February 12, 08:00-09:30 PM (@Peru) / February 13, 10:00-11:30 AM (@Korea)	
10:00-10:40	• **Introduction of Participants** - *All Participants both from Korea and Peru* • **2023 Work Progress and 2024 Work Plan** - *Presented by Dr. ChiUng Song (Senior Research Fellow, STEPI)*
10:40-11:20	• **Brief on a more in-depth Delphi Survey to improve the quality of the technology roadmap** - *Presented by Dr. Johng-Ihl Lee (Professor, Department of Technology and Society, SUNY Korea)* • **Discussion and Q&A** - Capacity Building Training Workshop in Korea - Dissemination Workshop in Peru - Role and Acitivity Scope of Local Consultant - Other issues
11:20-11:30	• **Closing Remarks** - *Dr. Agnes Franco (Director, Research & Studies, CONCYTEC)* - *Dr. ChiUng Song (Senior Research Fellow, STEPI)*

■ Photos

2. Capacity-Building Workshop (Invitational Training) in South Korea (May 18~24, 2024)

2.1 Capacity Building Workshop Overview

- Date: May 18-24 2024 (Including the day of entry and exit)
- Working Language: English
- Venue: Seoul, Goyang, and Incheon, etc. - The Republic of Korea

2.2 Objectives and Activities

- To cultivate practical skills in planning the R&D program based on a technology roadmap for the Prioritized Sectors in the 2022-2023
- To share the knowledge and experiences for the establishment of the Technology Foresight Center
- To draw up a draft for the R&D planning and for the proposal to establish Technology Foresight Center of CONCYTEC
- To apply and integrate analysis results on a two-stage Delphi survey
- To take a field trip and experience Korean culture

2.3 Program Schedule

May 18 Peruvian delegation' Arrival in Korea	
May 19 Free Time	
May 20 Technology Foresight Workshop(1): Planning the R&D Program and the Technology Foresight Center	
11:00-12:30	**Introduction of Participants** • ***All Participants both from Korea and Peru*** Introduction on the Capacity Building Workshop *(Dr. ChiUng Song, Senior Research Fellow of STEPI/ Sohyun Kwon, Senior Researcher of STEPI)* *Welcoming Remarks (Dr. Hwanil Park, Chief Director of STEPI)* *Congratulatory Remarks (Mr. Paul Duclos, Peruvian Ambassador to Korea)*
12:30-14:00	**Lunch**
14:00-16:00	**Hands-on Project (1)** • Analysis results of a one-stage Delphi Survey and Methods of a two-stage Delphi Survey (Dr. Sehee Park, CTO of Deliverdy) • History of Industrial Technology Development in Korea and Planning R&D Programs using the Delphi Methods (Dr. Johng-Ihl Lee, Professor of SUNY Korea) **Q&A and Discussion**
16:00-17:00	**Practice (1)**
18:00-19:30	**Welcome Dinner**
May 21 Technology Foresight Workshop(2): Planning the R&D Program and the Technology Foresight Center	
10:00-12:30	**Hands-on Project (2)** • *R&D Governance in Korea (Dr. ChiUng Song, Senior Research Fellow of STEPI)* • *R&D Planning (Dr. Johng-Ihl Lee, Professor of SUNY Korea)* **Q&A and Discussion**
12:30-14:00	**Lunch**
14:00-15:00	**Practice (2)**
16:00-17:00	**Field trip** - Global Knowledge Exchange and Development Center(GKEDC)
17:30-19:00	**Dinner**
19:00-20:00	**Korean Culture Experience** - Lotte World Tower - Seoul Sky

May 22 Technology Foresight Workshop(3): Planning the R&D Program and the Technology Foresight Center	
10:00-12:30	**Hands-on Project (3)** • *Experiences from operating and managing Technology Foresight Center in Korea (Dr. Hyun Yim, Senior Research Fellow of KISTEP)* **Q&A and Discussion**
12:30-14:00	**Lunch**
14:00-15:00	**Practice (3)**
15:00-18:00	**Korean Culture Experience** - Blue House(Chungwadae), Gyeongbok Palace
18:30-19:30	**Dinner**
May 23 Technology Foresight Workshop(4): Planning the R&D Program and the Technology Foresight Center	
10:00-12:30	**Hands-on Project (4)** • *Delphi Method and Policy Formulation (Dr. Ahreum Hong, Professor of Kyung Hee University)* Presentation on Output by Invited Peruvian Participants • *To draw up a draft for the R&D program planning (Peruvian Experts (6))* • *To create a proposal for planning the Technology Foresight Center of CONCYTEC (Peruvian Experts (6))*
12:30-14:00	**Lunch**
14:00-18:00	**Field Trip** - Hyundai Goyang Motor Studio
18:30-19:30	**Dinner**
May 24 Technology Foresight Workshop(5): Planning the R&D Program and the Technology Foresight Center	
09:00-09:30	**Wrap-up and Certificate Ceremony**
10:00-12:00	**Go to Incheon**
12:00-13:30	**Lunch**
13:30-16:00	**Field trip (3)** (Incheon Technopark, G-Tower, Incheon Free Economic Zone Tour Korea, SUNY)
16:30-	**Go to Incheon Airport and Back to Lima**

2.4 Results

2.4.1 Analysis results of a one-stage Delphi Survey and Methods of a two stage Delphi Survey

The session led by Dr. Sehee Park reviewed the results from the first round of Delphi surveys, which focused on Peru's strategic sectors: Agribusiness, Advanced Manufacturing, Mining and New Materials, Textile and Apparel, Creative Industry, and Forestry. The Delphi method was introduced as a structured approach for gathering expert insights, emphasizing the principles of anonymity, controlled feedback, and iterative rounds. This approach is particularly effective in minimizing the influence of dominant participants, promoting consensus, and reducing risks in technology roadmap planning. By leveraging the Delphi method, Peru aims to harness informed predictions and build alignment among diverse stakeholders within these priority sectors.

For the second round of the Delphi survey, several guidelines were set to refine and deepen the insights gathered. Only those experts who participated in the initial survey will be invited to the second round, ensuring continuity and commitment. Additionally, statistical data from the first round will be shared, allowing participants to review and possibly adjust their prior responses, fostering a more robust consensus. Measures will be taken to restrict survey access to authorized participants only, ensuring the integrity of the expert feedback. Success in this round will be gauged by a reduction in response variability, as indicated by a lower standard deviation in the answers.

During the session, the survey questions were closely examined to enhance clarity and precision. This review is crucial for collecting reliable responses in areas such as technological capacity, technology readiness levels, and acquisition strategies. Experts will assess these areas across various sectors, applying consistent evaluation criteria to ensure meaningful comparisons. With a focus on both international collaboration and local R&D, the Delphi survey is expected to provide strategic insights that support Peru’s roadmap for technological development in these vital industries.

2.4.2 History of Industrial Technology Development in Korea and Planning R&D Programs using the Delphi Methods (Dr. Johng-Ihl Lee)

Dr. Johng-Ihl Lee, a professor from SUNY, delivered an insightful presentation on the critical factors contributing to South Korea's industrial development, using the nation's economic trajectory as a case study. Central to his discussion was the role of technology as a key driver of economic growth and the importance of R&D investment. Dr. Lee highlighted that South Korea's economic progress over the last 50 years was largely due to a sustained commitment to science, technology, and innovation (STI). This approach has enabled South Korea to evolve from a labor-driven economy into a powerhouse of capital-intensive and knowledge-based industries, illustrating the transformative power of strategic technological investment over time.

Dr. Lee pointed out ten essential factors behind South Korea's success that could provide valuable lessons for other countries seeking to strengthen their own industrial landscapes. A primary factor is the establishment of a strong consensus around investment in STI, supported by a robust legal infrastructure that ensures stability and promotes long-term innovation. He also emphasized the significance of financial stability within local governments, which enabled a shared investment burden and cohesive alignment of regional and national development goals. Additionally, Dr. Lee discussed how strategic timing, adaptable legal frameworks, and responses to evolving economic needs were critical in fostering sustained growth and creating a supportive environment for industry to flourish.

A substantial portion of Dr. Lee's analysis focused on the necessity for coordinated efforts across various sectors, including the government, private industry, and academia. He underscored that while policy direction plays a key role, it is the collaboration between these stakeholders that reinforces the resilience of the industrial ecosystem. The case of South Korea demonstrates how a long-term focus on developing STI capacities can drive competitive advantage, with strategic policies and well-timed interventions allowing the country to build robust industries, from manufacturing to high-tech fields. This cooperative model allowed for diversified growth, adaptation to technological advancements, and the development of an export-oriented economy that has become globally influential.

Finally, Dr. Lee concluded that South Korea's journey to industrial maturity highlights the importance of patience and sustained investment in STI. Success in this domain, he explained, typically requires at least 20 years of dedicated effort and consistent policy application, underscoring the need for a long-term vision. The South Korean example illustrates that impactful STI development is rarely achieved in isolation but through cohesive, multi-sectoral efforts that involve active participation from government, the private sector, and educational institutions. By steadily investing in these collaborative frameworks, South Korea has managed to establish a dynamic innovation ecosystem and a competitive industrial base, serving as a potential model for other countries pursuing similar economic growth and technological progress.

2.4.3 R&D Governance in Korea (Dr. ChiUng Song)

Dr. ChiUng Song outlined South Korea's development strategy, beginning with light industry in the 1960s and advancing to high-tech industries by the 1990s. This growth was supported by a targeted "Select and Focus" strategy, where the government prioritized certain industries for development. Key legislation, such as the Technology Development Promotion Act of 1972, laid the groundwork for a structured R&D sector, which helped South Korea transition from basic manufacturing to more sophisticated technological domains. This approach underscored the importance of innovation as a catalyst for economic growth, with a gradual but intentional expansion from light goods to advanced electronic products.

The creation of government research institutes (GRIs) played a fundamental role in building South Korea's R&D infrastructure. Institutions such as the Korea Institute of Science and Technology (KIST), the Korea Atomic Energy Research Institute (KAERI), and the Agency for Defense Development (ADD) became central to the Seoul knowledge cluster, driving technological advances and supporting the growth of strategic industries. In 1982, South Korea launched national R&D programs that promoted partnerships between academia and industry, enabling both sectors to collaborate effectively on developing new technologies for emerging industries. This approach encouraged a collaborative environment that was essential for the country's industrial evolution.

As South Korea's technological aspirations grew, so did its governance structure for science, technology, and innovation (STI). By the 1990s, the Ministry of Science and ICT began to oversee these initiatives, shaping policies to support innovation across sectors. This governance model evolved into a multifaceted system involving various ministries and councils to guide the R&D agenda. Notably, initiatives like the Highly Advanced National (HAN) Project and the Creative Research Initiative helped to align public and private sector goals, ensuring that resources and policies were strategically directed towards areas with the highest potential for impact. The Ministry's coordination role allowed for greater coherence and adaptability in STI policies, addressing national needs effectively.

Dr. Song highlighted that South Korea's progress in science and technology relied on long-term, structured efforts and consistent investment in R&D. The establishment of a strong STI governance model, paired with a sustained emphasis on collaboration between public and private sectors, proved essential in building a resilient and competitive industrial ecosystem. The South Korean experience demonstrates how strategic investment in R&D infrastructure, combined with targeted governance and cross-sectoral cooperation, can drive transformative economic growth. This model offers valuable insights for other countries aspiring to achieve similar success in developing a dynamic, innovation-driven economy.

2.4.4 R&D Planning (Dr. Johng-Ihl Lee)

Dr. Johng-Ihl Lee's session focused on the core principles of R&D policy planning, particularly in the context of South Korea's successful development strategies. He emphasized the importance of prioritization in R&D to direct resources toward high-impact areas, ensuring efficient use of funds and alignment with national goals. The planning process was presented as a structured approach that considers both immediate economic needs and long-term development, facilitating a balance between innovative research and practical, market-driven applications. This planning model has been central to Korea's transformation into a leading technological and industrial power.

A key strategy discussed was the "Imitate to Innovate" approach, which encourages building on existing technologies as a pathway to innovation and market competitiveness. This method allows for a faster transition of ideas from R&D to commercialization by adapting proven technologies for new markets, supporting industry growth and encouraging innovation within a more controlled risk environment. Dr. Lee highlighted that this strategy, along with a focus on legal infrastructure, such as technoparks, has been essential in fostering a favorable environment for technology-driven industries to flourish.

Dr. Lee also pointed to the importance of a well-established legal framework and favorable working conditions for public servants involved in R&D governance. South Korea's legal infrastructure, supported by policies like the Act on Technoparks, has provided a stable foundation for collaborative innovation and industrial growth. Additionally, Dr. Lee noted that secure and competitive compensation for public officials, such as pensions, plays a role in attracting and retaining skilled professionals, which strengthens institutional capacity in R&D management and policy implementation.

Dr. Lee concluded with a focus on sustainable financial resources for R&D, including the strategic use of public funds and external financing sources such as Official Development Assistance (ODA) projects. By diversifying funding sources and establishing a consistent policy framework, South Korea has been able to support long-term R&D initiatives effectively. Dr. Lee's insights underscore the significance of coordinated financial and policy planning in achieving sustained technological advancement, providing a valuable model for other countries seeking to build a resilient R&D ecosystem and drive innovation through structured policy and strategic investments.

2.4.5 Experiences from operating and managing Technology Foresight Center in Korea (Dr. Hyun Yim)

Dr. Hyun Yim's presentation outlined the primary roles and functions of the Technology Foresight Center within South Korea's KISTEP (Korea Institute of Science and Technology Evaluation and Planning). He explained that technology foresight serves as a critical tool in S&T policy by providing structured, long-term analysis to identify emerging technologies with high economic and social potential. The Center's roles include technology monitoring, projecting key trends, and supporting national decision-making, all aimed at shaping South Korea's future technological landscape. These activities underscore the importance of foresight as a foundational pillar in national S&T policy.

The Center's specific tasks involve environmental scanning to detect early signs of emerging trends and challenges across multiple domains, from defense to environmental sustainability. By continuously assessing these areas, the foresight team can provide early warnings and strategic insights. Additionally, every five years, KISTEP conducts national foresight exercises to identify key technologies that align with Korea's strategic needs and growth objectives. This structured approach allows the Center to develop a national vision, establish R&D priorities, and create comprehensive S&T strategies.

Dr. Yim highlighted that effective foresight requires not only advanced methodologies, such as big data analysis and Delphi surveys, but also strategies to engage stakeholders and ensure robust participation. In this context, he discussed the challenges faced during the Delphi process, especially in achieving high response rates from experts. The Center found that incentivizing participation and establishing minimum response thresholds tailored to each study improved the quality and reliability of foresight outputs. This insight is particularly relevant for establishing similar practices in Peru, where consistent engagement from stakeholders is essential for actionable foresight results.

Finally, the presentation emphasized the importance of foresight as a strategic decision-support tool for S&T policy. In South Korea, foresight projects go beyond merely predicting trends; they actively shape the future through a proactive, collaborative approach involving public, private, and academic partners. For Peru, implementing similar foresight practices could significantly strengthen national S&T planning by establishing a structured approach to identify and monitor critical technologies, create a unified vision for S&T, and align efforts with long-term development goals.

2.4.6 Delphi Method and Policy Formulation (Dr. Ahreum Hong)

Dr. Ahreum Hong outlined the approach for the second round of the Delphi survey, which involves an in-depth re-evaluation of key technologies, acquisition strategies, and their economic, social, and competitive impacts. This round is designed to refine the initial feedback and prioritize policies that support technology adoption, re-assessing collaborative partnerships based on new insights. Additionally, experts are encouraged to re-evaluate future challenges and identify necessary technological strategies. This stage aims to gather more nuanced input to help shape policies that effectively support Peru's technological goals and align with strategic priorities.

The first-round analysis of Delphi survey responses provided insights into sector-specific needs, such as advanced technologies in agribusiness. For example, experts emphasized prioritizing investments in AI, blockchain, and big data to optimize decision-making and resource management in agriculture. Sustainable practices, precision agriculture, and technologies to reduce environmental impact were highlighted, along with the importance of collaborations between researchers, technology firms, and farmers. Training programs were also suggested to ensure that agricultural professionals can fully leverage these advanced technologies, promoting efficient and sustainable practices across the sector.

Dr. Hong emphasized several key policy implications from the Delphi survey, noting the need for targeted resource allocation to support critical technologies identified by experts. Policies should prioritize technologies ready for commercialization and promote technology transfer and acquisition strategies that align with expert recommendations. Emphasis was placed on fostering technologies with high economic and social returns, which would require strategic collaborations between academia, industry, and research centers.

Furthermore, policies must address both immediate and future challenges, preparing Peru to capitalize on emerging technologies and strengthen its industrial ecosystem.

The presentation also proposed a segmentation, targeting, and positioning (STP) framework to unify R&D efforts across six prioritized industries. This framework includes segmenting industries by technological adoption and environmental impact, targeting high-impact projects, and positioning Peru as a leader in sustainability and technology-driven transformation. Dr. Hong suggested that the Technology Roadmap for Peru should incorporate this STP approach to ensure that R&D investments foster innovation and inter-sectoral collaboration. The session concluded with a vision for establishing a permanent Technology Foresight Center and a set of KPIs to guide Peru's national development strategy, ensuring sustainable growth and technological competitiveness.

2.5 Evaluation Survey Results

■ Overview of Evaluation Survey

- **Survey Period and Method**: Conducted on the workshop's closing day (May 23) via an online form accessible through a QR code.
- **Participants and Response Rate:** Total of 5 participants, with a 100% response rate.
- **Survey Content:** Comprised of individual lecture evaluations and an overall assessment of the program.

■ Lecture Evaluation Survey

- **Survey Structure:** A total of 15 questions were included in the lecture evaluation form.
  - **13 multiple-choice questions** assessed the content, instructor performance, and lecture structure. Respondents rated each item on a 5-point Likert scale (Strongly Agree – Agree – Neutral – Disagree – Strongly Disagree).
  - **2 open-ended questions** asked participants to freely describe what they found most valuable and what could be improved in the lecture.

- **Relevance of Lecture Content:** Most participants found the lecture topics highly relevant. In particular, sessions on Korea's technology foresight and science & technology planning methodologies were evaluated as being practically helpful for policy development and R&D program planning in Peru.
- **Overall Evaluation of the Lecture:** The lectures were generally well-structured, and the instructors were praised for their effective knowledge delivery. Some participants noted that having access to lecture materials in advance would have allowed them to engage more thoroughly. Many participants also highlighted the usefulness of the Q&A sessions and group discussions following the lectures. These segments were especially helpful for reviewing content and exploring ways to apply it in practice. The opportunity to share experiences and compare Korean and Peruvian cases was seen as particularly beneficial.
- **Suggestions:** Participants requested more detailed materials on specific processes such as R&D planning and the establishment of a technology foresight center.

[Table 8-1] Lecture Evaluation Survey

Lecture 1	Title of Lecture		Lecturer	Name	
Questions	Strongly Agree	Agree	Neutral	Disagree	Strongly Disagree
1. The topic covered in the lecture was relevant to me.	○	○	○	○	○
2. The content was well-organized and easy to follow.	○	○	○	○	○
3. The lecture was helpful for understanding of the field.	○	○	○	○	○
4. The lectures would be helpful in strengthening the institutional capacity of my organization.	○	○	○	○	○
5. The time allotted to the lecture was sufficient.	○	○	○	○	○
6. The pace of the lecture was good.	○	○	○	○	○
7. The lecturer was sound in knowledge and speech.	○	○	○	○	○

Lecture 1	Title of Lecture		Lecturer		Name
Questions		**Strongly Agree**	**Agree**	**Neutral**	**Disagree**	**Strongly Disagree**

Questions	Strongly Agree	Agree	Neutral	Disagree	Strongly Disagree
8. The lecturer noticed the indications when trainees needed help.	○	○	○	○	○
9. The objectives of each lecture were stated clearly at the beginning of the lecture.	○	○	○	○	○
10. The lecturer conveyed in the topic in depth.	○	○	○	○	○
11. Q&A and discussion were adequate to enhance my understanding.	○	○	○	○	○
12. I will be able to apply the knowledge learned in my work/practice.	○	○	○	○	○
13. I will recommend this lecture to others.	○	○	○	○	○
14. What did you like most about the lecture?					
15. What aspects of the lecture can be improved?					

■ Program Evaluation Survey

- **Survey Structure:** The overall program evaluation included 12 questions.
  - **10 multiple-choice questions** assessed the overall effectiveness of the program, rated on a 5-point Likert scale (Strongly Agree – Agree – Neutral – Disagree – Strongly Disagree).
  - **2 open-ended questions** invited suggestions and comments, as well as specific examples of enhanced self-efficacy or application of the training in professional settings.
- **Overall Evaluation of the Program:** Participants gave highly positive feedback on the overall program. It was particularly valued for offering practical education and hands-on activities relevant to R&D program planning and the establishment of a technology foresight center. Nearly all participants indicated they would recommend the program to colleagues or other organizations. The program was seen as making a meaningful contribution to strengthening Peru’s science and technology innovation capacity.

- **Program Duration:** Most participants agreed that the 7-day schedule was appropriate. However, there were requests for continued support and additional learning opportunities after the program. Participants suggested that ongoing mentorship or follow-up sessions would help apply the knowledge gained more effectively in practice.
- **Suggestions:** To further enhance program effectiveness, participants proposed providing pre-training materials, increasing the number of hands-on activities, and adding online participation options. There were also calls for sustained follow-up support or advisory services to help translate the program outcomes into actual policy implementation.

[Table 8-2] Program Evaluation Survey

Questions	Strongly Disagree	Disagree	Neutral	Agree	Strongly Agree
1. I expected that this training would be helpful in performing my duties.	①	②	③	④	⑤
2. The program was helpful in enhancing my capacity to perform the duties.	①	②	③	④	⑤
3. The program met the objective of institutional capacity building.	①	②	③	④	⑤
4. The program was composed in accordance to its purpose.	①	②	③	④	⑤
5. The materials have been helpful in understanding the contents of the program.	①	②	③	④	⑤
6. The topics covered were relevant to my country's needs for Technology Roadmapping.	①	②	③	④	⑤
7. Enough participation opportunities were provided throughout the program.	①	②	③	④	⑤
8. The program has influenced in changing my mindset.	①	②	③	④	⑤
9. The duration of the program (7 days) was sufficient.	①	②	③	④	⑤
10. I would recommend this program to others.	①	②	③	④	⑤

Questions	Strongly Disagree	Disagree	Neutral	Agree	Strongly Agree
11. Are there any suggestions for improvement or requests on the overall procedure? If so, please explain in detail (Program process, contents, results, Sustainability, etc.).					
12. If you think your capacity improved through this project, please write your score compared to 2023 (e.g. Score rangs 1(low) – 5(high)) and describe your experience applying the knowledge learned in your work/practice (policy reflection, institutional work, manpower training, etc.) in detail.					

[Figure 8-1] Program Evaluation Survey Results

2.6 Photos and Media


2024
K-Innovation
Partnership
Program
with Perú


NETFLIX

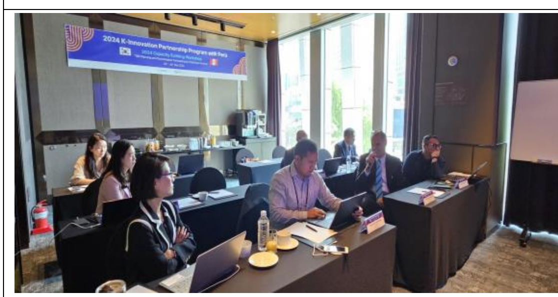

2024 K-Innovation Partnership Program with Perú
2024 Capacity Building Workshop


SKY GATE
2024 K-Innovation Partnership Program with Perú
2024 Capacity Building Workshop
"S&T Planning and Technological Forecasting for Prioritized Sectors"
20th - 24th May 2024

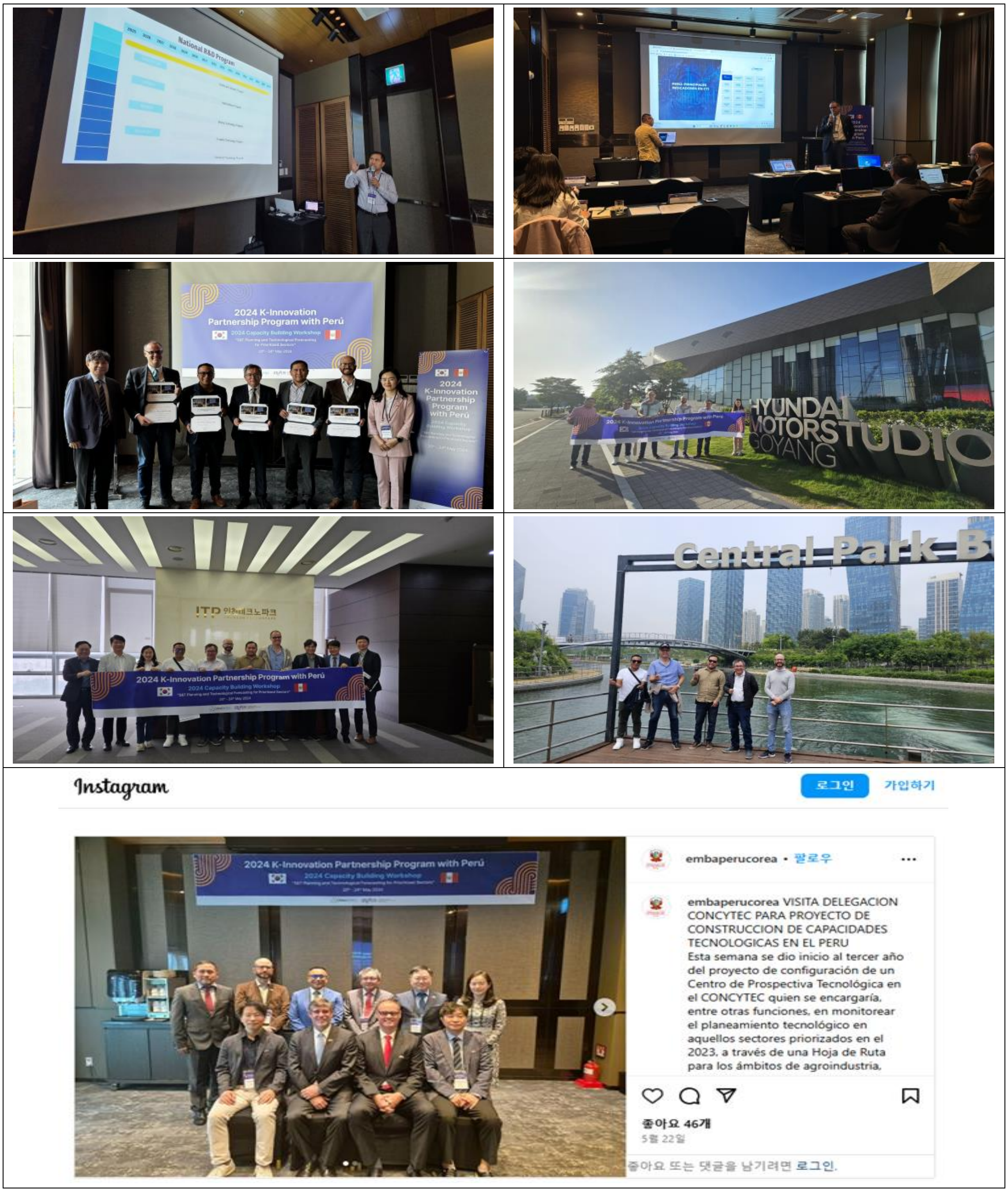

2024
K-Innovation
Partnership
Program
with Perú


청와대
국민품으로


2024 K-Innovation Partnership Program with Perú


2024 K-Innovation Partnership Program with Perú


2024 K-Innovation
Partnership Program with Perú
2024
K-Innovation
Partnership
Program
with Perú
2024 Capacity
Building Workshop



Source: Embajada del Perú en Corea Instagram (May 22, 2024), 「VISITA DELEGACION CONCYTEC PARA PROYECTO DE CONSTRUCCION DE CAPACIDADES TECNOLOGICAS EN EL PERU」, https://www.instagram.com/embaperucorea/p/C7QGSBkBGdt/?img_index=1 (Search date: November 1, 2024)

3. Dissemination Workshop & Media Day in Peru (October 5~13, 2024)

3.1 Dissemination Workshop and Media Day Overview

- Date: October 5-13, 2024 (Including the day of entry and exit)
- Working Language: English, Spanish
- Venue: CONCYTEC, PUCP, Lima, Peru

3.2 Objectives and Activities

- Organize a local capacity-building and dissemination workshop aimed at planning an R&D program for CONCYTEC practitioners and relevant stakeholders (government policymakers, research institutions, universities, private sector entities) focused on S&T planning and technology forecasting in six prioritized sectors.
- Facilitate in-depth discussions with participants from the second-stage Delphi technology foresight survey and the technology roadmap analysis team (a joint research team from STEPI and CONCYTEC) to align perspectives and refine insights.
- Develop a draft proposal for the R&D program and Technology Foresight Center through comprehensive theoretical and practical sessions.
- Host a Media Day, co-organized by CONCYTEC and STEPI, to present the draft R&D program and Technology Foresight Center establishment plan, and to publicize related research outcomes.
- Review the progress of the 2024 project and disseminate the achievements of the 2024 K-Innovation Partnership Program with Peru.
- Conclude the three-year 2022-2024 project and explore planning for future collaborative initiatives.

3.3 Program Schedule

Date	Activities
October 5 Korean delegation' Arrival in Peru	
October 6 Korean meeting	
October 7	
09: 30 - 10:30	Visit the KOICA Peru Office
11: 00 - 12:00	Visit the Korean Embassy in Peru
13: 00 - 14:00	Pre-meeting between STEPI and CONCYTEC
October 7, R&D planning and foresight center establishment: Workshop (1) - Venue: CONCYTEC in San Borja (Calle Morelli 150 - San Borja) - Participants: STEPI, CONCYTEC, Korean and Peruvian experts	
14:30 - 15:00	2nd Delphi survey results (by Dr. Sehee Park)
15:00 - 15:30	Korea's R&D governance (by Dr. ChiUng Song)
15:30 - 16:00	R&D planning guidelines and agency roles (by Prof. Johng-Ihl Lee)
16:00 - 16: 15	*Coffee Break*
16:15 - 16:45	Peruvian R&D budget and Process (by Peruvian expert)
16:45 - 17:15	Draft R&D and technology foresight center plan in Peru (by Peruvian experts)
October 8, Meeting of the Korean research team	
- Review of the results from the 2nd stage technology roadmap analysis and 2022-2024 project progress - Review of the Peru R&D program plan and the establishment of the technology foresight center plan. - Participants: STEPI and Korean experts	
October 9, R&D planning and foresight center establishment: Workshop (2) - Venue: CONCYTEC in San Borja (Calle Morelli 150 - San Borja) - Participants: STEPI, CONCYTEC, domestic and local experts	
08:30 - 09:30	Korea's experience about technology foresight centers establishment (by Dr. Hyun Yim)
09:30 - 10:30	Applying technology foresight for planning R&D program and technology foresight center (by Prof. Ahreum Hong).
10:30 - 10:45	*Coffee Break*
10:45 - 12:15	Finalizing Draft R&D and technology foresight center plan in Peru (by Peruvian experts) (1)

12:25 - 12:30	*Coffee Break*
12:30 - 14:30	Finalizing Draft R&D and technology foresight center plan in Peru (by Peruvian experts) (2)

October 10, Media Day to disseminate the draft R&D program and foresight center plans

- Venue: PUCP in San Isidro (Av. Camino Real 1075 – San Isidro)
- Jointly Hosted by: CONCYTEC and STEPI
- Participants: STEPI and the domestic expert team
- Target Audience: CONCYTEC, local experts in six key sectors, stakeholders from Peru's central and local governments in science and technology, companies, universities, research institutions, the general public, and members of the Peruvian media.

10:00 - 10:30	Opening Ceremony (by Dr. Sixto Sánchez, Presidente del CONCYTEC) (by Dr. ChiUng Song, Senior Research Fellow de STEPI) (by Dr. Eduardo Ismodes, Vicerrector de Investigación de la PUCP)
10:30 - 11:00	Sharing of key research outcomes from 2022-2024 (by Dr. ChiUng Song and STEPI team)
11:00 - 11:30	*Coffee Break*
11:30 - 12:20	Linking R&D and Technology Foresight with the Private Sector (by Marisela Benavides, Development Consultant) Proposal for the Establishment of a Technology Foresight Entity in Peru (by Carlos Hernández, Executive Director of the Graduate School – PUCP) (by Felipe Valentín, Technology Foresight Specialist – CONCYTEC) R&D Governance (by Jhon Collantes, Econometrics Specialist – CONCYTEC)
12:20 - 12:30	Closing remarks (Dr. Sixto Sánchez, Presidente del CONCYTEC)

October 11, Wrap-up meeting

Venue: CONCYTEC in San Borja (Calle Morelli 150 - San Borja)

10: 00 - 11:00	Review of the 2022-2024 project outcomes Final project evaluation	
11:00 - 11:30	*Coffee Break*	
11:30 - 12:30	Discussion on new projects.	
October 12 **October 13**	Departure from Lima → Arrival in Incheon	STEPI and Korean experts

3.4 Results

3.4.1 2nd Delphi survey results (Dr. Sehee Park)

Dr. Sehee Park presented the findings of a Delphi survey aimed at forecasting technology needs and priorities across six key sectors in Peru: Advanced Manufacturing, Agribusiness, Creative Industries, Forestry, Mining and Materials, and Textile and Apparel. The survey, conducted in two rounds, gathered insights from experts to identify critical technologies, preferred acquisition strategies, and investment priorities for each sector. Consensus was measured through methods like the t-test and Kendall's W, with increased agreement observed in the second round, reflecting a refined understanding of technological priorities and the necessary infrastructure for each industry.

For Agribusiness, the survey identified food processing as the most capable and immediate priority, while agrobiology and smart farming showed strong future potential. Policies recommended for this sector include enhancing infrastructure and fostering international R&D collaborations with countries like the US and the EU. In the Textile and Apparel sector, experts highlighted textile manufacturing as a short-term priority, with digital design anticipated to gain importance in the long term. The survey suggested that domestic R&D and infrastructure improvements would be essential for supporting growth in this industry.

In the Creative Industries, digital content and marketing services emerged as key focus areas, with financial services gaining importance for future development. Experts recommended policies to enhance manpower training and international partnerships for effective technology acquisition. Advanced Manufacturing, on the other hand, highlighted the importance of automation, clean manufacturing, and robotics, with an emphasis on training programs and collaboration with global partners outside the traditional markets of the US, EU, and Asia.

The survey also addressed Forestry, where environmental management and precision machinery were prioritized, suggesting the need for collaboration with international research institutes. Across all sectors, the findings underscored the importance of targeted policies and strategic partnerships for technology transfer and innovation. These insights are intended to guide Peru's R&D and technology adoption strategies, helping the country build a sustainable and competitive technological infrastructure.

3.4.2 Korea's R&D governance (Dr. ChiUng Song)

Dr. ChiUng Song's presentation covered the structure and purpose of Korea's Science, Technology, and Innovation (STI) governance model, highlighting the processes and institutions involved in managing and allocating R&D resources. The presentation outlined the roles and functions of key organizations, including the Ministry of Science and ICT (MSIT) and the Presidential Advisory Council on Science and Technology (PACST), which work in tandem to coordinate national STI policies and set the R&D budget. This governance structure aims to foster a systematic approach to technological advancement and innovation, supporting Korea's competitive position in the global technology landscape.

The STI governance system in Korea involves a hierarchical, well-coordinated structure that integrates government agencies, advisory councils, and research institutions to align national R&D objectives with strategic economic goals. The National Science and Technology Council (NSTC), chaired by the President, plays a pivotal role in setting the agenda and overseeing the draft of the national R&D budget. The NSTC's agenda is developed in collaboration with PACST, which provides specialized advice on science and technology priorities, helping shape policies that address both immediate and long-term needs.

A significant part of the presentation focused on the R&D budgeting process, where MSIT serves as the central coordinating body, working closely with other ministries and stakeholders. This process ensures that R&D funding aligns with Korea's broader development goals, with budget allocations reviewed and approved based on national priorities. The budgeting approach fosters transparency and accountability, aiming to maximize the impact of R&D investments and encourage sustainable innovation across multiple sectors.

Dr. Song also touched on how elements of Korea's STI governance model could be applied to the Peruvian context, suggesting that establishing a similar structure could help Peru streamline its STI policies and promote collaborative efforts across government, academia, and industry. By adopting aspects of Korea's model, Peru could enhance its R&D management capabilities, improve coordination among stakeholders, and drive strategic investments in science and technology, ultimately boosting the country's innovation capacity and economic growth.

3.4.3 R&D planning guidelines and agency roles (Dr. Johng-Ihl Lee)

Prof. Johng-Ihl Lee's presentation provided a comprehensive overview of key factors, principles, and practices essential to effective R&D program development, particularly focusing on the roles of agencies and stakeholders. He emphasized that R&D planning is crucial for organizing efforts to advance technology and meet national strategic goals, addressing factors like priority setting, legal and financial infrastructure, and collaboration across sectors. He explained that prioritizing R&D areas, ensuring stable legal frameworks, and maintaining timely resource allocation are fundamental to driving impactful research outcomes.

The presentation introduced ten principles guiding successful R&D programs, including clear goal setting, legal and financial infrastructure, demand-based program design, intermediary roles, and dissemination of results. Prof. Lee stressed that each program should be demand-driven, responding to needs from both industry and public stakeholders, with agencies playing a central role in planning and managing projects. Evaluation mechanisms are essential for assessing program success, with methods such as peer reviews and Delphi surveys providing objective measurements of progress and impact.

Prof. Lee discussed the roles of government and agency stakeholders in the R&D lifecycle, detailing how central and local governments provide regulatory support, funding, and guidelines to facilitate agency-led initiatives. Agencies, on their part, are tasked with program planning, managing relations with stakeholders, and ensuring that programs align with national goals. Intermediaries like universities and public research institutes are critical partners, often serving as the bridge between research efforts and industry applications.

The presentation also highlighted specific R&D program structures, including examples like the Industrial Technology Development Program for SMEs (ITDP) and the Collaborative R&D Program between Peru and Korea. These programs illustrate the need for tailored frameworks that include goal setting, budget allocation, evaluation criteria, and legal guidelines to support cross-border research initiatives. Prof. Lee concluded by reinforcing the importance of systematic planning, continuous evaluation, and strategic partnerships in establishing successful R&D programs that meet both domestic and international technological needs.

3.4.4 Korea's experience about technology foresight centers establishment (Dr. Hyun Yim)

Dr. Hyun Yim's presentation outlined the structure, roles, and importance of the Technology Foresight Center at KISTEP (Korea Institute of Science and Technology Evaluation and Planning), a key organization for science and technology policy in South Korea. The Center's purpose is to provide strategic guidance through activities such as environmental scanning, identifying key technologies, building a national S&T vision, and supporting decision-making. This comprehensive foresight system aids Korea's STI governance by anticipating technological trends, managing R&D resources, and aligning long-term goals with national priorities.

A major role of the Technology Foresight Center is conducting environmental scanning, which involves monitoring and analyzing technology-related issues to anticipate future trends and provide early warnings. The Center also identifies emerging technologies with high potential, conducting foresight exercises every five years to establish priorities. These exercises aim to match Korea’s strengths with national needs, ensuring that resources are directed toward impactful areas. Through systematic assessment and priority setting, the Center helps enhance Korea’s technological competitiveness.

Vision-building is another key activity, wherein the Center develops long-term strategies and visions for Korea's future. Every decade, a strategic vision is crafted, outlining societal, economic, and technological goals, as seen in Korea's S&T Future Strategy for 2045. This vision guides policy direction and ensures that S&T development aligns with broader societal aspirations, such as creating a sustainable, equitable, and technologically advanced society. The Center's work in vision-building thus plays a pivotal role in shaping the national S&T agenda.

Additionally, the Center supports decision-making through targeted foresight projects and technology assessments on specific issues, such as energy consumption and digital technology's impact. These activities assist government agencies in navigating complex policy landscapes by providing insights into emerging challenges and potential solutions. The presentation concluded with insights on how a similar foresight center in Peru, supported by CONCYTEC, could adapt these practices to strengthen national S&T policy, prioritize critical technologies, and foster innovation.

3.4.5 2nd Delphi survey results (Prof. Ahreum Hong)

Prof. Ahreum Hong's presentation emphasized the transformative role of technology foresight in Peru's national development, particularly with the establishment of a Technology Foresight Center. This center aims to strengthen Peru's capabilities in technology monitoring, scenario development, and policy advisory. Recognizing the impact of digital transformation and Industry 4.0, Dr. Hong highlighted the need for Peru to develop a robust foresight infrastructure that leverages AI to enhance decision-making in six key ministries. The center's goals align with positioning Peru as a leader in technology standards and digital innovation within Latin America.

The foresight center's activities will focus on environmental scanning to anticipate technological trends, a practice crucial for addressing rapid technological changes. AI-driven tools like LDA/BERT topic modeling will assist in identifying critical KPIs for tracking technology advancements across sectors. The center will not only monitor emerging trends but also establish a structured framework for integrating AI research, which will provide insights for public R&D initiatives. Through these methods, the center is expected to provide scalable evaluation and personalized learning tools, significantly improving Peru's response to technological challenges.

Prof. Hong outlined a comprehensive R&D program proposal designed to support Peru's priority industries, including agribusiness, mining, forestry, textiles, advanced manufacturing, and creative industries. The program is structured to develop and implement technology roadmaps (TRMs) for each sector, focusing on both immediate and long-term needs. Each sector's TRM will identify the top-priority products and processes, with AI tools applied to enhance planning and evaluation. The proposal also calls for a foundation of IT infrastructure to support inter-ministerial collaboration, making it easier for these sectors to adopt emerging technologies and meet innovation targets.

To operationalize the foresight center, Prof. Hong proposed a phased approach with short, mid, and long-term KPIs. The short-term phase will focus on establishing foundational capabilities and collaborative networks, while the mid-term phase will emphasize digital innovation and Industry 4.0 alignment. In the long term, the center is expected to lead global foresight efforts by harnessing advanced AI and green technologies. The KPIs developed will guide the foresight center's progress and ensure alignment with Peru's broader S&T objectives, fostering sustainable and inclusive growth.

Prof. Hong concluded with policy recommendations to support the foresight center's integration with Peru's R&D ecosystem. Key proposals included developing an AI research function within the center, creating robust IT infrastructure for multi-ministerial cooperation, and using generative AI in public R&D for TRM and evaluation. By implementing these strategies, Peru aims to establish itself as a digital innovation leader in Latin America, with enhanced foresight and decision-making capabilities that will drive efficiency and position the country as a global standard-bearer in technology-driven development.

3.4.6 R&D planning and foresight center establishment (Draft)

During a collaborative workshop, Peruvian and Korean experts gathered to design a comprehensive plan for establishing a technology foresight and R&D center. Organized into three groups, each team explored critical aspects of the center's structure and objectives to guide its establishment and strategic vision.

The first group focused on strengthening the relationship between R&D, the Technology Foresight Center, and private sector stakeholders. Their approach emphasized fostering technology transfer, facilitating information exchange, and leveraging big data analytics. Highlighting Peru's rich biodiversity and regional diversity, the group advocated for a strategic framework that integrates the private sector, academia, and regional governments to drive sustainable development and decentralization. To enhance competitiveness, they proposed a collaborative approach to identify and address bottlenecks within key value chains and regional clusters, enabling targeted, technology-driven solutions. Drawing on Peru's experience with Regional Development Agencies, the group underscored the importance of collaboration among chambers of commerce, regional governments, academia, and civil society. They also emphasized that cooperation from South Korea would be essential in facilitating technology transfer, advanced data analysis, and monitoring systems to align foresight initiatives with evolving market demands and environmental challenges.

The second group focused on defining the center's mission, functions, staffing, budget, and a phased roadmap, envisioning a five-year transition from an internal unit within CONCYTEC to an autonomous entity. The proposed mission for the Peruvian Technology Foresight Center (CPTP) is to anticipate technological shifts, identify emerging key technologies, and construct long-term scenarios to support strategic decision-making for Peru's sustainable development. Serving as a foundation for long-term technology planning, the center will align with national priorities outlined in the National Strategic Development Plan 2050. Key functions will include continuous environmental scanning, periodic foresight exercises, and scenario development to inform Peru's science and technology policies. Initial staffing and budget are planned to grow gradually to support this phased approach, ensuring the center's capacity to meet its expanding mandate.

The third group examined Peru's governance framework for science, technology, and innovation, focusing on refining the roles of government bodies such as CONCYTEC and exploring the potential establishment of a Ministry of Science and Technology or a national advisory committee on R&D. They highlighted the strategic role of the Multisectoral Science, Technology, and Innovation Commission (CMCTI) in overseeing the National Science, Technology, and Innovation Policy (POLCTI), which aims to align STI activities with national goals and strengthen governance to support the foresight center.

These results were subsequently presented during Media Day, where the collaborative efforts and strategic insights gained through the workshop were shared. The presentation highlighted the vision for Peru's Technology Foresight Center, emphasizing its role in fostering sustainable development and innovation, enhancing private sector integration, and strengthening R&D governance

3.5 Photos & Media


Contents
Part A
I. The Factors of R&D Planning
II. The Principles of R&D Programs
III. The Roles and Responsibilities
Part B
IV. The Practices of Developing Programs


Concytec Políticos
Concytec Estudios
Concytec Data
Concytec Fondos


Programas de I+D
Hojas de Ruta Tecnológicas
Preliminares


10 de octubre
9:30 a.m.
Centro Cultural PUCP
Iniciamos la transmisión en breve

gob.pe

Inicio > El Estado > PCM > CONCYTEC > Noticias > CONCYTEC impulsará creación de Unidad de Prospe

Consejo Nacional de Ciencia, Tecnología e Innovación

CONCYTEC impulsará creación de Unidad de Prospectiva Tecnológica con apoyo de STEPI de Corea del Sur

Nota de prensa

15 de octubre de 2024 - 3:31 p. m.

El Consejo Nacional de Ciencia, Tecnología e Innovación (CONCYTEC), el Instituto de Políticas en Ciencia y Tecnología STEPI (siglas en inglés) de Corea del Sur y la Escuela de Posgrado de la Pontificia Universidad Católica del Perú (PUCP) llevaron a cabo el **"Seminario de Difusión: Programas I+D Hojas de Ruta Tecnológicas Preliminares"**, con el objetivo de impulsar la prospectiva tecnológica que permita ser una herramienta de desarrollo para el futuro de la ciencia, tecnología y la sociedad en el país.

Source: Plataforma digital única del Estado Peruano (15 de Octubre de 2024), 「CONCYTEC impulsará creación de Unidad de Prospectiva Tecnológica con apoyo de STEPI de Corea del Sur」, https://www.gob.pe/institucion/concytec/noticias/1040158-concytec-impulsara-creacion-de-unidad-de-prospectiva-tecnologica-con-apoyo-de-stepi-de-corea-del-sur
(Search date: November 1, 2024.)

Source: Concytec Perú Facebook(1 de Octubre de 2024), 「Seminario de Difusión: Programas de I+D Hojas de Ruta Tecnológicas Preliminares」, https://www.facebook.com/concytec/photos/concytec-y-stepi-de-corea-del-sur-te-invitan-al-seminario-de-difusi%C3%B3n-de-program/936922415136624/ (Search date: November 1, 2024)

Source: Concytec Perú Facebook: Viedeo(10 de Octubre de 2024.), 「Estamos en vivo con el seminario de difusión "Programas de I+D"」, https://www.facebook.com/concytec/videos/564774029412911 (Search date: November 1, 2024)

CHAPTER 9

Remarks & Implication

S&T Planning and Technological Forecasting for Prioritized Sectors in Perú

Chapter 9. Remarks & Implication

1. Wrap up of 3rd Year Activities

Historically, the technology has been playing a major role in economic growth by creating additional economic values from producing new goods and services. As a result, almost every country has tried to build technological capacity by investing on higher education and R&D activities. However, it has not been so easy for most countires, especially developing countries due to lack of financial resources. Thus, the government tried to secure certain level of financial resources and figure out the priority in resource operation. That is, the government needs to have well-designed methodologies and very specific guidlelines in setting the R&D program as well as making R&D budget. That's why we need to conduct the technology foresight, and construct the roadmap.

During the last two years. STEPI team and CONCYTEC have worked together in order to build capacity in implementing technology foresight and constructing technology roadmap for Peruvian experts. In 2022, STEPI team visited Lima and held the Disseminationa Workshop on October. During that span, STEPI team and CONCYTEC were able to transfer sufficient amount of knowledge on technology foresight, and Peruvian experts came to be familiar with concept and practice. In 2023, STEPI team and CONCYTEC held the Capacity Building Workshop in Seoul and the Dissemination Workshop in Lima respectively. During the courses, Peruvian experts representing 6 priortized sectors has made the draft of technology roadmaps successfully. They presented the main outcome by themselves in front of Peruvian public at the Media day.

In 2024, STEPI team and CONCYTEC more focused on R&D planning and introduction of technology foresight center. For the R&D planning, STEPI team has presented the current R&D system in Korea focusing on R&D governance. That is, STEPI team has shared the process of R&D planning along with the budget process. Then, STEPI team has suggested the number of policy implications, and CONCYTEC has accepted the main idea to prepare a tentative version of Peruvian R&D governance. At the same time, STEPI team explained about the Center for Technology Foresight at KISTEP including personnels, operation, mission and function so on. It would be helpful to CONCYTEC that prepares the launching of technology foresight unit with its organization.

2. Follow-up Management and New Project Planning

During the stay at Lima on October 2024, STEPI team and CONCYTEC held closed meeting at the headquarter of CONCYTEC. In this meeting, STEPI team and CONCYTEC discussed about the plan of launching follow-up project. On the basis of in-depth discussion, STEPI team and CONCYTEC decided to pursue the following three themes. First, both sides are going to establish the long-term masterplan for CONCYTEC. In this context, STEPI team will support CONCYTEC in developing organizational capabilities, introducing new sections, setting a new legal position and preparing R&D plans. Two sides are going to address the issue of launching a unit for technology foresight here. Second, STEPI team is going to support CONCYTEC in establishing policy measures to promote trilateral cooperation among academia, research and industry. It is expected that STEPI team would be able to provide bunch of policy implications from Korean experiences. Finally, STEPI team is going to support CONCYTEC in designing R&D infrastructure specialized in bio-technology and bio-engineering. The follow-up project has been discussed as a three-year plan from 2026 to 2028 with the theme of bio R&D infrastructure.

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APPENDICES

Attachment 1	A. Technology Capability : T-test results for Technology Capability

Category	P-value	Mean of 1st Year	Mean of 2nd Year	Statistical Significance
Agribusiness - Agro Chemicals				
HR	0.6725502	2.751852	2.713514	FALSE
Infra	0.6898077	2.492593	2.459459	FALSE
Invest	0.4447510	2.455556	2.389189	FALSE
Eco	0.4742842	3.192593	3.264865	FALSE
Agribusiness - Agro Mechanics				
HR	0.5048652	2.588889	2.535135	FALSE
Infra	0.5499724	2.381481	2.335135	FALSE
Invest	0.5462998	2.348148	2.297297	FALSE
Eco	0.4380749	3.100000	3.178378	FALSE
Agribusiness - Agro Biology				
HR	0.3041923	2.622222	2.529730	FALSE
Infra	0.4400858	2.388889	2.324324	FALSE
Invest	0.4673313	2.362963	2.302703	FALSE
Eco	0.5034094	3.122222	3.194595	FALSE
Agribusiness - Smart Farm				
HR	0.2624764	2.344444	2.243243	FALSE
Infra	0.3595516	2.185185	2.108108	FALSE
Invest	0.2595012	2.181481	2.081081	FALSE
Eco	0.5060463	3.077778	3.156757	FALSE
Agribusiness - Food Processing				
HR	0.5494899	3.018519	2.962162	FALSE
Infra	0.8192448	2.777778	2.756757	FALSE
Invest	0.5885330	2.681481	2.632432	FALSE
Eco	0.4702367	3.448148	3.524324	FALSE

Forestry - Saw Milling				
HR	0.5181332	2.659259	2.605405	FALSE
Infra	0.3501039	2.381481	2.308108	FALSE
Invest	0.5469864	2.296296	2.248649	FALSE
Eco	0.6592403	2.833333	2.875676	FALSE
Forestry - Environment Management				
HR	0.4872479	2.570370	2.508108	FALSE
Infra	0.5161042	2.344444	2.291892	FALSE
Invest	0.5250628	2.351852	2.297297	FALSE
Eco	0.5877233	2.988889	3.048649	FALSE
Forestry - Precision Machineries				
HR	0.1677459	2.411111	2.286486	FALSE
Infra	0.2034660	2.262963	2.156757	FALSE
Invest	0.3669637	2.237037	2.156757	FALSE
Eco	0.6203340	2.951852	3.005405	FALSE
Forestry - Surface Treatment				
HR	0.2584069	2.474074	2.378378	FALSE
Infra	0.2688694	2.348148	2.259459	FALSE
Invest	0.2253143	2.244444	2.145946	FALSE
Eco	0.7666403	2.929630	2.962162	FALSE
Forestry - Furniture Manufacturing				
HR	0.6893483	2.803704	2.767568	FALSE
Infra	0.9274287	2.500000	2.491892	FALSE
Invest	0.6213264	2.385185	2.340541	FALSE
Eco	0.7282965	2.937037	2.972973	FALSE
Textile And Apparel - Textile Materials				
HR	0.8558566	2.988889	2.972973	FALSE
Infra	0.9129768	2.785185	2.794595	FALSE
Invest	0.9348071	2.677778	2.670270	FALSE
Eco	0.5646354	3.362963	3.421622	FALSE
Textile And Apparel - Textile Manufacturing				
HR	0.8759788	3.181481	3.167568	FALSE

Infra	0.7783605	2.840741	2.864865	FALSE
Invest	0.7725126	2.718519	2.691892	FALSE
Eco	0.5365923	3.448148	3.513514	FALSE
Textile And Apparel - Apparel Manufacturing				
HR	0.7903477	3.225926	3.200000	FALSE
Infra	0.7254856	2.918519	2.951351	FALSE
Invest	0.9865110	2.725926	2.724324	FALSE
Eco	0.4585217	3.377778	3.454054	FALSE
Textile And Apparel - Digital Design				
HR	0.6449935	3.081481	3.037838	FALSE
Infra	0.9260273	2.807407	2.816216	FALSE
Invest	0.7517000	2.711111	2.681081	FALSE
Eco	0.5956280	3.285185	3.340541	FALSE
Textile And Apparel - Logistics				
HR	0.7478672	3.081481	3.054054	FALSE
Infra	0.8492638	2.822222	2.805405	FALSE
Invest	0.6091066	2.666667	2.621622	FALSE
Eco	0.4907573	3.303704	3.372973	FALSE
Mining And Materials - Resource Exploration				
HR	0.6246670	3.344444	3.389189	FALSE
Infra	0.5110664	3.085185	3.145946	FALSE
Invest	0.3904738	3.048148	3.129730	FALSE
Eco	0.5605962	3.629630	3.691892	FALSE
Mining And Materials - Clean Energy Mining				
HR	0.9318467	2.770370	2.778378	FALSE
Infra	0.8279067	2.633333	2.654054	FALSE
Invest	0.9974681	2.611111	2.610811	FALSE
Eco	0.3684612	3.411111	3.513514	FALSE
Mining And Materials - Geotechnical Engineering				
HR	0.8625161	3.011111	3.027027	FALSE
Infra	0.8659374	2.822222	2.837838	FALSE
Invest	0.8263803	2.785185	2.805405	FALSE

Eco	0.3959767	3.433333	3.524324	FALSE
Mining And Materials - Casting And Welding				
HR	0.9764515	3.029630	3.027027	FALSE
Infra	0.7404817	2.829630	2.800000	FALSE
Invest	0.9974746	2.788889	2.789189	FALSE
Eco	0.3360778	3.385185	3.486486	FALSE
Mining And Materials - Material Processing				
HR	0.9767630	3.029630	3.032432	FALSE
Infra	0.6840523	2.900000	2.859459	FALSE
Invest	0.9718021	2.862963	2.859459	FALSE
Eco	0.4685051	3.548148	3.627027	FALSE
Advanced Manufacturing - Digital Production				
HR	0.6102940	2.688889	2.643243	FALSE
Infra	0.7315242	2.511111	2.481081	FALSE
Invest	0.7152128	2.496296	2.464865	FALSE
Eco	0.9890431	3.144444	3.145946	FALSE
Advanced Manufacturing - Automation				
HR	0.6843576	2.625926	2.589189	FALSE
Infra	0.4810688	2.451852	2.389189	FALSE
Invest	0.4025997	2.459259	2.383784	FALSE
Eco	0.9511394	3.203704	3.210811	FALSE
Advanced Manufacturing - Robotics				
HR	0.6452980	2.329630	2.286486	FALSE
Infra	0.4474786	2.240741	2.172973	FALSE
Invest	0.5376637	2.285185	2.227027	FALSE
Eco	0.8852013	3.114815	3.097297	FALSE
Advanced Manufacturing - Clean Manufacturing				
HR	0.5058819	2.403704	2.345946	FALSE
Infra	0.3894389	2.296296	2.221622	FALSE
Invest	0.5756551	2.244444	2.194595	FALSE
Eco	0.6588021	3.159259	3.210811	FALSE
Advanced Manufacturing - Engineering Service				

HR	0.6052840	2.711111	2.664865	FALSE
Infra	0.6769145	2.544444	2.508108	FALSE
Invest	0.7231786	2.462963	2.432432	FALSE
Eco	0.6496538	3.244444	3.291892	FALSE
Creative Industry - Financial Service				
HR	0.7947281	2.925926	2.902703	FALSE
Infra	0.7188496	2.751852	2.718919	FALSE
Invest	0.6001723	2.729630	2.681081	FALSE
Eco	0.7983565	3.307407	3.281081	FALSE
Creative Industry - Marketing Service				
HR	0.8599330	3.048148	3.032432	FALSE
Infra	0.7158213	2.729630	2.697297	FALSE
Invest	0.6171007	2.703704	2.659459	FALSE
Eco	0.9269625	3.214815	3.205405	FALSE
Creative Industry - Media Service				
HR	0.8073764	2.977778	2.956757	FALSE
Infra	0.5922417	2.700000	2.654054	FALSE
Invest	0.4489181	2.677778	2.610811	FALSE
Eco	0.9250457	3.185185	3.194595	FALSE
Creative Industry - Digital Contents				
HR	0.8246119	3.048148	3.027027	FALSE
Infra	0.6084156	2.748148	2.702703	FALSE
Invest	0.5002960	2.737037	2.675676	FALSE
Eco	0.9684109	3.244444	3.248649	FALSE
Creative Industry - Copyright System				
HR	0.4520648	2.625926	2.562162	FALSE
Infra	0.4993594	2.459259	2.400000	FALSE
Invest	0.5022567	2.451852	2.394595	FALSE
Eco	0.9904455	3.114815	3.113514	FALSE

Attachment 2	B. Leading Country : Leading Country Responses

Country	1st Year	2nd Year	1st Year %	Rank 1yr	2nd Year %	Rank 2nd	Same Rank	Consensus	Difference %
Agribusiness - Agro Chemicals									
C/J/K	141	101	0.2611111	2	0.2729730	2	TRUE	FALSE	0.0118618619
Others	125	84	0.2314815	1	0.2270270	1	TRUE	FALSE	-0.0044544545
US/EU	274	185	0.5074074	3	0.5000000	3	TRUE	TRUE	-0.0074074074
Agribusiness - Agro Mechanics									
C/J/K	156	108	0.2888889	2	0.2918919	2	TRUE	FALSE	0.0030030030
Others	145	97	0.2685185	1	0.2621622	1	TRUE	FALSE	-0.0063563564
US/EU	239	165	0.4425926	3	0.4459459	3	TRUE	TRUE	0.0033533534
Agribusiness - Agro Biology									
C/J/K	158	113	0.2925926	2	0.3054054	2	TRUE	FALSE	0.0128128128
Others	136	83	0.2518519	1	0.2243243	1	TRUE	FALSE	-0.0275275275
US/EU	246	174	0.4555556	3	0.4702703	3	TRUE	TRUE	0.0147147147
Agribusiness - Smart Farm									
C/J/K	154	104	0.2851852	1	0.2810811	2	FALSE	FALSE	-0.0041041041
Others	156	103	0.2888889	2	0.2783784	1	FALSE	FALSE	-0.0105105105
US/EU	230	163	0.4259259	3	0.4405405	3	TRUE	TRUE	0.0146146146
Agribusiness - Food Processing									
C/J/K	124	93	0.2296296	1	0.2513514	2	FALSE	FALSE	0.0217217217
Others	128	80	0.2370370	2	0.2162162	1	FALSE	FALSE	-0.0208208208
US/EU	288	197	0.5333333	3	0.5324324	3	TRUE	TRUE	-0.0009009009
Forestry - Saw Milling									

C/J/K	157	108	0.2918216	2	0.2934783	2	TRUE	FALSE	0.0016566995
Others	127	82	0.2360595	1	0.2228261	1	TRUE	FALSE	-0.0132333926
US/EU	254	178	0.4721190	3	0.4836957	3	TRUE	TRUE	0.0115766931
Forestry - Environment Management									
C/J/K	113	70	0.2100372	1	0.1902174	1	TRUE	FALSE	-0.0198197834
Others	202	138	0.3754647	2	0.3750000	2	TRUE	FALSE	-0.0004646840
US/EU	223	160	0.4144981	3	0.4347826	3	TRUE	TRUE	0.0202844674
Forestry - Precision Machineries									
C/J/K	138	89	0.2565056	1	0.2418478	1	TRUE	FALSE	-0.0146577501
Others	176	120	0.3271375	2	0.3260870	2	TRUE	FALSE	-0.0010505899
US/EU	224	159	0.4163569	3	0.4320652	3	TRUE	TRUE	0.0157083401
Forestry - Surface Treatment									
C/J/K	139	91	0.2583643	1	0.2472826	1	TRUE	FALSE	-0.0110817036
Others	151	100	0.2806691	2	0.2717391	2	TRUE	FALSE	-0.0089300145
US/EU	248	177	0.4609665	3	0.4809783	3	TRUE	TRUE	0.0200117181
Forestry - Furniture Manufacturing									
C/J/K	176	118	0.3271375	2	0.3206522	2	TRUE	FALSE	-0.0064853726
Others	138	88	0.2565056	1	0.2391304	1	TRUE	FALSE	-0.0173751414
US/EU	224	162	0.4163569	3	0.4402174	3	TRUE	TRUE	0.0238605140
Textile And Apparel - Textile Materials									
C/J/K	227	163	0.4219331	3	0.4429348	3	TRUE	TRUE	0.0210016971
Others	125	77	0.2323420	1	0.2092391	1	TRUE	FALSE	-0.0231028770
US/EU	186	128	0.3457249	2	0.3478261	2	TRUE	FALSE	0.0021011799
Textile And Apparel - Textile Manufacturing									
C/J/K	231	167	0.4293680	3	0.4538043	3	TRUE	TRUE	0.0244363181

Others	122	73	0.2267658	1	0.1983696	1	TRUE	FALSE	-0.0283962340
US/EU	185	128	0.3438662	2	0.3478261	2	TRUE	FALSE	0.0039599160
Textile And Apparel - Apparel Manufacturing									
C/J/K	223	160	0.4144981	3	0.4347826	3	TRUE	TRUE	0.0202844674
Others	121	74	0.2249071	1	0.2010870	1	TRUE	FALSE	-0.0238201067
US/EU	194	134	0.3605948	2	0.3641304	2	TRUE	FALSE	0.0035356392
Textile And Apparel - Digital Design									
C/J/K	148	101	0.2750929	1	0.2744565	2	FALSE	FALSE	-0.0006364151
Others	149	98	0.2769517	2	0.2663043	1	FALSE	FALSE	-0.0106473250
US/EU	241	169	0.4479554	3	0.4592391	3	TRUE	TRUE	0.0112837401
Textile And Apparel - Logistics									
C/J/K	185	131	0.3438662	2	0.3559783	2	TRUE	FALSE	0.0121120899
Others	120	78	0.2230483	1	0.2119565	1	TRUE	FALSE	-0.0110918054
US/EU	233	159	0.4330855	3	0.4320652	3	TRUE	TRUE	-0.0010202845
Mining And Materials - Resource Exploration									
C/J/K	163	115	0.3029740	2	0.3125000	2	TRUE	FALSE	0.0095260223
Others	130	91	0.2416357	1	0.2472826	1	TRUE	FALSE	0.0056469210
US/EU	245	162	0.4553903	3	0.4402174	3	TRUE	TRUE	-0.0151729433
Mining And Materials - Clean Energy Mining									
C/J/K	129	88	0.2397770	1	0.2391304	1	TRUE	FALSE	-0.0006465169
Others	178	126	0.3308550	2	0.3423913	2	TRUE	FALSE	0.0115362858
US/EU	231	154	0.4293680	3	0.4184783	3	TRUE	TRUE	-0.0108897689
Mining And Materials - Geotechnical Engineering									
C/J/K	119	77	0.2211896	1	0.2092391	1	TRUE	FALSE	-0.0119504606
Others	171	113	0.3178439	2	0.3070652	2	TRUE	FALSE	-0.0107786488

US/EU	248	178	0.4609665	3	0.4836957	3	TRUE	TRUE	0.0227291094
Mining And Materials - Casting And Welding									
C/J/K	155	113	0.2881041	2	0.3070652	2	TRUE	FALSE	0.0189611282
Others	128	84	0.2379182	1	0.2282609	1	TRUE	FALSE	-0.0096573460
US/EU	255	171	0.4739777	3	0.4646739	3	TRUE	TRUE	-0.0093037821
Mining And Materials - Material Processing									
C/J/K	167	126	0.3104089	2	0.3423913	2	TRUE	FALSE	0.0319823824
Others	133	83	0.2472119	1	0.2255435	1	TRUE	FALSE	-0.0216684176
US/EU	238	159	0.4423792	3	0.4320652	3	TRUE	TRUE	-0.0103139648
Advanced Manufacturing - Digital Production									
C/J/K	143	97	0.2657993	1	0.2635870	1	TRUE	FALSE	-0.0022123000
Others	158	109	0.2936803	2	0.2961957	2	TRUE	FALSE	0.0025153548
US/EU	237	162	0.4405204	3	0.4402174	3	TRUE	TRUE	-0.0003030548
Advanced Manufacturing - Automation									
C/J/K	139	96	0.2583643	1	0.2608696	1	TRUE	FALSE	0.0025052529
Others	192	132	0.3568773	2	0.3586957	2	TRUE	FALSE	0.0018183288
US/EU	207	140	0.3847584	3	0.3804348	3	TRUE	TRUE	-0.0043235817
Advanced Manufacturing - Robotics									
C/J/K	136	87	0.2527881	1	0.2364130	1	TRUE	FALSE	-0.0163750606
Others	215	156	0.3996283	3	0.4239130	3	TRUE	TRUE	0.0242847907
US/EU	187	125	0.3475836	2	0.3396739	2	TRUE	FALSE	-0.0079097301
Advanced Manufacturing - Clean Manufacturing									
C/J/K	124	83	0.2304833	1	0.2255435	1	TRUE	FALSE	-0.0049397931
Others	207	144	0.3847584	2	0.3913043	3	FALSE	TRUE	0.0065459835
US/EU	207	141	0.3847584	2	0.3831522	2	TRUE	FALSE	-0.0016061904

Advanced Manufacturing - Engineering Service									
C/J/K	138	93	0.2565056	1	0.2527174	1	TRUE	FALSE	-0.0037881849
Others	164	114	0.3048327	2	0.3097826	2	TRUE	FALSE	0.0049498949
US/EU	236	161	0.4386617	3	0.4375000	3	TRUE	TRUE	-0.0011617100
Creative Industry - Financial Service									
C/J/K	103	66	0.1914498	1	0.1793478	1	TRUE	FALSE	-0.0121019880
Others	127	89	0.2360595	2	0.2418478	2	TRUE	FALSE	0.0057883465
US/EU	308	213	0.5724907	3	0.5788043	3	TRUE	TRUE	0.0063136415
Creative Industry - Marketing Service									
C/J/K	102	65	0.1895911	1	0.1766304	1	TRUE	FALSE	-0.0129606433
Others	131	90	0.2434944	2	0.2445652	2	TRUE	FALSE	0.0010707936
US/EU	305	213	0.5669145	3	0.5788043	3	TRUE	TRUE	0.0118898497
Creative Industry - Media Service									
C/J/K	104	70	0.1933086	1	0.1902174	1	TRUE	FALSE	-0.0030911589
Others	129	88	0.2397770	2	0.2391304	2	TRUE	FALSE	-0.0006465169
US/EU	305	210	0.5669145	3	0.5706522	3	TRUE	TRUE	0.0037376758
Creative Industry - Digital Contents									
C/J/K	112	72	0.2081784	1	0.1956522	1	TRUE	FALSE	-0.0125262647
Others	145	104	0.2695167	2	0.2826087	2	TRUE	FALSE	0.0130919670
US/EU	281	192	0.5223048	3	0.5217391	3	TRUE	TRUE	-0.0005657023
Creative Industry - Copyright System									
C/J/K	93	60	0.1728625	1	0.1630435	1	TRUE	FALSE	-0.0098189753
Others	113	76	0.2100372	2	0.2065217	2	TRUE	FALSE	-0.0035154356
US/EU	332	232	0.6171004	3	0.6304348	3	TRUE	TRUE	0.0133344109

Attachment 3	C. Strategic R&D Investment : Strategic R&D Investment in Peru

Industry	technology	year	short	mid	long
Advanced Manufacturing	Automation	first	3.353160	3.334572	3.275093
Advanced Manufacturing	Automation	second	3.380435	3.418478	3.255435
Advanced Manufacturing	Clean Manufacturing	first	3.256506	3.256506	3.405204
Advanced Manufacturing	Clean Manufacturing	second	3.342391	3.298913	3.451087
Advanced Manufacturing	Digital Production	first	2.509294	2.579926	2.397770
Advanced Manufacturing	Digital Production	second	2.445652	2.489130	2.304348
Advanced Manufacturing	Engineering Service	first	2.888476	2.617100	2.713755
Advanced Manufacturing	Engineering Service	second	2.864130	2.586957	2.744565
Advanced Manufacturing	Robotics	first	2.992565	3.211896	3.208178
Advanced Manufacturing	Robotics	second	2.967391	3.206522	3.244565
Agribusiness	Agro Biology	first	3.052045	3.107807	3.152416
Agribusiness	Agro Biology	second	3.043478	3.125000	3.146739
Agribusiness	Agro Chemicals	first	2.676580	2.661710	2.687732
Agribusiness	Agro Chemicals	second	2.695652	2.695652	2.614130
Agribusiness	Agro Mechanics	first	2.724907	2.661710	2.728625
Agribusiness	Agro Mechanics	second	2.815217	2.722826	2.815217
Agribusiness	Food Processing	first	3.494424	3.486989	3.315985
Agribusiness	Food Processing	second	3.434783	3.391304	3.326087
Agribusiness	Smart Farm	first	3.052045	3.081784	3.115242
Agribusiness	Smart Farm	second	3.010870	3.065217	3.097826
Creative Industry	Copyright System	first	2.773234	2.988848	2.907063
Creative Industry	Copyright System	second	2.793478	2.983696	2.809783

Creative Industry	Digital Contents	first	3.245353	3.163569	3.208178
Creative Industry	Digital Contents	second	3.211957	3.114130	3.157609
Creative Industry	Financial Service	first	3.029740	3.003717	2.985130
Creative Industry	Financial Service	second	3.043478	2.978261	3.032609
Creative Industry	Marketing Service	first	3.048327	3.011152	2.955390
Creative Industry	Marketing Service	second	3.070652	3.076087	3.054348
Creative Industry	Media Service	first	2.903346	2.832714	2.944238
Creative Industry	Media Service	second	2.880435	2.847826	2.945652
Forestry	Environment Management	first	3.505576	3.576208	3.568773
Forestry	Environment Management	second	3.619565	3.625000	3.630435
Forestry	Furniture Manufacturing	first	2.966543	2.795539	2.825279
Forestry	Furniture Manufacturing	second	2.885870	2.750000	2.880435
Forestry	Precision Machineries	first	3.267658	3.319703	3.230483
Forestry	Precision Machineries	second	3.266304	3.375000	3.233696
Forestry	Saw Milling	first	2.197026	2.171004	2.252788
Forestry	Saw Milling	second	2.228261	2.119565	2.163043
Forestry	Surface Treatment	first	3.063197	3.137546	3.122677
Forestry	Surface Treatment	second	3.000000	3.130435	3.092391
Mining And Materials	Casting And Welding	first	2.405204	2.379182	2.375465
Mining And Materials	Casting And Welding	second	2.391304	2.429348	2.315217
Mining And Materials	Clean Energy Mining	first	3.397770	3.516729	3.568773
Mining And Materials	Clean Energy Mining	second	3.440217	3.500000	3.625000
Mining And Materials	Geotechnical Engineering	first	3.011152	2.977695	2.951673
Mining And Materials	Geotechnical Engineering	second	3.059783	3.021739	2.978261
Mining And Materials	Material Processing	first	3.014870	3.085502	3.104089

Mining And Materials	Material Processing	second	2.983696	3.097826	3.081522
Mining And Materials	Resource Exploration	first	3.171004	3.040892	3.000000
Mining And Materials	Resource Exploration	second	3.125000	2.951087	3.000000
Textile And Apparel	Apparel Manufacturing	first	3.089219	3.003717	3.022305
Textile And Apparel	Apparel Manufacturing	second	3.108696	3.000000	3.016304
Textile And Apparel	Digital Design	first	2.825279	2.840149	2.996283
Textile And Apparel	Digital Design	second	2.788043	2.809783	3.016304
Textile And Apparel	Logistics	first	2.602230	2.594796	2.743494
Textile And Apparel	Logistics	second	2.630435	2.614130	2.733696
Textile And Apparel	Textile Manufacturing	first	3.371747	3.386617	3.237918
Textile And Apparel	Textile Manufacturing	second	3.407609	3.380435	3.239130
Textile And Apparel	Textile Materials	first	3.111524	3.174721	3.000000
Textile And Apparel	Textile Materials	second	3.065217	3.195652	2.994565